

Period Tracker App

By Billie Tinniswood

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Analysis

Background

The struggles of having a period affects half the population and there is no doubt it can leave strong imprints on the women affected. Symptoms can vary between every woman, but common factors include lower stomach pains, nausea or psychological changes such as emotional shifts. Consequently, this leads to anger, distress and embarrassment within the female community. These feelings can be further reinforced due to the lack of education within feminine hygiene, how to use products and the disregard of some doctors when it comes to concerns on period lengths, symptoms or what is considered to be normal.

My passion for women's health and wellbeing has led me to want to design an app which not only empowers all types of period owners but also allows them to track their cycle. An understanding of their cycle and symptoms can leave significant impacts on the daily lifestyles of women: it can decrease the chances of a period coming "unexpectedly" as the app will clearly show when the period is due, therefore allowing women to feel more relaxed and comfortable doing their everyday activities.

I am aiming to create an easy to use and well accessorized app to enhance a women's confidence and ability. I also aim to make women feel more supported and relaxed.

My clients include a few of my friends, who are ages 16 through to 18, and my mum who is 43. The reason I have clients in a vast age range is because I will design my app to help women of all ages; this will allow me to target my audience better.

Description of the current system

Undoubtedly, other apps do exist. Some of the well know ones include Flo, Clue and luna. Here I will take a closer look at these apps and why they are so popular. Ill also look at some of the features included in them but the possible limitations of these apps. My aim is to overcome their limitations and to use their strengths to further adapt my app.

Flo

Flo is worlds most used app to track female health. The main aim of the app is to "build a better future for female health by helping to harness the power of body signals" [Flo Health UK Limited. (2024).]. Flo uses medical articles and a team of doctors to improve its advice to women. It incorporates a tracker for your period to help with ovulation and future cycles. It does this by using AI. Flo also is good as it offers a pregnancy mode. Women also have the chance within the app to talk to and get support from other women within the Flo community.

Flo promises to keep your data safe, however, they have broken this promise. Many articles state that Flo actually sells women's personal data to third parties for its promotional and marketing purposes. An investigative piece by the Wall Street Journal found that the app sells the users personal data to Facebook: it told Facebook when the user was on their period or intended to get pregnant [John D. McKinnon. (2021).].

This opens a massive window for a need of a new app. Many women deleted Flo after finding out that it sells their data to third parties. This furthers my want for creating a similar app, that does not sell data. Women deserve a safe app that does not sell their data, an app where they do not need to worry. This is morally wrong since Flo has brought shame and embarrassment to many women by doing this.

Another issue with Flo is that to unlock all the features, you must pay for "Flo premium". Features such as "health assistant", "symptom patterns and predictions" and "cycle trends" are unlocked only when the user pays. Should women have to pay for features like these? There's a whole thing on that women should not have to pay for sanitary products. So why should they have to pay to help themselves track their cycle trends and reach further insights on how their body's health works? Should women have to pay to learn more about their symptoms and consequently how to cope with them?

No, is the answer. Why create an app for women's health, yet make women pay for deeper insights into their health and period, that is not fair. Knowledge into periods should be free since it is something women cannot help. It is a natural thing in a women's life, that it is also vital to be aware of and have knowledge about.

Flo is rated a 4.1/5 star on trust pilot, which emphasizes that some work needs to be done on this app and women are not 100% satisfied with it. A one star review from the trust pilot website read "I can't push anything or get any information without it pushing to pay for a subscription. It's ridiculous. Women just want help to understand their body's" [Trustpilot A/S. (2024).]. Which suggests that paying for a subscription is genuinely something women do not want to be doing. Another one star review read "I will NEVER pay for something that is supposed to "help me" or "educate me" when it's false." [Trustpilot A/S. (2024).]. This suggests that women are not happy with the content that Flo is putting out and may feel it is incorrect and not helpful to them .

Clue

Clue is a Berlin based female health app which helps women understand the inner workings of their body, beyond just bleeding. It's a science-based app which allows women to live in sync with their period, not just track it. What's great about it is that it allows women to enter in a confirmed diagnosis of 21 different health issues (such as endometriosis and PCOS) to help women to work around these.

Whilst Clue is free to download, it requires you to pay for Clue so you can unlock all the features. These include things such as personalized modes (e.g. Clue Perimenopause, Clue Conceive and Clue pregnancy) [Clue by Biowink GmbH. (2024).]. Other features included with Clue Plus are things such as planning ahead with 12 upcoming cycle predictions. The issue with this, again is that women should not have to pay for features that are period related, as again, this is a vital part of female health.

Clue is rated a 4.7/5 stars on the app store, showing that it is particularly popular. A one star review suggested it was angry at the app where you “Can’t use it offline” [Apple Inc. (2024).]. This feature of needing Wi-Fi or data to use the app may be frustrating for others to as women don’t pick when they get periods. It would be helpful not be able to use the app at any place and at any time, no matter of Wi-Fi connection.

Luna

Lunas app is designed to help teenage girls with their periods. It allows them to track their mood, skin, sleep and pain on top of their periods aswell. The app again is free to download but requires you to pay for all the features. The features included with payments include:

- Ability to ask anonymous questions
- Being able to read other teens questions
- Learning from articles and videos

Whilst the idea of this app is good, the payment to get “luna premium” [Luna Life Limited. (2023).] limits it. Also, whilst the app being aimed at teenagers may be good, this also limits the app. What do you do if you’ve been using this app since you got your first period and now you’re not a teen anymore?

Luna is rated a 3.8/5 star on google play, many people suggesting its great and helpful for “tweens” [Google Play. (2024).] and teenagers. However one review read “I think this is a great app but there’s not much information on the *learn button*” [Google Play. (2024).] . Considering that this app is made for teenagers, that probably are getting use to experiencing periods, the arguably the app should expand on this. Another review suggested that having a symptom tracker would make the app more worth it and that it seems pointless not having it: “seems kinda useless to me personally. Being able to see symptom suggestions is what makes a tracking app worth it for me” [Google Play. (2024).] .

Overall, Luna is a fantastic app for teenagers, but multiple users agree it lacks detail into advice and symptoms.

What will I take away from these apps?

Now that I have carefully considered the positive and negative aspects of these apps (based on user reviews and various articles) I can use the information to aid the development of my app.

1. I'd like to incorporate a help or advice page to display support to the user as all the apps have used something like this,
2. I'd like to incorporate something similar to Clues cycle predictions to help women predict and work around their upcoming period.
3. I'd like to be able to add in a mood tracker, similar to Lunas.
4. I'll use necessary security measures to ensure that the users data is protected.
5. If I was to publish this app, it would not include any in app purchases, unlike all of the above as information and support for periods should be free.
6. It will be available to women of all ages, which gives it an advantage over Luna.

identification of the prospective users

The app I am designing is aimed at women and girls of all ages, hence why my clients are of different ages. However, there will only be one type of user because women should be allowed to access all the information they would like. No one will have to pay for features, therefore there is only a need for one type of user as everyone already has all features unlocked for free.

identification of users' needs and acceptable limitations

After asking my friends their ideas for my project, they suggested that a period tracker app would be good since many women struggle with this aspect of their life. They initially came up with this project idea for me and I agreed it would be a great thing to develop. Once I had a strong project idea, I interviewed them to get a better idea of what they'd like the app to include.

Full details of the interviews (questions and answers) will be listed at the end of the analysis in the "evidence of appropriate analysis techniques" section, however, here is a summary of the things my clients have said:

I asked about what features they would like to be incorporated. They would like a feature to track their symptoms and mood. This would be helpful for them to be able to understand why they might feel a way on a certain day. They also would like a help page incorporated to ensure they feel supported and able to seek advice when needed. They potentially want a feature added in which they can personally record their mood and symptoms during the day, storing it as a kind of diary for them to read back through at any time. This can again offer them support as it gives them more insight into the future about why they may feel a certain way on a particular day.

I asked my mum about how she felt about this idea for my project and agreed to help by being interviewed aswell. She explained how, growing up, she didn't get much support and periods were not talked about and seen as gross. Because of this, she wants the help page to be age specific. For example, if you use this app and are age 13, the help page should be aimed around dealing with your first period, what to expect and who to talk to if you're

struggling. Whereas if you are 50 the help page will be aimed around support for menopause. This will help to ensure young girls and women both feel equally supported and normal.

Both my mum and my friends agreed on a period tracker app as this is an easy way to quickly access everything you need to ensure you feel supported. It is a vital part of a woman's life that should not be shameful.

To summarise, the key suggestions from my clients are as follows:

1. A cycle tracker
2. Symptom and mood tracker
3. Age specific help / advice page
4. Option to add personalized notes
 - The tracker for the period cycle is arguably the most important feature for my application as this is the whole point of the project, so this will be my main priority to get working in my project.
 - I will aim to incorporate a symptom and mood tracker as these are key features that my clients believe will be important to make the app stand out.
 - I will also aim to add an advice page as many of my clients suggested this will be extremely useful to use within the app.

However, I am giving myself a couple of acceptable limitations. These are the things I will try to include if I get enough time to. The incorporation of a symptom and mood tracker will be complex and significantly increase the volume of data in my project. Therefore this will only be included if I have enough time. Along with the user's ability to add personalized notes.

realistic appraisal of the feasibility of potential solutions

I asked my clients about their opinions on alternative methods of tracking their period and the advantages and disadvantages they thought of for these. This was an overview of what they said:

Solution 1:

Using pen and paper to track their cycle

Advantages of this:

1. Can be simple
2. Might be easier to notice mistakes that an app may not pick up on
3. Potentially more visual tracking

Disadvantages of this:

1. don't receive reminders and notifications
2. inconvenient having to carry around paper or a diary

3. may be difficult to predict next cycle, especially if your period is irregular
4. may lose paper as it may not all be in the same place, may just get confusing

Solution 2:

Using online software such as excel or a spreadsheet to track cycle

Advantages of this:

1. Could be helpful for overviews of your period
2. More straight forward than using pen and paper
3. Easier to remember for some as online a lot of the time
4. All the information in one space

Disadvantages of this:

1. Hard to comprehend and take in
2. A lot of work to create and keep up to date
3. Could be complicated for some people
4. Difficult to load up and update every time
5. “unnecessary” and “unneeded”
6. Not very convenient

Solution 3:

using an app to track cycle

Advantages of this:

1. Useful and easy option
2. Notifications, updates and reminders
3. Quick to access and does everything for you
4. Typically has other software that girls find helpful
5. Interactive and all the information is in one place for you

Disadvantages of this:

1. Sometimes not completely accurate, depending on the app

justification of the chosen solution

Overall, my clients seemed to find their experiences with an app a lot better than using any other way of tracking their period. They all agreed that its more convenient than using any other method for them. Here is an overview of their answers to the questions specifically on using an app:

How do you feel about using an app to track your period ?

[More Details](#)

4
Responses

Latest Responses

"Depending on the app I think it's the best way to track it "

"a lot easier and i would feel more comfortable using an app so would say is the best option "

"I would happily use an app, I do use an app now "

6. What advantages and disadvantages would you think would come from using an app to track your period ?

[More Details](#)

4
Responses

Latest Responses

"it's interactive, all the information is accessible and it's set up for you"

"Apps typically have other software in addition to tracking which many girls find useful, it would be much e..."

"It's quick to access, it's easy to use, it does everything for you, and it's private"

7. What would you pick to use to track your period ?

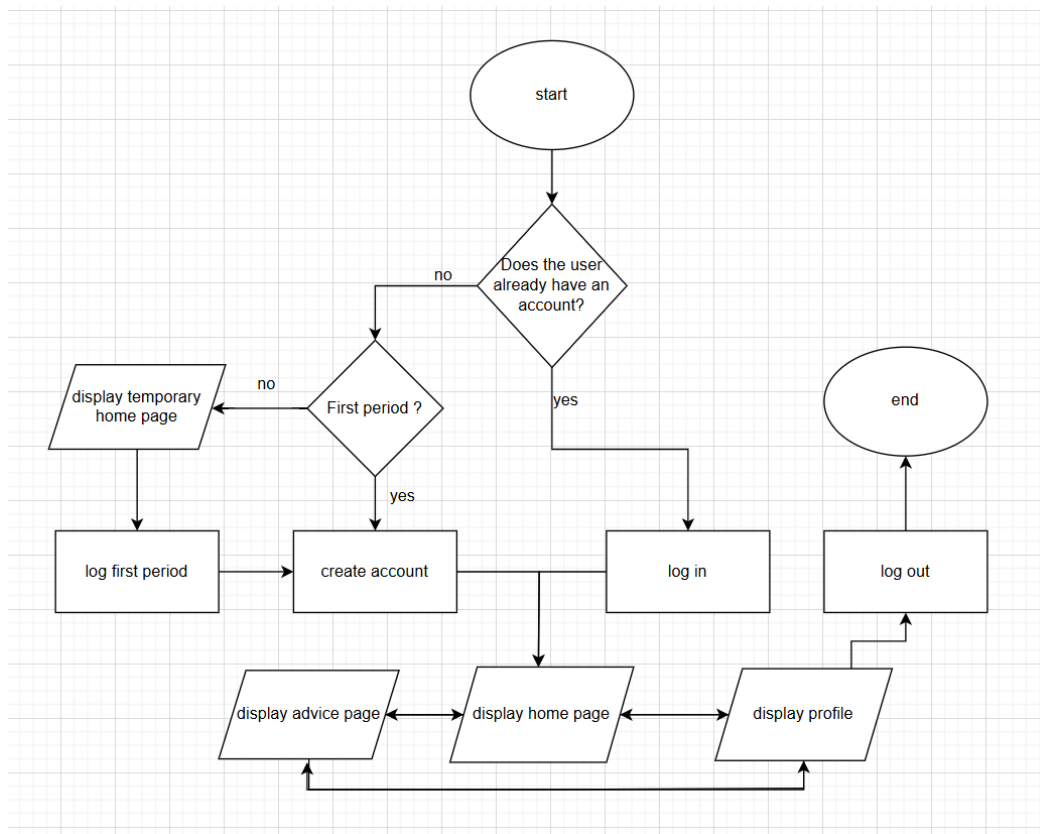
[More Details](#)

● Pen and paper	0
● spreadsheet/ excel	0
● An app	4



As you can see, they all preferred to use an app which suggests to me that this project idea is good.

data flow diagram for my program



This is a simple data flow diagram of how my program will work.

data sources and destinations

data will either be input into the system by me (the developer) or by the user. For example, the advice page will be input by me and displayed to the user, but the cycle tracker information will be input by the user since it is specific to them.

Users profile:

The questions are outputted onto the screen; these will be created by me and asks every user the same. Then the user inputs their information (this is dependent on if they are signing up or logging in). Their details from when they sign up are to be stored into a database so that they can always log back in.

Cyle tracker:

The user will enter what day of their cycle they are on, and roughly how long their period lasts for. This data is used to make a prediction on their upcoming cycle which will be output back to them on the screen. For example, if they say their cycle is roughly 30 days, their period is predicted every 30 days for the next upcoming cycles. If their period is late, the app will display that their period is late by however many days it is. The data they input here is stored in a database, so they do not have to reenter their information again unless they change it.

Help page:

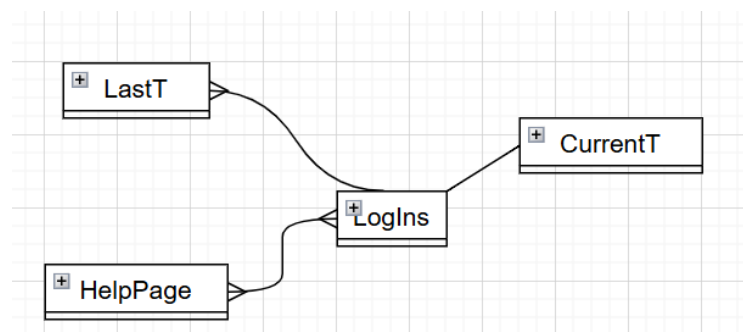
The contents of the help/advice page are input by me and displayed to the user based on different values they have stored in their database for example how old they are and what phase of their cycle they are in.

Personal profile:

The users personal profile page will be an output of their personal data. For example, the page will state their name, age, password etc. The user has the option to change the data in these fields if they wish to. The original data stored in the database will be overridden with this new data to avoid a buildup requiring lots of data.

entity-relationship model

Here is a basic idea of my database and how it will link together:



The LogIns table contains the primary key to link to the rest of the tables.

The attributes of the entities stored in the LogIns table are the data that are displayed in the user profile and are required for the user to log in.

The attributes of the entities stored in the CurrentT table are the data that is used to display the information on the Tracker and also to display appropriate information on the advice page.

The attributes of the entities stored in the LastT are used to display an overview of the user's previous cycle.

data analysis dictionary

Name	Data Type	Description	Example
UserName	string	Linked to the users ID and personal details. This is input by the user and checked to ensure its unique to every other user.	Username12
FirstName	string	Used within the app to make it feel more personal to the user.	Jennifer

UserID	Int	Linked to all the tables to ensure personalization for this one user. This is generated by the program.	172
Password	string	Generated by the user and must meet the requirements. Must be entered whenever the user turns starts the app.	Password12!
Birthday	DateTime	Used to add personalisation to the advice page.	12/09/2003
AvCycLen	Int	The average length that their period lasts, this is input by the user when making the account and can be changed. Cycle predictions are made upon this	28
CCycLen	Int	The current length of the user's cycle allows the app to say how many days late they are. Starts at 1 and increments by 1 every 24 hours	13
AvPLen	Int	The average number of days the users period lasts for. This is input by the user and used to make cycle predictions,	5
CPLen	Int	This is the current length of days that the user's period has lasted for. This is used to make cycle predictions	3
DaysLate	Int	This is calculated by subtracting the AvCycLen from the CCycLen when this value gets larger than the average. If the current is greater than the number of days late is 0.	2
StartDate	DateTime	This is calculated by using the datetime now and how long ago the user started their cycle. It holds the day their current cycle started.	03/02/2024
OnPeriod	bool	Is true if the user is on their period currently and is false if the user is not on their period currently.	true
Recent1	Date	This is the date that the recent period started. This is overridden by each cycle.	03/02/2024
Recent2	Date	This is the date that the recent period ended. This is overridden by each cycle.	08/02/2024
Length	Int	Holds the number of days that their period lasted within their previous cycle.	6

LastDaysLate	Int	Holds the number of days the user was late to start their period during their last period.	3
FirstPeriod	Bool	If the user has never had their first period this is used to direct them to a different homepage.	True
phase	int	Holds 1,2,3 or 4 to determine what phase of their period the user is on. This is used in the advice page to display relevant information to the user.	4
BC	Boolean	This is a true or false statement to represent if someone is on birth control or not, this will determine information on the advice page and on the tracker.	true

Core and advanced objectives

Core objectives:

1. Connect to a database
 - 1.1. Have a database coded into my project
 - 1.2. Have a connection string connecting this to the rest of the code
 - 1.3. Query to the database including inserts, updates, reads and deletes
2. Use of a sign up and log in system
 - 2.1. Data must be checked and validated when the user signs up (e.g. username cannot have been used before, password must meet requirements for password)
 - 2.2. Must be easy for the user to understand what data they should input
 - 2.3. When an account is created, they must be added to the database
 - 2.4. Log in only allowed once been checked against the database to see if the user exists
3. The system must predict when the user will get their next period
 - 3.1. Uses data stored in database to make a prediction of when their next period will be
 - 3.2. When a full cycle has passed, the system must use the data in the database and the data from the cycle that's just ended to update its predictions
4. Displays data about the user's current cycle
 - 4.1. Must display a model to show their cycle in some way
 - 4.2. Must display what day the user is on and their current phase
 - 4.3. Must display if the user is late to start their period
5. Displays data about the user's previous cycle

- 5.1. Uses data stored in database to display to user how long their cycle and period lasted, and if they were, how many days late they were (if they were)
6. Use of a help page to display advice to the user
 - 6.1. Use the user's birthday to work out their age and display appropriate information (age specific)
 - 6.2. Display appropriate advice for if they are on birth control or not
 - 6.3. Display information based upon what phase of their period they are in currently
7. Users are able to view their personal details
 - 7.1. Have user be able to view their username, first name and password stored on the database
8. Users are able to change their personal details
 - 8.1. Have user be able to change their username, first name and password that's stored on the database
9. Users are able to use a simplified version of the app if they have never had a period
 - 9.1. Have the app ask the user if they have had their first period when they create an account
 - 9.2. If they have not, the app is directed to a temporary homepage
 - 9.3. It also creates a temporary account for them
 - 9.4. Have a button that leads the user to finish creating their account
10. User able to delete account
 - 10.1. If they press the button to delete the account, a warning is displayed to check they definitely want to
 - 10.2. All their data is permanently deleted from the database
11. Able to log out without reloading the app
 - 11.1. The user is given the option to log out via a button
12. Navigation between all the pages works
 - 12.1. Navigation in log in and sign up system works
 - 12.2. Navigation between the tracker, advice page and user profile works
 - 12.3. Navigation for a user who has not experienced their first period works
13. Users use the app with ease and like the interface
 - 13.1. My clients feel like the app is easy to understand in terms of questions the system asks and the outputs displayed to them etc
 - 13.2. My clients can easily navigate between the pages in the app

Advanced objectives:

1. Implement a merge sort and binary search
 - a. Use of a merge sort to order the user ids that have been used
 - b. Use of a binary search to check a randomly generated user id doesn't exist in the sorted list
2. Implement a hash code for the password

- a. Hash code works
 - b. Stored instead of the password for security purposes
- 3. Symptom tracker
 - a. User can view common symptoms they may experience currently in their cycle
 - b. User can log symptoms
 - c. User can view their previously logged symptoms
- 4. Mood tracker
 - a. User can view common moods they may experience currently in their cycle
 - b. User can log moods
 - c. User can view their previously logged moods

evidence of appropriate analysis techniques

To gather information with my clients I used structured interviews. Within this I had a set list of questions which I will list below. I interviewed 4 people, 3 of which were my friends from college and the other being my mum. Their names will remain anonymous. Below I will write out what was said in them. Here is the preset list of questions:

1. Why do you think tracking your period is useful?
2. What key feature would you like included in an ideal period tracking app?
3. Within your period cycle, what do you struggle with the most?

Interview 1:

Interviewee: anonymous

Interviewer: Billie Tinniswood (BT)

Date: July 27th 2024

BT: "Why do you think tracking your period is useful?"

A: "So I can be prepared with products I need. Um also so that I can quickly see if I am late on my period for early signs of pregnancy or health issues."

BT: "What key feature would you like included in an ideal period tracking app?"

A: "I would like a section where I can track my moods, maybe as a diary sort of thing. This would be helpful as I can prepare myself for upcoming negative moods and try to avoid them by doing things that make me calm or happy."

BT: "Within your period cycle, what do you struggle with the most?"

A: "I think mainly just dealing with my moods as I get very irritable and angry at times because of my cycle. Also getting it at inconvenient times or inconveniently having to go out of lessons to change tampons or at the beach when I have to try find toilets."

Interview 2:

Interviewee: anonymous

Interviewer: Billie Tinniswood (BT)

Date: 9th August 2024

BT: "Why do you think tracking your period is useful?"

A: "It's helpful because you can prepare for it so you're not in the middle of nowhere when it starts with no products on you and it also helps me understand myself better."

BT: "What key feature would you like included in an ideal period tracking app?"

A: "A key feature could be explaining the different stages of your cycle like your follicular stage and stuff because it's interesting and not many people know about it well enough. I think having this in the form of a help page would be good."

BT: "Within your period cycle, what do you struggle with the most?"

A: "I struggle with pms so things like depression and anxiety during my period and I lose all my motivation due to my hormones. I think having more help and advice with this stuff would be more helpful for me."

Interview 3:

Interviewee: anonymous

Interviewer: Billie Tinniswood (BT)

Date: 9th August 2024

BT: "Why do you think tracking your period is useful?"

A: *"It honestly just helps me prepare to ensure I have enough products for myself beforehand and that I am ready and expecting it."*

BT: *"What key feature would you like included in an ideal period tracking app?"*

A: *"I'd like it to include an advice page in which girls of all ages can seek advice and help about their period. This would be especially useful for young girls that feel as if they can't talk about it with anyone else."*

BT: *"Within your period cycle, what do you struggle with the most?"*

A: *"When I first started my period, I wasn't taught fully at school or by my parents, so I sort of had to learn myself which was dangerous because of things like toxic shock syndrome and how to keep clean. Periods were seen as a very taboo and unspoken about subject and that's why I lacked knowledge on it growing up."*

Interview 4:

Interviewee: anonymous

Interviewer: Billie Tinniswood (BT)

Date: 20th August 2024

BT: *"Why do you think tracking your period is useful?"*

A: *"It's useful to know when roughly my period would be coming as it helps me to plan when I am able to do certain things, especially gym and fitness related things, because I get very strong symptoms like very bad cramps and pains and headaches. I also like tracking my cycle as its often very irregular."*

BT: *"What key feature would you like included in an ideal period tracking app?"*

A: *"I'd want to be able to track and record my symptoms. This would be extremely helpful because like I said I suffer with really extreme symptoms. Tracking these would help me*

understand my period more and learn how to prevent someone of them or at least be able to cope with them better.”

BT: “Within your period cycle, what do you struggle with the most?”

A: “Mainly just the symptoms I get like really bad cramping, headaches, fatigues, tiredness and feeling sick all the time.”

I also used google forms to survey my friends on their preferred ways of tracking their periods. This information is used in the “realistic appraisal of the feasibility of potential solutions” section and the “justification of the chosen solution” section.

These were the results of their responses:

1. How would you feel tracking your period using pen and paper ? *

It's not something I've ever done but it's a good option if I don't have an app to use

2. What advantages and disadvantages do you think would come from using pen and paper for tracking your period? *

Using pen and paper is more simple but unlike an app you won't receive reminders

3. How would you feel about using an online software such as excel and spreadsheets to track your period ? *

It seems complicated for something that could be noted down more simply

4. What advantages and disadvantages would you think would come from using things such as excel and spreadsheets to track your period ? *

It would be helpful for overviews of your period but it isn't very convenient

5. How do you feel about using an app to track your period ? *

It seems the most useful/easy option

6. What advantages and disadvantages would you think would come from using an app to track your period ? *

It's all tracked in one place and sends you notifications but they're not always very accurate

7. What would you pick to use to track your period ? *

- ☐ Pen and paper
- ☐ spreadsheet/ excel
- ☒ An app

1. How would you feel tracking your period using pen and paper ? *

I would feel weird.

2. What advantages and disadvantages do you thin would come from using pen and paper for tracking your period? *

I would forget to do it as it's a bit of an inconvenience to have to carry round pen and paper all day and find somewhere to write it. An advantage could be you won't make a mistake and not notice like you can digitally.

3. How would you feel about using an online software such as excel and spreadsheets to track your period ? *

It would be strange. There's no way I could do that, it wouldn't make sense and it's unnecessary and there would barely be a time I'd be able to fill it out

4. What advantages and disadvantages would you think would come from using things such as excel and spreadsheets to track your period ? *

It would be very difficult as you'd have to load it up every time and it's not quick and it's not needed.

5. How do you feel about using an app to track your period ? *

I would happily use an app, I do use an app now

6. What advantages and disadvantages would you think would come from using an app to track your period ? *

It's quick to access, it's easy to use, it does everything for you, and it's private

7. What would you pick to use to track your period ? *

- ☐ Pen and paper
- ☐ spreadsheet/ excel
- ☒ An app

1. How would you feel tracking your period using pen and paper ? *

Would find it very inconvenient

2. What advantages and disadvantages do you thin would come from using pen and paper for tracking your period? *

pen and paper would give a more potentially visual tracking however would be more difficult to predict next period etc if your cycle is irregular or just in general

3. How would you feel about using an online software such as excel and spreadsheets to track your period ? *

would be better than pen and paper as able to view on your phone on a day to day basis, but still wouldn't choose to do that

4. What advantages and disadvantages would you think would come from using things such as excel and spreadsheets to track your period ? *

would be more straightforward and possibly more likely to remember to track if online as most day to day uses are now virtual but then again having to load excel and create a spreadsheet could be quite complicated for some

5. How do you feel about using an app to track your period ? *

a lot easier and i would feel more comfortable using an app so would say is the best option

6. What advantages and disadvantages would you think would come from using an app to track your period ? *

Apps typically have other software in addition to tracking which many girls find useful, it would be much easier to use an app then have to open a spreadsheet or handwrite tracking. But, an app still might not be completely accurate on tracking

7. What would you pick to use to track your period ? *

- ☐ Pen and paper
- ☐ spreadsheet/ excel
- ☒ An app

1. How would you feel tracking your period using pen and paper ? *

It feels pointless as you'd lose track, lose the paper or it would be unorganised and unstructured

2. What advantages and disadvantages do you think would come from using pen and paper for tracking your period? *

I wouldn't find any advantages I like to have access to all information in one area where as I'd have several sheets of paper

3. How would you feel about using an online software such as excel and spreadsheets to track your period ? *

It's better than paper but I would find it hard to comprehend and take in

4. What advantages and disadvantages would you think would come from using things such as excel and spreadsheets to track your period ? *

As it's online it's easy to have all the information in one space but it's a lot of work to create

5. How do you feel about using an app to track your period ? *

Depending on the app I think it's the best way to track it

6. What advantages and disadvantages would you think would come from using an app to track your period ? *

it's interactive, all the information is accessible and it's set up for you

7. What would you pick to use to track your period ? *

- ☐ Pen and paper
- ☐ spreadsheet/ excel
- ☒ An app

Design Document

Overview

My app will be created in visual studio using the C# programming language. I will develop a mobile app using Xamarin forms and an emulator to run the project.

IPSO Chart

Inputs	Processes	Storage	Outputs
Sign Up details Input: Username, password, first name, birthday	generates random user id. Checks details haven't been used before. Checks all in correct format.	Store into database	No output of correct, error messages if wrong
Log in details Input:	Checks these are valid and are already in database to	Don't store as already in database	Welcome "" Or

Username, password	allow them to get into the system		Error message if incorrect
Tracker details Input: cycle length, current day, period length, birth control	Checks the inputs are valid answers. Creates a prediction for them.	Store into database	Displays tracker Or Error message if incorrect
Update personal profile: Username, password, first name, birth control	Updates database, overrides previous storage	Store in database	Shows new data on tracker and personal profile
Delete profile	Asks for conformation the removes all their data from database	No storage as it removes data	N/A
Log period	Calculates new average cycle, new period length, new ovulation date etc	Updates data in database	New tracker

Modular Design / Plan

I will be splitting my project into separate sections to make it easier to code. Here is the plan of how I am going to complete it:

- 1. Create a new project with the template Mobile App (Xamarin.Forms) and give it an appropriate name.**
- 2. Download the appropriate NuGet packages for the code.**
I will need to download multiple packages. Microcharts.Forms and SkiaSharp.Views.Forms will be downloaded to ensure that I can use a model in the main tracker of the project. sqlite-net-pcl will be downloaded so that I can create my database.
- 3. Create my data base**
I will be using SQLite to create my database, it will be coded manually into a form in the project. I will also need to code a connection from the database so that it can accessed elsewhere.
- 4. Create the pages**
The pages will be added next with appropriate names so that I can switch page easily whilst coding and be aware which page is which. Navigation between pages will be coded as I go through as I will be using things such as buttons
- 5. Code the user interface for the sign up and log in pages.**

The user interface will be created first so I can ensure all the entries fit on the page.

6. Make the Insert to create a user.

This will be connected to the database. Only users that don't already exist will be created.

7. Code the Log In page.

This will also be connected to the database so that only users that already exist can login and their username must be linked to their password.

8. Design the user interface and navigation between the main pages (tracker page, advice page and user profile)

9. Create the code behind the main pages

These will need to use the database in order to access all the data required for them

10. Create the homepage for the users who have not got a period.

This will navigate to create a user page when they log their first period and their account is deleted (as not a full account was created for them)

11. Create the home page for users who have got their period

This page will allow the users to see all their info for their current period and their most recent period. It will predict when they start their period next, display how late they are etc.

12. Create the user profile

This page will allow the user to view and update their personal details

13. Create the advice page

This page will display advice and help to the user about their period. The advice displayed will be based on data stored in the database

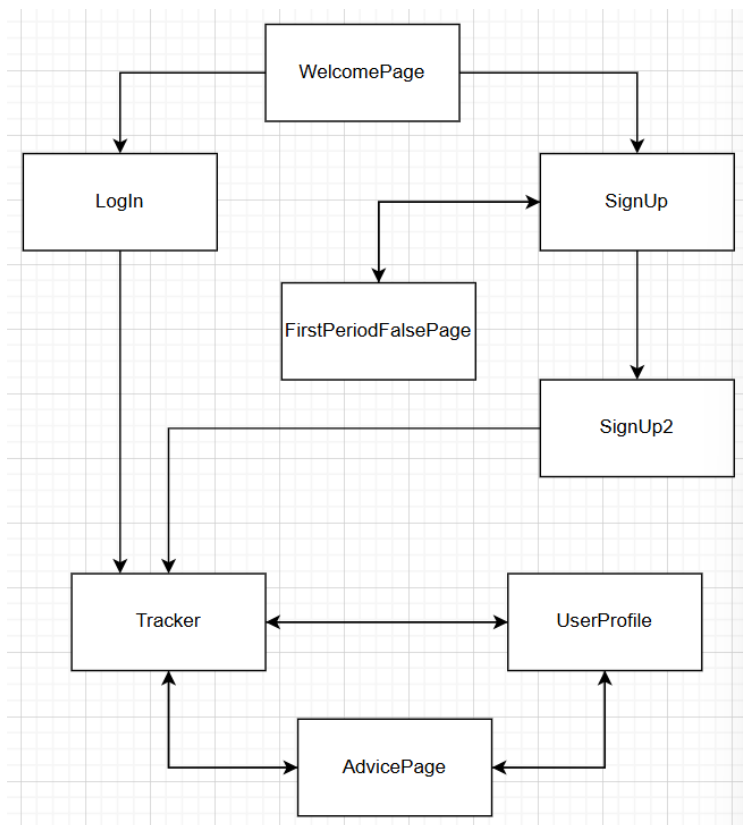
Navigation Document

My app will have multiple pages that can be navigated to in different ways. This is a description of each page, where it navigates to as well as the entry's, the output and the buttons that will be included on the page.

- WelcomePage
 - The user can select log in button (navigates to LogIn page)
 - The user can select Create an account button (navigates to SignUpPage)
- SignUp
 - First name entry
 - Username entry
 - Birthday entry
 - Password entry
 - If user says have had a period the navigates to SignUp2 page
 - If user says they haven't had a period, then navigates to FirstPeriodFalsePage
- SignUp2
 - Average cycle length entry

- Average period length entry
 - Current day entry
 - On birth control entry
 - Button pressed when info filled out and navigated to Tracker
- Login
 - Username entry
 - Password entry
 - Log in button that works if the username and password linked in database
- FirstPeriodFalsePage
 - Displays help for people who haven't had their period before
 - Button pressed to create full account (navigates to SignUp)
- Tracker
 - Button to navigate to AdvicePage
 - Button to navigate to UserProfile
 - Displays current day
 - Displays when next period due
 - Displays what phase the user is in
 - Displays Previous period information
- AdvicePage
 - Button to navigate to Tracker
 - Button to navigate to UserProfile
 - Uses database info to display advice to the user including symptoms and moods due to age or current phase
- UserProfile
 - Button to navigate to Tracker
 - Button to navigate to AdvicePage
 - Displays name as entry you can change
 - Displays username as entry you can change
 - Displays password as entry you can change
 - Displays birthcontrol as entry you can change

This is a diagram of how my pages will be linked:



The WelcomePage acts as a parent page for the log in and sign up pages. The user will access whichever of these is necessary for them. After logging into their account or creating an account, they will instantly be directed to the Tracker in which they can access the other pages from.

My GUI design meets the basic principles for good navigation design. It is consistent because it meets the users' expectations. My clients, who are all women (my target audience), wanted a period tracker, a symptom and mood tracker and an age specific help page. My design incorporates all of these. For example, the Tracker predicts when their next cycle will be and the advice page displays common symptoms, moods and is based off of age. It is simplistic, meaning it does not include anything more than it needs to currently. I can always add more features if I have time, but for now this is sufficient. It has clear paths and options; nothing too complicated. The navigation between the pages may change slightly when coding but for the most part it will remain like this.

Data Dictionary

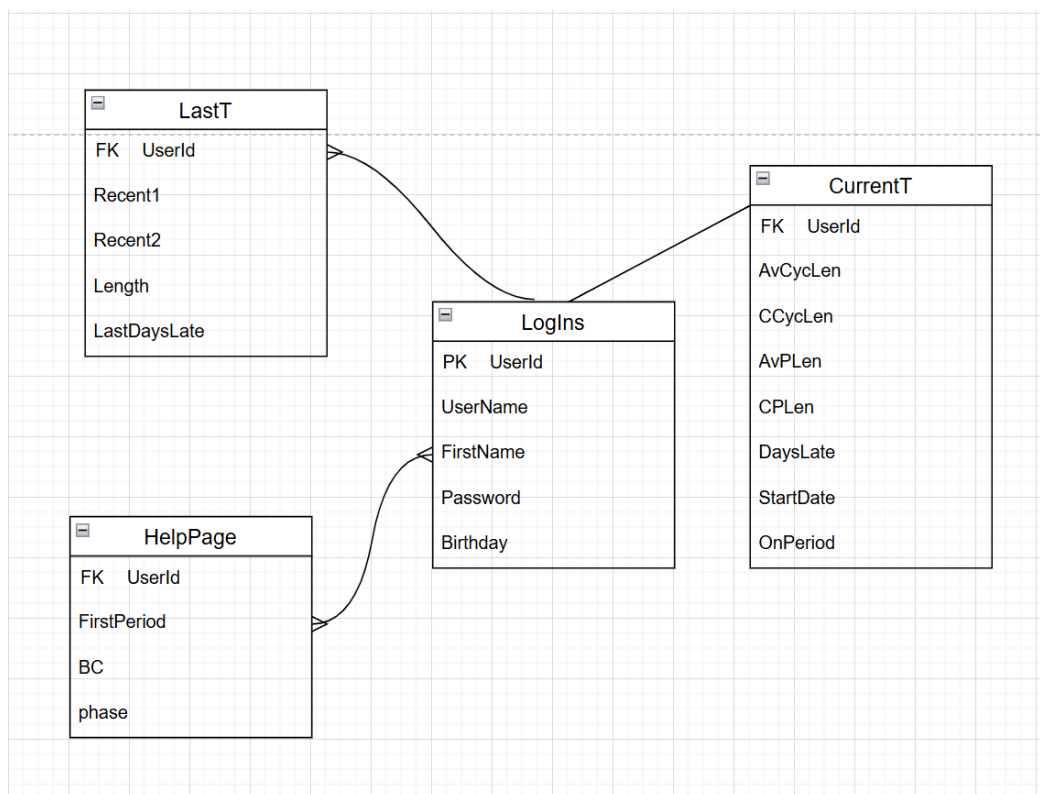
Name	Data Type	Description	Example
UserName	string	Linked to the users ID and personal details. This is input by the user and checked to ensure its unique to every other user.	Username12
FirstName	string	Used within the app to make it feel more personal to the user.	Jennifer

UserID	Int	Linked to all the tables to ensure personalization for this one user. This is generated by the program.	172
Password	string	Generated by the user and must meet the requirements. Must be entered whenever the user turns starts the app.	Password12!
Birthday	DateTime	Used to add personalisation to the advice page.	12/09/2003
AvCycLen	Int	The average length that their period lasts, this is input by the user when making the account and can be changed. Cycle predictions are made upon this	28
CCycLen	Int	The current length of the user's cycle allows the app to say how many days late they are. Starts at 1 and increments by 1 every 24 hours	13
AvPLen	Int	The average number of days the users period lasts for. This is input by the user and used to make cycle predictions,	5
CPLen	Int	This is the current length of days that the user's period has lasted for. This is used to make cycle predictions	3
DaysLate	Int	This is calculated by subtracting the AvCycLen from the CCycLen when this value gets larger than the average. If the current is greater than the number of days late is 0.	2
StartDate	DateTime	This is calculated by using the datetime now and how long ago the user started their cycle. It holds the day their current cycle started.	03/02/2024
OnPeriod	bool	Is true if the user is on their period currently and is false if the user is not on their period currently.	true
Recent1	Date	This is the date that the recent period started. This is overridden by each cycle.	03/02/2024
Recent2	Date	This is the date that the recent period ended. This is overridden by each cycle.	08/02/2024
Length	Int	Holds the number of days that their period lasted within their previous cycle.	6

LastDaysLate	Int	Holds the number of days the user was late to start their period during their last period.	3
FirstPeriod	Bool	If the user has never had their first period this is used to direct them to a different homepage.	True
phase	int	Holds 1,2,3 or 4 to determine what phase of their period the user is on. This is used in the advice page to display relevant information to the user.	4
BC	Boolean	This is a true or false statement to represent if someone is on birth control or not, this will determine information on the advice page and on the tracker.	true

Database Design

Here is an entity relationship diagram of my database:



- The primary key UserId is stored in the LogIns table, this is a foreign key in all the other tables.
- There is a many-to-many relationship between the LogIns and HelpPage tables because multiple users may be able to see if the same advice, and the page can change for one individual user as well.

- There is a one-to-one relationship between the LogIns and CurrentT page because a user only has one cycle at a time.
- There is one-to-many relationship between LogIns and LastT because a user can have multiple different previous cycle stored.

Validation of data

Field Name	Validation Checks	Description	Error Message	Data	Caught
Password	20>Length>= 6. Contains special character. Contains a number. Contains both lowercase and capital letter.	Only allow passwords that contain all of these and are the correct length.	Sorry, invalid password! Please enter another one that contains " ____".	Passwo rd12	Yes, because no special character
UserName	5<Length<13 String Format.	Need to be the correct length and be a string, however if they match another username already in the database it cannot be used.	Invalid sorry! Please make it longer/ Shorter. Sorry this username already exists!	User3	Yes, It is too short
FirstName	1<=Length<=10. Only characters.	Correct length is needed and correct format	Sorry, please enter a valid name. Sorry, please use a shorter name! Your nickname maybe?	jess4	Yes, contains number
Birthday	DataType(Date)	Make sure they have input a correct date of birthday.	Invalid! Please enter again as ##/##/####	7/2/203 2	Yes, as they cannot have been born then

AvCycLen	14<= value <= 90. Is an Integer.	Ensure they have input an integer that is an appropriate value	Sorry, please enter a valid number!	700	Yes, as this is not an appropriate size
CCycLen	1<= value <= AvCycLen. Is an Integer.	Ensure that they input a valid integer that is a valid size and is either smaller than or equal to the inputted AvCycLen	Sorry, please enter a valid number!	990	Yes, as this is not an appropriate size
AvPLen	1<= value <= 21 Is an Integer.	Make sure that the user inputs an integer and it within the correct range.	Sorry please ensure you entered a number!	three	Yes, as this is a string not an integer.
BC	Boolean. True or False. User selects Yes or No.	Ensure it is a yes or no answer or else its invalid.	I am unsure if you are on birth control please re answer !	yep	Yes, since it is not just yes or no
FirstPeriod	"yes" or "no" entered. Boolean.	Ensure it is a yes or no answer or else its invalid.	I am unsure if you have had your period before so please answer again	gdchj	Yes because it is not yes or no

SQL statements

Display information for users profile:

```
SELECT LogIn.UserName, LogIn.FirstName, CurrentT.AvCycLen, CurrentT.AvPLen
FROM LogIn, CurrentT
WHERE LogIn.UserId == 'current users id' && CurrentT.UserId == 'current users id'
```

Update LastT when user logs period

```
UPDATE LastT
```

```
SET Recent1 = DateTime.Now && Recent2 = StartDate
```

```
WHERE UserId = 'current users id'
```

Reset values when user logs period:

```
UPDATE CurrentT
```

```
SET CCycLen = '0' and CPLen = '0'
```

```
WHERE UserId = 'current users id'
```

Reset values when a user's period ends

```
UPDATE CurrentT
```

```
SET AvPLen = CCycLen
```

```
WHERE UserId = 'current users id'
```

Display users first name on advice page:

```
SELECT FirstName
```

```
FROM LogIn
```

```
WHERE UsersId = 'current users id'
```

Insert log in details when they sign up:

```
INSERT INTO Login (UserId, UserName, FirstName, Password, Birthday)
```

```
VALUES (random number, users input, user input, user input, user input)
```

Insert into HelpPage when they sign up:

```
INSERT INTO HelpPage( FirstPeriod, BC, phase)
```

```
VALUES (Boolean, Boolean, integer)
```

Insert into CurrentT when they sign up:

```
INSERT INTO CurrentT(AvCycLen, CCycLen, AvPLen, CPLen, StartDate)
```

VALUES (user input, user input, user input, calculated value, calculated value)

Update Users information:

UPDATE HelpPage, LogIns

SET HelpPage.BC = 'user entry' && LogIns.UserName = 'user entry' && LogIns.FirstName
= 'user entry' && LogIns.Password = 'user entry'

WHERE HelpPage.UserId == current user id && LogIns.UserId == current user id

Delete all of users information when the user deletes their account:

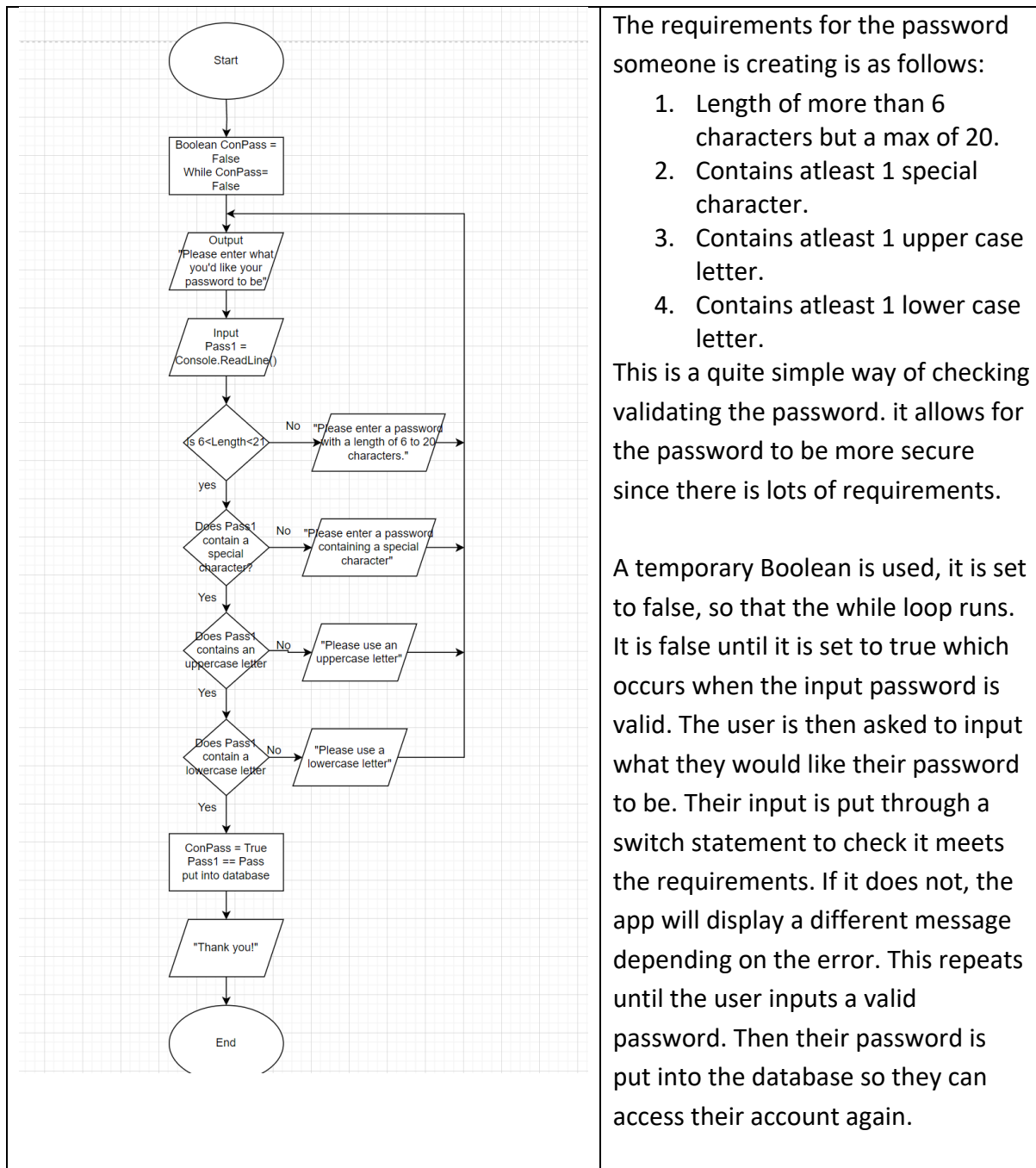
DELETE * FROM LogIn, CurrentT, LastT, AdvicePage

WHERE UserID = current users id

Algorithms

I am using a range of different algorithms to calculate different values in my program. Here are some examples.

Validation for password



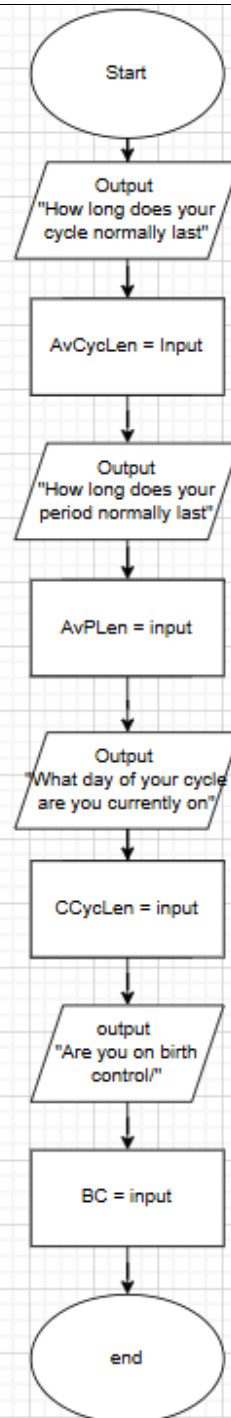
Gathering data about current cycle (when signing up)

This shows the very basics of how data is put into the database when the women initially sign up. Whilst it is considerably basic, it lays the foundations of data needed for lots of calculations to make predictions within the cycle.

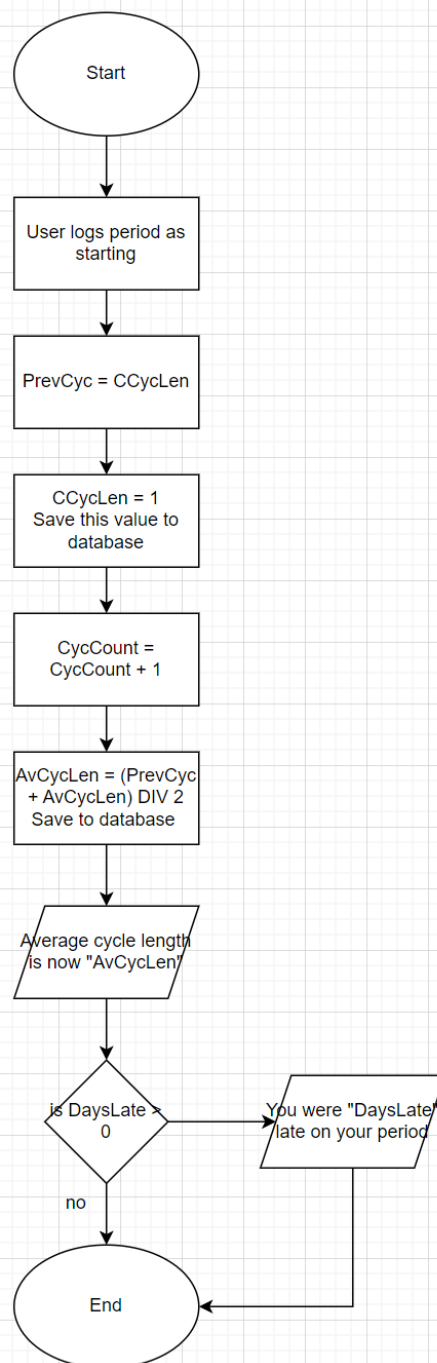
The user is asked to input their averages and this is stored into the database to be used to display the database.

Data needed to be gathered includes:

- Their average cycle length
- Their average period length
- The day they are currently on
- If they are on any birth control



Calculating their average cycle



This part of the program basically used this original input data by the user and updates it as necessary to make more accurate predictions.

When the user logs day 1 of their period, this is what happens.

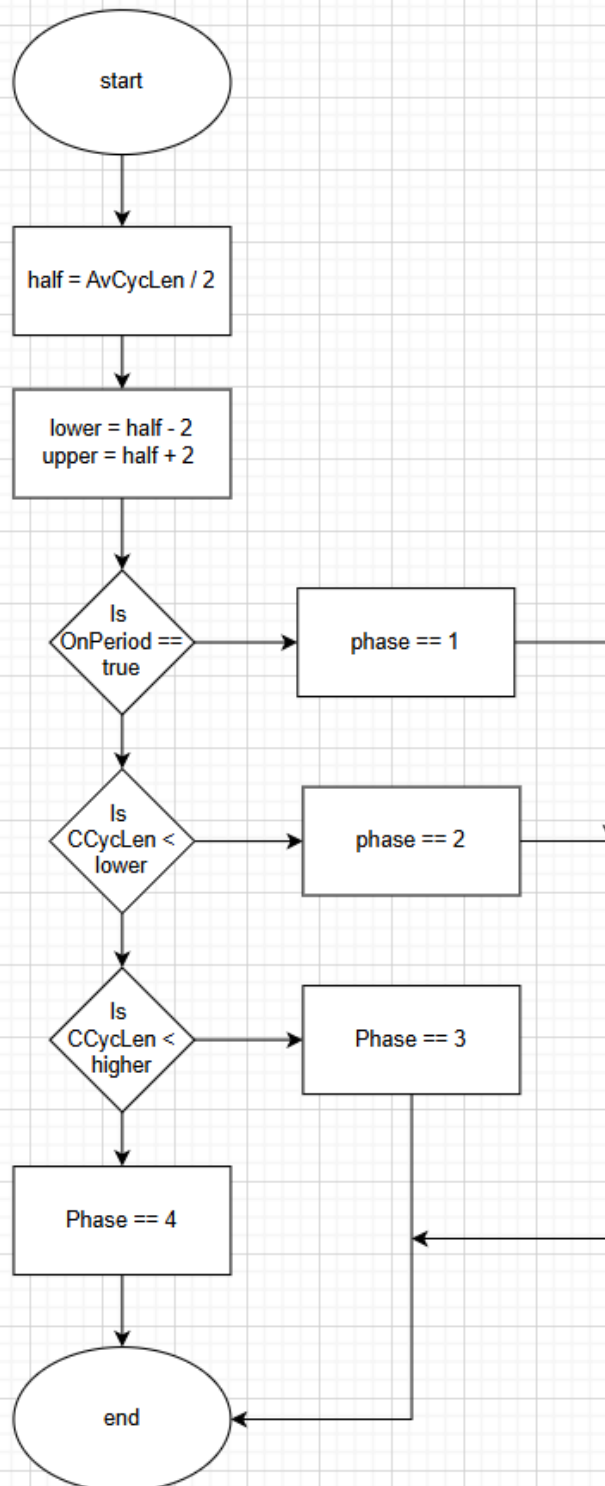
A temporary variable is used (PrevCyc) to hold the value of the current cycle (the one that has just ended). Then the current cycle can be reset back to day number 1 and thus is changed in the database. The cycle count is increased by 1 to ensure that it can display which cycle (for example the 3rd or the 50th cycle) they have successfully got through whilst using the app. A calculation is performed to work out their average cycle length, this is to ensure the app can make more successful predictions and help women feel more at ease. If the women was late onto their period, the app makes them aware of this.

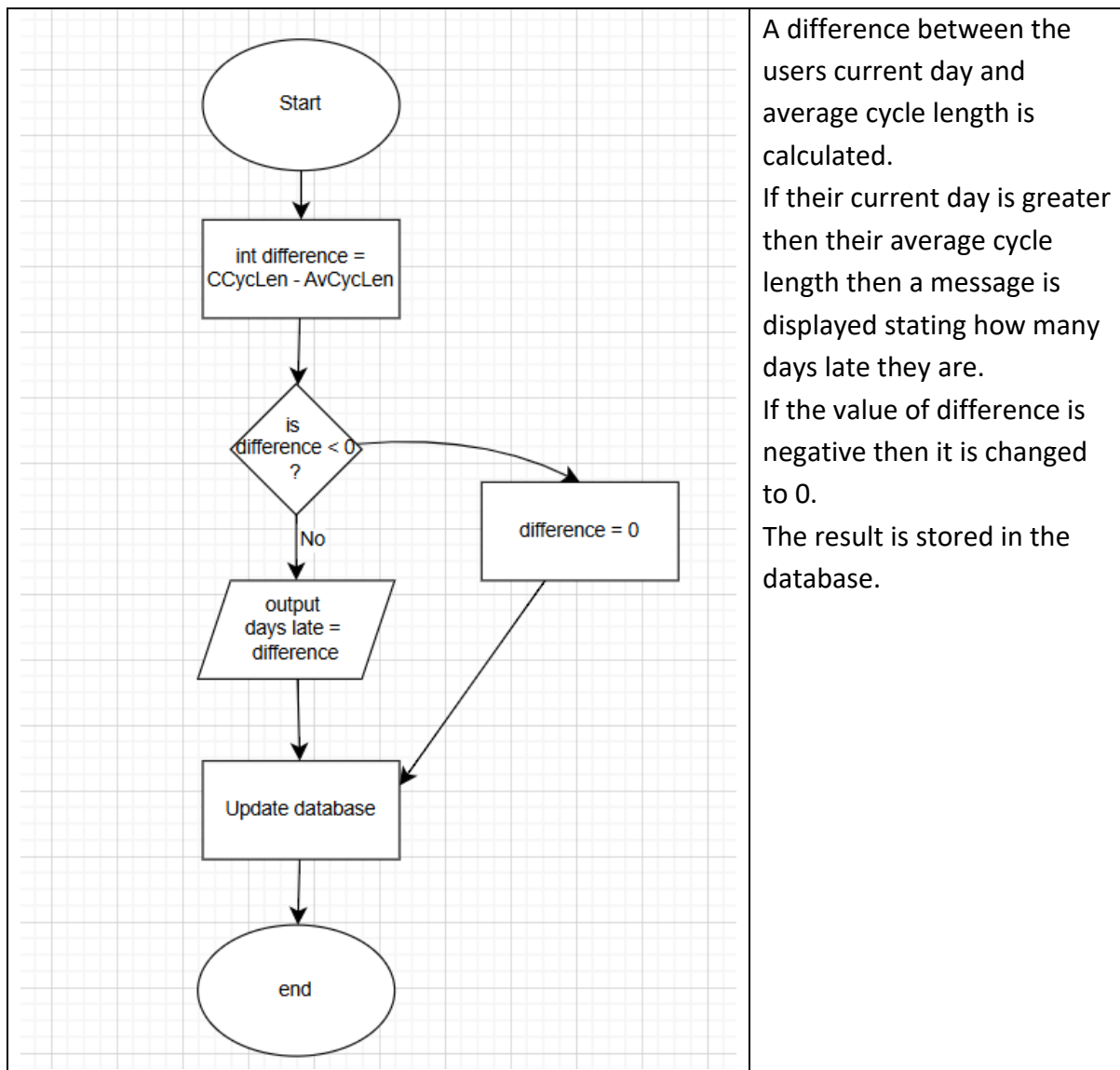
Calculating the phase that the user is currently in

I will use an if statement to determine what phase of their cycle the user is in. The day that marks half of their cycle will be used. This plus or minus two will be the period of time in which the person is ovulating

What each number represents:

- 1 = menstruating
- 2 = follicular
- 3 = ovulating
- 4 = luteal



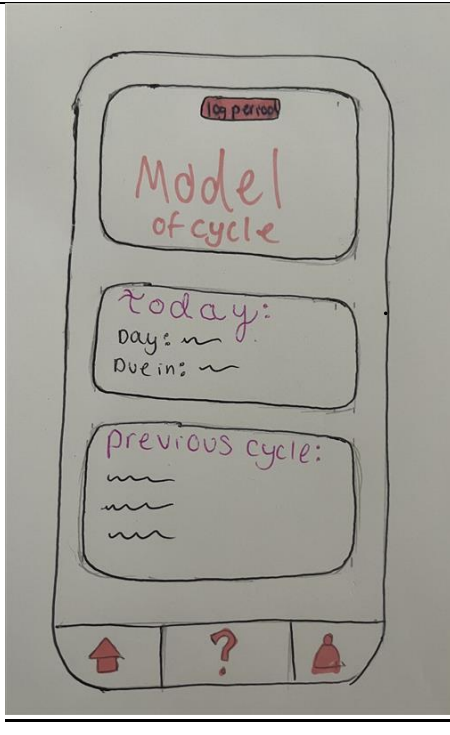
Calculating how late the user is**Interface Design**

I have hand drawn the idea for how I would like my app to look. I will explain what each part is and how they all link together:

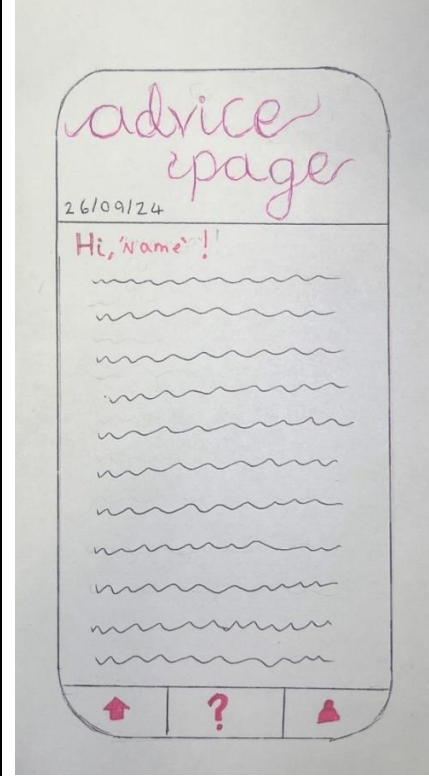
Log in / sign up page:

	This is the initial page the user will see, here they can log into a preexisting account or choose to sign up. The user will be able to enter their username, password or select to create an account.
---	--

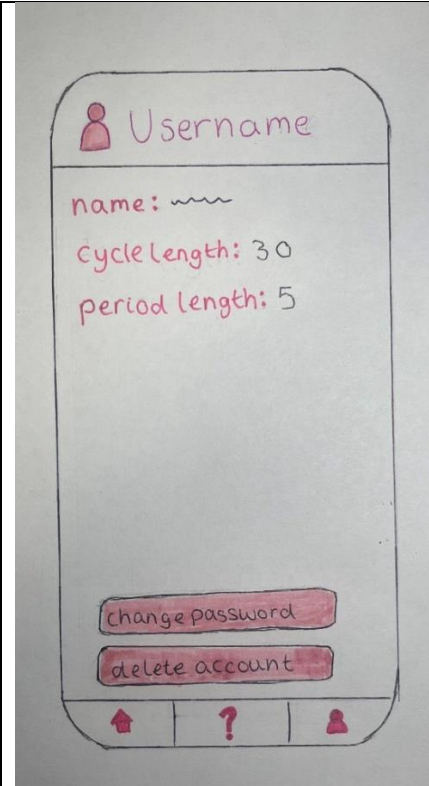
The menu page / the tracker

	This is where the user is directed to after creating their account or logging in, here they can view their current cycle and previous cycle data. A model is used to display their current cycle visually. The user can select to log a period using the button. The bar at the bottom allows navigation between the other two main pages
---	---

The advice page

	This is the advice page. Here users can read information on the current cycle and it'll display commonly asked questions with answers for their age group. It'll offer help and easy to understand explanations for the women and girls using the app. The advice displayed will be based on data stored in the database for each individual user.
---	---

The user profile page

	This page allows users to view their personal information. It also will allow them to change it. User have the option to delete their account of they wish to.
---	---

Security

I have thought of a few security measures to include within this app so that women's personal data is protected.

1. Use of a username

A username is used and linked to a unique ID number that is connected to the program. usernames are also unique to the user so that women cannot use the same username as this would cause them to share data as it would be linked to one ID.

2. Use of a password.

Since it is easy to find out someone's username (as it will be displayed within their user profile) a password is used to add extra protection to ensure that no one else can access their account without this password.

3. Log in every time they open the app.

This is required so that if someone gets hold of their phone, they cannot go onto the app and view this extremely personal data without requiring the username and password.

4. Able to delete account.

If they wish to delete their account for whatever reason, they can choose to delete it on their user profile. This will delete all the data stored for them within the database and they cannot get this back. They do not need to worry about doing this on accident as there will be a conformation question after they press the button.

5. Able to change password

This ensures that if someone managed to log into their account they could then change the password to ensure that the other person could not log back in

6. Hashing algorithm used with a password

This ensures extra security as if someone was able to access the database, they would need to be able to unscramble the hash to be able to get the password.

Testing Plan

My testing plan consists of two main parts. The first part includes testing general validation of data, different functions and pages within the database. The second part consists of a 16 day plan to check the function of the main tracker and advice page.

Part 1 – general validation of data, functions, pages and navigation testing

Creating a user:

Test Case	Test Data	Reason	Expected result
1	First name entry: "Linda1"	Contains a number	Error message
2	First name entry: ""	Is blank	Error message

3	First name entry: "Linda"	Is a valid first name	Should be accepted
4	First name entry: "a"	Is a valid string length on the boundary	Should be accepted
5	Username entry: "User#@"	Contains special characters	Error message
6	Username entry: "Username1234ABCDE "	Is too long	Error message
7	Username entry: "Username1234BC"	Is a valid string length on the boundary	Should be accepted
8	Username entry: ""	Is blank	Error message
9	Username entry: "Username12"	Is valid	Should be accepted
10	Birthday entry: "33/02/2001"	DD is not valid	Error message
11	Birthday entry: "13/00/2001"	MM is not valid	Error message
12	Birthday entry: "/07- 2007"	Format is wrong	Error message
13	Birthday entry: ""	Is blank	Error message
14	Birthday entry: "31/07/2015"	DD is on the boundary	Should be accepted
15	Password entry: "Yay1!"	Password is too short	Error message
16	Password entry: "Password*"	No number	Error message
17	Password entry: ""	Is blank	Error message
18	Password entry: "Pass1!"	Is on boundary for length	Should be accepted
19	Password entry: "Hello1234!"	Is valid	Should be accepted
20	Average cycle length entry: "c"	Is a string not an integer	Error message
21	Average cycle length entry: ""	Is blank	Error message
22	Average cycle length entry: "32"	Is a valid integer	Should be accepted

23	Average period length entry: ""	Is blank	Error message
24	Average period length entry: "700"	Is an integer but is invalid	Error message
25	Average period length entry: "6"	Is a valid integer	Should be accepted
26	Current day entry: ""	Is blank	Error message
27	Current day entry: "18"	Is valid	Should be accepted
28	Birth control entry: "3"	Is an integer	Error message
29	Birth control entry: ""	Is blank but user is allowed to leave blank	Should be accepted
30	Birth control entry: "Y"	Is valid string	Should be accepted
31	Birth control entry: "no"	Is valid string	Should be accepted

I will also carry out two tests, including all valid data, to ensure that the data is correctly being input into the database when the user creates their account.

Tests to check the functions the code is doing what its suppose to within the main pages will also take place.

Part 2 – checking the functionality of the tracker and advice page

I will do this by creating 2 different users. I will log into each account every day and ensure the app displays the expected data.

User 1:

- The first user will start on day one of their cycle and will follow the data input by them originally. For example, they will say their period usually lasts 4 days so on the fourth day the will log their period as done.
- This allows me to test the functionality of the tracker and the advice page if their period is consistent.
- The following data for their account will be entered:
 - First name: Userone
 - Username: userone1
 - Password: Hello1234!
 - Birth control: true
 - First period: true
 - Average cycle length: 16
 - Average period length: 5

- Current day: 1
- The period will be logged on day 5 and then again on day 16 to ensure that they program runs how its suppose to.

User 2:

- The second user will begin later on in their cycle and will not follow the data input by them originally. For example, if their period is supposed to last 5 days it actually lasts for 3.
- This will allow me to test the functionality of the tracker and the advice page if their period is not consistent.
- The following data for their account will be entered:
 - First name: Usertwo
 - Username: usertwo2
 - Password: Hello1234!
 - Birth control: false
 - First period: true
 - Average cycle length: 14
 - Average period length: 4
 - Current day: 8
- Their cycle ending will be logged on day 16 and then their period will be logged on day . This way it checks that the new average length are calculated correctly.
- Below is a detailed explanation of the exact values that will be expected of the outputs every day for each user:

Day number	User 1 expected outputs	User 2 expected outputs
1	Current day = 1 Phase = menstruating Previous period = N/A	Current day = 8 Phase = Ovulation Due in = 6 days Previous period = N/A
2	Current day = 2 Phase = menstruating Previous period = N/A	Current day = 9 Phase = Luteal Due in = 5 days Previous period = N/A
3	Current day = 3 Phase = menstruating Previous period = N/A	Current day = 10 Phase = Luteal Due in = 4 days Previous period = N/A
4	Current day = 4 Phase = menstruating	Current day = 11 Phase = Luteal

	Previous period = N/A	Due in = 3 days Previous period = N/A
5	Current day = 5 Phase = menstruating Previous period = N/A WHEN BUTTON PRESSED Current day = 5 Phase = follicular Due in = 11 days Previous period = N/A	Current day = 12 Phase = Luteal Due in = 2 days Previous period = N/A
6	Current day = 6 Phase = follicular Due in = 10 days Previous period = N/A	Current day = 13 Phase = Luteal Due in = due today or tomorrow Previous period = N/A
7	Current day = 7 Phase = ovulation Due in = 9 days Previous period = N/A	Current day = 14 Phase = Luteal Due in = today or tomorrow Previous period = N/A
8	Current day = 8 Phase = ovulation Due in = 8 days Previous period = N/A	Current day = 15 Phase = Luteal Late = 1 day Previous period = N/A
9	Current day = 9 Phase = ovulation Due in = 7 days Previous period = N/A	Current day = 16 Phase = Luteal Late = 2 days Previous period = N/A WHEN BUTTON PRESSED Current day = 1 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days
10	Current day = 10 Phase = luteal Due in = 6 days	Current day = 2 Phase = menstruating <u>Previous period:</u>

	Previous period = N/A	Cycle length = 16 days Period length = 4 days Late by= 2 days
11	Current day = 11 Phase = luteal Due in = 5 days Previous period = N/A	Current day = 3 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days
12	Current day = 12 Phase = luteal Due in = 4 days Previous period = N/A	Current day = 4 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days
13	Current day = 13 Phase = luteal Due in = 3 days Previous period = N/A	Current day = 5 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days
14	Current day = 14 Phase = luteal Due in = 2 days Previous period = N/A	Current day = 6 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days WHEN BUTTON PRESSED Current day = 6 Phase = follicular Due in = 9 days <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days
15	Current day = 15 Phase = luteal Due in = 1 days	Current day = 7 Phase = ovulating Due in = 8 days

	Previous period = N/A	<u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days
16	Current day = 16 Phase = luteal Due in = today or tomorrow Previous period = N/A WHEN BUTTON PRESSED Current day = 1 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 5 days	Current day = 8 Phase = ovulating Due in = 7 days <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days

Technical Solution

Overview

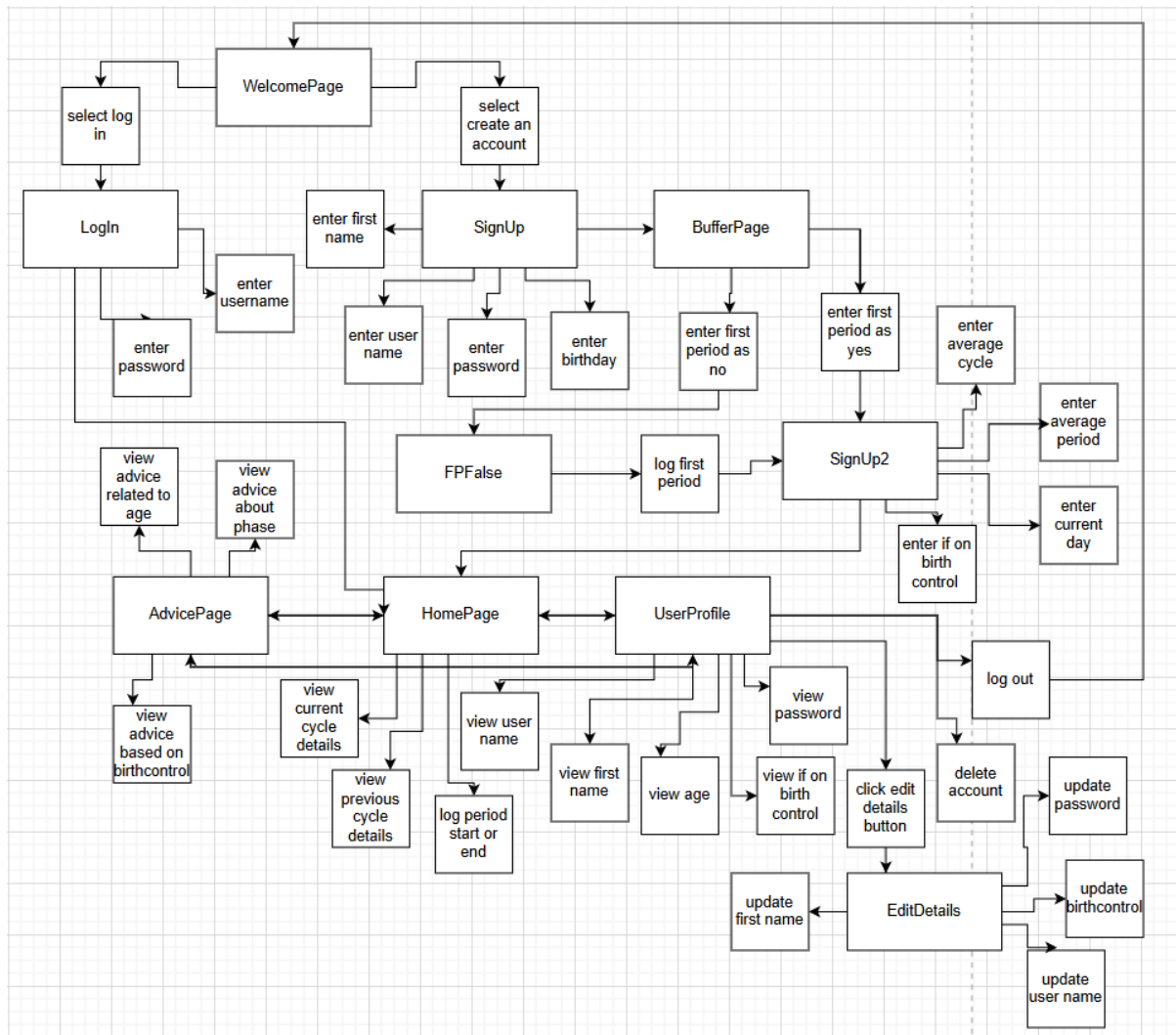
My app has been created in Visual Studio using the C# programming language and Xamarin.Forms. I downloaded multiple packages to enable different features to be used whilst coding. These included Microcharts and SkiaSharp packages to allow for a doughnut pie chart and sqlite packages to allow for a database to be used.

I developed the mobile app to run on an android and ran my solution on the built in android emulator in visual studio (using Pixel 4 – API 29).

My app allows for users to be created, which are stored in the sqlite database. A user must use their log in details to access the main pages of the app. The main pages include the HomePage, AdvicePage and UserProfile. The HomePage displays relevant details to the user that are to do with their current and previous period cycle. The AdvicePage displays appropriate information to the user regarding advice and help based upon their current cycle. The UserProfile allows the user to view and change their personal details. It also contains buttons for the user to log out or delete their account.

Navigation of the app

This diagram shows how all the navigation between the pages works and what each page does.



Key parts of the solution

Heading	Form name in solution	Page No.
Database connection	Querying.cs	75
Insert into database	SignUp2.xaml.cs	84
Hash code	LogIn.xaml.cs & SignUp.xaml.cs	73, 78
Model for the tracker	HomePage.xaml.cs	65
Merge sort	SignUp.xaml.cs	80
Binary search	SignUp.xaml.cs	79

Database connection

This part of my code provides a direct connection to the database programmed in LogIn.cs. It means that the database can be accessed from anywhere in the entire code by using Querying.Connection(). This allows data stored to be retrieved, inserted, updated and deleted within my code with ease.

- The public class allows this to be called for anywhere in the code.
- Connection can also be called for when Querying has been first (Querying.Connection()) as it is public.
- databsepath holds the connection string for the database.
- All the tables are created if the database does not yet exist.

```
public class Querying
{
    //This provides the connection to the database and the rest of the code
    private static SQLiteConnection database;
    public static SQLiteConnection Connection()
    {
        if (database == null)
        {
            var databsepath =
Path.Combine(Environment.GetFolderPath(Environment.SpecialFolder.LocalApplicationData),
                "App8.db3");
            database = new SQLiteConnection(databsepath);
            database.CreateTable<LogIns>();
            database.CreateTable<HelpPage>();
            database.CreateTable<CurrentT>();
            database.CreateTable<LastT>();
        }
        return database;
    }
}
```

Insert into database

This section of code is how a user gets created and inserted into the database.

- Each table is created individually using the same UserId as the one stored in the LogIns table so that it can be queried to correctly in other sections (all linked). Also, so that when the user logs in, their UserId can just be retrieved from the LogIns tables for ease.
- Querying.Connection() is how the database is accessed.
- Insert() stores the data into the correct table.

```
var AddUserLI = new LogIns
{
    UserId = userid2,
    UserName = username2,
    FirstName = firstName2,
```

```

        Password = password2,
        Birthday = birthday2,
    };
    Querying.Connection().Insert(AddUserLI);

    var AddUserHP = new HelpPage
    {
        UserId = AddUserLI.UserId,
        Id = AddUserLI.UserId,

        FirstPeriod = true,
        BC = bc,
    };

    var AddUserCT = new CurrentT
    {
        UserId = AddUserLI.UserId,
        Id = AddUserLI.UserId,

        AvCycLen = AverageCycle,
        CCycLen = CurrentDay,
        AvPLen = AveragePeriod,
        CPLen = AveragePeriod,
        DaysLate = Late,
        StartDate = startdate,
        OnPeriod = onperiod,
    };

    var AddUserLT = new LastT
    {
        UserId = AddUserLI.UserId,
        Id = AddUserLI.UserId,
        Length = 0,
        Recent1 = DateTime.MinValue,
        Recent2 = DateTime.MinValue,
        LastDaysLate = 0,
    };

    Querying.Connection().Insert(AddUserHP);
    Querying.Connection().Insert(AddUserCT);
    Querying.Connection().Insert(AddUserLT);

```

Hash code

I created a hash code to store a user's password in the database instead of the actual password for security purposes. When a user logs in, a hash of their inputted password is created and checked against the hash stored in the database to ensure that they are the same (they have entered the correct password). The hash code works by:

- The GetHashCode() function is called within the code. The users entered password is passed in.

- A integer named count is set to zero, then for each character in the users entered password, the count increases by the ASCII value of that character.
- The integer count is turned into a string (string2). This would be a string of numbers.
- Then for each character within in string2, count increases by the ASCII value again.
- The final count is now converted back to a string and returned.

```
public string GetHashCode(string password1)
{
    //This is used every time a password is inserted into the database or the password is retrieved
    int count = 0;
    foreach (char a in password1)
    {
        count = count + a;
    }

    string string2 = Convert.ToString(count);
    foreach (char a in string2)
    {
        count = count + a;
    }

    return Convert.ToString(count);
}
```

Model for the tracker

I made use of Microcharts to represent the users cycle as a model in my code.

- Querying.Connection() is used to call the database and find the details of the user based upon their UserId
- The users average period length is stored as an integer called len1 and the remaining length of their cycle is stored as an integer called len2.
- These are both then inserted as ChartEntry
- I used a doughnut pie chart, with a transparent background and large inside radius to allow for a circular model that I could display the Current Day label inside

```
var database = Querying.Connection();
var table3 = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
if (table3 != null)
{
    int len1 = table3.AvPLen;
    int len2 = table3.AvCycLen - len1;

    var entries = new List<ChartEntry>
    {
        new ChartEntry(len1) { Color = SKColor.Parse("#FF6384") },
        new ChartEntry(len2) { Color = SKColor.Parse("#36A2EB") },
    };
};
```

```

//Used a doughnut chart to make a thin circular diagram that displays proportionally how
many days they bleed for compared to the rest of their period
chartView.Chart = new DonutChart
{
    Entries = entries,
    BackgroundColor = SKColors.Transparent,
    HoleRadius = 0.9f,
};
}

```

Merge sort

I used a merge sort within my code to allow for the UserId's within the database to be sorted so that a binary search can then be used on them. This is used when a user is created to ensure that the user being created does not have the same UserId stored in the database.

- The CreateUserId() function is called, the database is called for using Querying.Connection() and an integer list of the UserIds is gathered
- This is stored as an array called intlist and passed into the MergeSort function
- The positions of the start and the end of the list are also passed into this, a mid-position of the list is also found.
- The list is then repeatedly split down over and over and passed back into the MergeSort() until r == l (the list is split into single elements)
- Then these are passed into the MergeMethod() function repeatedly until fully merged back together.
- Here they are merged back together into groups double their current size by comparing the which integer is smaller.

```

public int CreateUserId()
{
    var database = Querying.Connection();
    List<int> intlist1 = database.Table<LogIns>().Select(u => u.UserId).ToList();
    int[] intlist = intlist1.ToArray();

    //merge sorts the array of current ids already taken
    MergeSort(intlist, 0, intlist.Length - 1);

    //Keeps rerandomising until a number is found that isn't used in the database already
    Random random = new Random();
    int a;
    do
    {
        a = random.Next(1, 901);
    }
    //Uses binary search to check that its not contained in the list already
    while (!BinarySearch(intlist, a));
    return a;
}
//Merge sort algorithm

```

```
static void MergeSort(int[] intlist, int l, int r)
{
    int mid;
    if (r > l)
    {
        mid = (r + l) / 2;
        MergeSort(intlist, l, mid);
        MergeSort(intlist, (mid + 1), r);
        MergeMethod(intlist, l, (mid + 1), r);
    }
}
static void MergeMethod(int[] intlist, int l, int mid, int r)
{
    int[] temp = new int[901];
    int c, lEnd, numelements, temp1;
    lEnd = mid - 1;
    temp1 = l;
    numelements = (r - l + 1);
    while ((mid <= r) && (l <= lEnd) )
    {
        if (intlist[l] <= intlist[mid])
        {
            temp[temp1++] = intlist[l++];
        }
        else
        {
            temp[temp1++] = intlist[mid++];
        }
    }
    while (l <= lEnd)
    {
        temp[temp1++] = intlist[l++];
    }
    while (mid <= r)
    {
        temp[temp1] = intlist[mid++];
    }
    for (c = 0; c < numelements; c++)
    {
        intlist[r] = temp[r];
        r--;
    }
}
```

Binary search

The binary search is used within my code to check that the list of UserId's does not contain a randomly generated one. This is to ensure that two users don't get stored with the same Id.

- The list is arranged by the merge sort (see above) first.

- Then a random integer called a is created. This is repeatedly until it is found to be not in the list by calling the BinarySearch() function.
- First of all, the boundaries of the list are created by setting min and max. A mid value for the list is created from these. This is compared with the value of a.
- If a is bigger than mid then the right half of the list is searched (the min value becomes equal to the mid value plus one)
- If a is smaller than mid then the left half of the list is searched (the max value becomes equal to mid minus one).
- If a is equal to mid then true is returned to indicate that a is in the list and a needs to be re randomised.
- If not, the loop repeats until min is greater than max and false is returned, indicating that a can be used.

```
public int CreateUserId()
{
    var database = Querying.Connection();
    List<int> intlist1 = database.Table<LogIns>().Select(u => u.UserId).ToList();
    int[] intlist = intlist1.ToArray();

    //merge sorts the array of current Ids already taken
    MergeSort(intlist, 0, intlist.Length - 1);

    //Keeps rerandomising until a number is found that isnt used in the database already
    Random random = new Random();
    int a;
    do
    {
        a = random.Next(1, 901);
    }
    //Uses binary search to check that its not contained in the list already
    while (!BinarySearch(intlist, a));
    return a;
}

//The binary search
static bool BinarySearch(int[] intlist, int a)
{
    int min = 0;
    int max = intlist.Length - 1;
    while (min <= max)
    {
        int mid = (min + max) / 2;
        if (intlist[mid] == a)
        {
            return true;
        }
        else if (a < intlist[mid])
        {
            max = mid - 1;
        }
        else
    }
```

```

        {
            min = mid + 1;
        }
    }
    return false;
}

```

Full solution

LogIns.cs

```

using SQLite;
using System;

//This page holds my database
namespace App8.Model
{
    //Table 1
    public class LogIns
    {
        [PrimaryKey, AutoIncrement]
        //This is the primary key that connects to the other tables:
        public int UserId { get; set; }
        public string UserName { get; set; }
        public string FirstName { get; set; }
        public string Password { get; set; }
        public DateTime Birthday { get; set; }
    }

    //Table 2
    public class HelpPage
    {
        [PrimaryKey, AutoIncrement]
        public int Id { get; set; }
        public bool FirstPeriod { get; set; }
        public bool BC { get; set; }
        public int phase { get; set; }

        public int UserId { get; set; }
    }

    //Table 3
    public class CurrentT
    {
        [PrimaryKey, AutoIncrement]
        public int Id { get; set; }
        public int AvCycLen { get; set; }
        public int CCycLen { get; set; }
        public int AvPLen { get; set; }
        public int CPLen { get; set; }
        public int DaysLate { get; set; }
        public DateTime StartDate { get; set; }
        public bool OnPeriod { get; set; }
    }
}

```

```

        public int UserId { get; set; }
    }

    //Table 4
    public class LastT
    {
        [PrimaryKey, AutoIncrement]
        public int Id { get; set; }
        public DateTime Recent1 { get; set; }
        public DateTime Recent2 { get; set; }
        public int Length { get; set; }
        public int LastDaysLate { get; set; }

        public int UserId { get; set; }
    }
}

```

AdvicePage.xaml

```

<?xml version="1.0" encoding="utf-8" ?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    x:Class="App8.AdvicePage"
    Title="Help Page">
    <ContentPage.ToolbarItems>
        <ToolbarItem Text="Tracker"
            Priority="0"
            Order="Primary"
            Clicked="T"/>
        <ToolbarItem Text="Profile"
            Priority="1"
            Order="Primary"
            Clicked="UP"/>
    </ContentPage.ToolbarItems>
    <StackLayout>
        <Label x:Name="Header" FontSize="35" Margin="5" TextColor="HotPink"></Label>
        <Frame Margin="10" BorderColor="DeepPink">
            <Label x:Name="phase" FontSize="17" TextColor="Black"></Label>
        </Frame>
        <Frame Margin="10" BorderColor="DeepPink">
            <Label x:Name="Age" FontSize="17" TextColor="Black"></Label>
        </Frame>
        <Frame Margin="10" BorderColor="DeepPink">
            <Label x:Name="birthcon" FontSize="17" TextColor="Black"></Label>
        </Frame>
    </StackLayout>
</ContentPage>

```

AdvicePage.xaml.cs

```

using App8.Model;
using System;
using Xamarin.Forms;

```

```

using Xamarin.Forms.Xaml;

namespace App8
{
    [XamlCompilation(XamlCompilationOptions.Compile)]
    public partial class AdvicePage : ContentPage
    {
        public int id;
        //This page displays necessary advice to the user depending different factors.
        //The users id is passed into this page to allow it to access data stored in the database
        public AdvicePage(int userid1)
        {
            InitializeComponent();
            id = userid1;

            //Uses users Id to gather the required information from the database
            var database = Querying.Connection();
            var log = database.Table<LogIns>().FirstOrDefault(u => u.UserId == id);
            var help = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == id);

            //Checks that the database actually has data stored for the particular id
            if(help!=null && log != null)
            {
                //Uses the name the user has unput
                Header.Text = "Hi, " + log.FirstName;

                //Uses the stored phase of their period they are in to display appropriate information and
                advice
                //Displays it to a text box (called phase) for the user to read
                int p = help.phase;
                switch (p)
                {
                    case 1:
                        phase.Text = "You are in the menstrual phase of your cycle ! This is the first out of 4
phases. You may experience symptoms such as cramps, blating, irritability, tiredness, lower back pain.
These are all completely normal but if you have any concerns please speak to a doctor!";
                        break;
                    case 2:
                        phase.Text = "Welcome to the second stage of your cycle, the follicular phase. You may
be experiencing a boost of energy and your mood may even improve! This is a good time to take action
and try new things.";
                        break;
                    case 3:
                        phase.Text = "You are now ovulating which is the third phase of your period. Your energy
levels are now at their highest and you will be feeling most confident. Now is a good time to enjoy
physical activity and challenge yourself !";
                        break;
                    case 4:
                        phase.Text = "You are in the final phase of your period, the luteal phase. Common
symptoms here include bloating, moodiness, food cravings, weight gain, trouble sleeping and
headaches. Now is a good time to take care of yourself by eating healthy and doing light activity to help
push through this phase.";
                        break;
                }
            }
        }
    }
}

```

```

//Uses the date now and their birtchday to work out how old they are
DateTime birth = log.Birthday;
DateTime a = DateTime.Now;
int age = a.Year - birth.Year;
if (birth.Month > a.Month)
{
    age--;
}

//Then their age is used to determine what advice is appropriate
//Displays this to a textbox (called Age) for the user to read
if (age < 15)
{
    Age.Text = "If you are experiencing ongoing irregular periods, extremely bad cramps, very
heavy bleeding, consider speaking to a professional for help!";
}
else if (age < 25)
{
    Age.Text = "If you are experiencing anything abnormal from your regular period, consider
speaking to someone as this could be a big indicator for a health issue!";
}
else if (age < 50)
{
    Age.Text = "Lucky you ! your periods may be coming to an end yay ! Symptoms of
menopause include irregular periods or a general change in your cycle, hot flushes, palpitations, difficulty
sleeping, brain fog, changes in mood, dry and itchy skin and migraines. If you are concerned with any of
your symptoms please seek professional help.";
}
else
{
    Age.Text = "";
}

//Uses wheter they are on birtch control or not to display the appropriate advice
//Displays this to a textbox (called birthcon) for the user to read
bool BirthCon = help.BC;
if (BirthCon == true)
{
    birthcon.Text = "If you are experiencing any of the follow symptoms whilst on your birth
control please speak to a professional as these are not a good sign: depression, migraines, issues with
dry eye, aura, difficulty beathing, severe stomach pain or chest pain.";
}
else
{
    birthcon.Text = "Thank you for using Pearl!";
}
}

//Makes use of the tool bar to navigate to other pages (home page ot user profile)

//If the homepage button in toolbar is pressed then it passes the users id into the homepage and
displays this to the user
public void T(object obj, EventArgs args)

```

```

    {
        Navigation.PushAsync(new HomePage(id));
    }
    //If the userprofile button in toolbar is pressed then it passes the users id into the homepage and
    displays this to the user
    public void UP(object obj, EventArgs args)
    {
        Navigation.PushAsync(new UserProfile(id));
    }
}
}

```

App.xaml

```

<?xml version="1.0" encoding="utf-8" ?>
<Application xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    x:Class="App8.App">
    <Application.Resources>

    </Application.Resources>
</Application>

```

App.xaml.cs

```

using System;
using Application = Xamarin.Forms.Application;
using NavigationPage = Xamarin.Forms.NavigationPage;

namespace App8
{
    public partial class App : Application
    {
        //Allow for the pages to be navigated between:
        public NavigationPage SignUp { get; }
        public NavigationPage BufferPage { get; }
        public NavigationPage SignUp2 { get; }
        public NavigationPage LogIn { get; }
        public NavigationPage HomePage { get; }
        public NavigationPage AdvicePage { get; }
        public NavigationPage UserProfile { get; }
        public NavigationPage EditDetails { get; }
        public App()
        {
            InitializeComponent();

            //This allows the MainPage to be clicked and navigate to different pages:
            MainPage = new NavigationPage(new MainPage());

            //Allows the SignUp page to navigate to different pages:
            SignUp = new NavigationPage(new SignUp());

            //Allows the BufferPage page to navigate to different pages:
            BufferPage = new NavigationPage(new BufferPage("a", "a", DateTime.Now, "a", 1));
        }
    }
}

```

```

//Allows the SignUp2 page to navigate to different pages:
SignUp2 = new NavigationPage(new SignUp2("a", "a", DateTime.Now,"a",1,true));

//Allows the LogIn page to navigate to different pages:
LogIn = new NavigationPage(new LogIn());

//Allows HomePage to navigate to different pages:
HomePage = new NavigationPage(new HomePage(1));

//Allows AdvicePage to navigate to different pages:
AdvicePage = new NavigationPage(new AdvicePage(1));

//Allows UserProfile to navigate to different pages:
UserProfile = new NavigationPage(new UserProfile(1));

//Allows for EditDetails to navigate to differet pages
EditDetails = new NavigationPage(new EditDetails(1,"a"));

//The data passed into each page above is just temporary and isnt the actual value that will be
passed in. They act as placeholders for the real data.
}

protected override void OnStart()
{
}

protected override void OnSleep()
{
}

protected override void OnResume()
{
}
}
}

```

BufferPage.xaml

```

<?xml version="1.0" encoding="utf-8" ?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    x:Class="App8.BufferPage">
    <ContentPage.Content>
        <StackLayout BackgroundColor="FloralWhite">
            <Label Text="Have you had your period
                before?" FontSize="30" Margin="0,130" FontAttributes="Italic"
                HorizontalTextAlignment="Center" VerticalTextAlignment="End"
                TextColor="Black"/>
            <Button Text="Yes" TextColor="White" Margin="20"
                BackgroundColor="Pink" Clicked="SecondSignUpPage"> </Button>
            <Button Text="No" TextColor="White" Margin="20"
                Clicked="AppHomePage" BackgroundColor="Pink"> </Button>
        </StackLayout>
    </ContentPage.Content>

```

`</ContentPage>`

BufferPage.xaml.cs

```

using App8.Model;
using System;
using Xamarin.Forms;
using Xamarin.Forms.Xaml;

namespace App8
{
    [XamlCompilation(XamlCompilationOptions.Compile)]
    public partial class BufferPage : ContentPage
    {
        //Initialising variables that can be used in this page and passed into others:
        public bool period;
        private string username1;
        private string firstName1;
        private DateTime birthday1;
        private string password1;
        private int userid1;

        public BufferPage(string username, string firstName, DateTime birthday2, string password, int
userid)
        {
            InitializeComponent();

            //Assigning the data passed in to variables to allow it them to be used anywhere on the page
            username1 = username;
            firstName1 = firstName;
            birthday1 = birthday2;
            password1 = password;
            userid1 = userid;
        }

        //If the user enters that they have had their period before,
        //this part of the code directs them to continue setting up parts of their account
        public void SecondSignUpPage(object ob, EventArgs arg)
        {
            period = true;
            Navigation.PushAsync(new SignUp2(username1, firstName1, birthday1, password1, userid1,
period));
        }

        //If the user selects no:
        public void AppHomePage(object obj, EventArgs args)
        {
            //Creates a user and directs them to the homepage for users who havent started their period
            AddNewUserBuffer();
            Navigation.PushAsync(new FPFalse(userid1));
        }

        //This is how the data for a new user who has not had a period yet is inserted into my database:
        void AddNewUserBuffer()
        {
            var AddUserLI = new LogIns

```



```

    {
        UserId = userid1,
        UserName = username1,
        FirstName = firstName1,
        Password = password1,
        Birthday = birthday1,
    };
    //This inserts the data into the LogIns table
    Querying.Connection().Insert(AddUserLI);

    var AddUserHP = new HelpPage
    {
        UserId = AddUserLI.UserId,
        Id = AddUserLI.UserId,
        FirstPeriod = false
    };
    //Tis inserts the data into the HelpPage table
    Querying.Connection().Insert(AddUserHP);
    }
}
}

```

EditDetails.xaml

```

<?xml version="1.0" encoding="utf-8" ?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    x:Class="App8.EditDetails">
    <ContentPage.Content>
        <StackLayout>
            <Label Margin="20" FontSize="35" TextColor="HotPink" HorizontalTextAlignment="Center"
Text="Change Details"> </Label>
            <Entry x:Name="UN" Margin="15" FontSize="20"> </Entry>
            <Entry x:Name="Pass" Margin="15" FontSize="20"> </Entry>
            <Entry x:Name="FN" Margin="15" FontSize="20"> </Entry>
            <Entry x:Name="Bc" Margin="15" FontSize="20"> </Entry>
            <Button Text="Save Changes" Clicked="Change" Margin="40" BackgroundColor="LightPink"
TextColor="White"> </Button>
        </StackLayout>
    </ContentPage.Content>
</ContentPage>

```

EditDetails.xaml.cs

```

using App8.Model;
using System;
using System.Text.RegularExpressions;
using Xamarin.Forms;
using Xamarin.Forms.Xaml;

namespace App8
{
    [XamlCompilation(XamlCompilationOptions.Compile)]
    public partial class EditDetails : ContentPage
    {

```

```

//Initializes variables to be used anywhere in the page
public int userid;
public string un;
public string p;
public string fn;
public string bc;

//The purpose of this page is to allow the user to see their details, type in their new details, check
that its valid it and update the database
public EditDetails(int id, string birthcon)
{
    InitializeComponent();
    userid = id;

    //Calls the database tables needed to display the users information
    var database = Querying.Connection();
    var LI = database.Table<LogIns>().FirstOrDefault(u => u.UserId == id);
    var HP = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == id);
    if (LI != null)
    {
        //Displays the username to the user using what is stored in the database, using an entry that
they are able to change
        UN.Placeholder = "Username: " + LI.UserName;
        Pass.Placeholder = "Password: ***** ";
        //Displays the name of the user using what is stored in the database, using an entry that they
are able change
        FN.Placeholder = "Name: " + LI.FirstName;
    }
    if (HP != null)
    {
        bool b = HP.BC;
        //if the database says the user is on birthcontrol then it displays this back to them.
        if (b == true)
        {
            Bc.Placeholder = "On birth control: yes";
        }
        //However if this is stored as false then it displays no back to them
        else
        {
            Bc.Placeholder = "On birth control: no";
        }
    }
}

//When they press the button to save their details:
public void Change(object obj, EventArgs Args)
{
    un = UN.Text;
    p = Pass.Text;
    fn = FN.Text;
    bc = Bc.Text;
    //Goes through each entry individually and saves changes to database by using functions
    CheckUN();
    CheckPass();
    CheckFN();
}

```

```

    CheckBirthcon();
    //The it navigates back to the user profile (and passes in their id)
    Navigation.PushAsync(new UserProfile(userid));
}
public void CheckUN()
{
    //Uses regex to check that their new username is valid
    string un1 = @"^[a-zA-Z0-9]*$";
    if (!string.IsNullOrEmpty(un) && Regex.IsMatch(un, un1) && un.Length <= 15)
    {
        //If it is valid then it overrides the original username stored
        var database = Querying.Connection();
        var a = database.Table<LogIns>().FirstOrDefault(u => u.UserId == userid);
        if (a != null)
        {
            a.UserName = un;
            database.Update(a);
        }
    }
    //If the entry is empty nothing happens as not all the feilds need to be filled out
}
public void CheckPass()
{
    //Uses regex to check that the new password meets the requirements
    string pass1 = @"^(?=.*[a-zA-Z])(?=.*\d)(?=.*[*&$@!%?])";
    if (!string.IsNullOrEmpty(p) && Regex.IsMatch(p, pass1) && p.Length > 7)
    {
        //If it is valid then the new password overrides the old one
        //It calls the database using the Id
        var database = Querying.Connection();
        var a = database.Table<LogIns>().FirstOrDefault(u => u.UserId == userid);
        if (a != null)
        {
            string pass = a.Password;
            //Passes in password to this function to get a string of digits to be stored (to avoid the
            actual password being stored in the database)
            string Password = GetHashCode(pass);
            a.Password = Password;
            //Inserts the string of digits into the database
            database.Update(a);
        }
    }
    //If the entry is empty nothing happens as not all the feilds need to be filled out
}

//This function converts the password to a string of digits to be stored in the database (for security
purposes)
public string GetHashCode(string password1)
{
    //This is used every time a password is inserted into the database or the password is retrieved
    int count = 0;
    foreach (char a in password1)
    {
        count = count + a;
    }
}

```

```

    string string2 = Convert.ToString(count);
    foreach (char a in string2)
    {
        count = count + a;
    }

    return Convert.ToString(count);
}

//This function checks that the first name entry is valid
public void CheckFN()
{
    //Uses regex to ensure it only contains letters and that its not blank
    string fn1 = @"^[a-zA-Z]*$";
    if (!string.IsNullOrEmpty(fn) && Regex.IsMatch(fn, fn1) && fn.Length <= 15)
    {
        var database = Querying.Connection();
        var a = database.Table<LogIns>().FirstOrDefault(u => u.UserId == userid);
        if (a != null)
        {
            //If it is valid then the firstname is updated in the database
            a.FirstName = fn;
            database.Update(a);
        }
    }
    //If the entry is empty nothing happens as not all the fields need to be filled out
}

//This function checks that the birth control entry is valid
public void CheckBirthcon()
{
    var database = Querying.Connection();
    if (!string.IsNullOrEmpty(bc) && (bc == "yes" || bc == "Yes" || bc == "Y" || bc == "y"))
    {
        //If they have entered yes then the database is updated to true
        var a = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == userid);
        if (a != null)
        {
            a.BC = true;
            database.Update(a);
        }
    }
    else if(!string.IsNullOrEmpty(bc) && (bc == "no" || bc == "No" || bc == "N" || bc == "n"))
    {
        //If they have entered no then the database is updated to false
        var b = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == userid);
        if (b != null)
        {
            b.BC = false;
            database.Update(b);
        }
    }
    //If the entry is empty nothing happens as not all the fields need to be filled out
}

```

```
}
}
```

FPFalse.xaml

```
<?xml version="1.0" encoding="utf-8" ?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    x:Class="App8.FPFalse">
    <ContentPage.Content>
        <StackLayout>
            <Label Text="Welcome to Pearl, the period app aimed to help all girls!" FontSize="30"
HorizontalTextAlignment="Center" Margin="10" TextColor="DeepPink"/>
            <Label Text="Please click the button below when you get your first period!"
HorizontalTextAlignment="Center" Margin="5"></Label>
            <Button Clicked="FP" Margin="50" Text="Click here!" BackgroundColor="Pink"></Button>
            <Label FontSize="25" Margin="10" TextColor="DeepPink" Text="Tips for your first
period:"></Label>
            <Label FontSize="15" Margin="10" TextColor="black" Text="- Dont panic! it is
normal."></Label>
            <Label FontSize="15" Margin="10" TextColor="black" Text="- Create a period kit"></Label>
            <Label FontSize="15" Margin="10" TextColor="black" Text="- Ask questions !"></Label>
            <Label Margin="10" TextColor="black" Text="Feeling like everyone else has theres and you
dont, or yours is different to everyone elses ? It is best to talk to a trusted adult! Please dont be
scared."></Label>
        </StackLayout>
    </ContentPage.Content>
</ContentPage>
```

FPFalse.xaml.cs

```
using App8.Model;
using System;
using Xamarin.Forms;
using Xamarin.Forms.Xaml;

namespace App8
{
    [XamlCompilation(XamlCompilationOptions.Compile)]
    public partial class FPFalse : ContentPage
    {
        //Initialises the id so it can be used anywhere in the program
        public int id;
        //Tjis page is navigated to if the user selects that they haven't had a period before
        public FPFalse (int userid)
        {
            //retrives the users id
            id = userid;
            InitializeComponent();

            //If they then select that they have had their period, it naviagtes them ti the SignUp2
        }
    }

    Page
    public void FP(object obj, EventArgs Args)
    {
        var database = Querying.Connection();
    }
}
```

```

        var log = database.Table<LogIns>().FirstOrDefault(u => u.UserId == id);
        //Retrives the needed variables from the database to pass into the SignUp2
        page so a full account can be created
        string user = log.UserName;
        string firstn = log.FirstName;
        DateTime birth = log.Birthday;
        string pass = log.Password;
        Navigation.PushAsync(new SignUp2(user, firstn, birth, pass, id, true));
    }

}

}

```

HomePage.xaml

```

<?xml version="1.0" encoding="utf-8" ?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    xmlns:microcharts="clr-namespace:Microcharts.Forms;assembly=Microcharts.Forms"
    xmlns:local="clr-namespace:App8"
    x:Class="App8.HomePage"
    Title="Tracker">

    <ContentPage.ToolbarItems>
        <ToolbarItem Text="Help Page"
            Priority="0"
            Order="Primary"
            Clicked="AP"/>
        <ToolbarItem Text="Profile"
            Priority="1"
            Order="Primary"
            Clicked="UP"/>
    </ContentPage.ToolbarItems>
    <StackLayout>
        <Label x:Name="Header1" Margin="15" TextColor="DeepPink" FontSize="25"></Label>
        <Grid>
            <microcharts:ChartView x:Name="chartView" HeightRequest="200" WidthRequest="200"/>
            <Label x:Name="DayHold" FontSize="25" TextColor="LightCoral" VerticalOptions="Center"
                HorizontalOptions="Center" HorizontalTextAlignment="Center"></Label>
        </Grid>
        <Label x:Name="PhaseHold" FontSize="20" TextColor="Black"
            HorizontalTextAlignment="Center"></Label>
        <Label x:Name="DueOrLate" FontSize="20" TextColor="Black"
            HorizontalTextAlignment="Center"></Label>
        <Button x:Name="LogPeriod" Clicked="PeriodLogged" Margin="20" BackgroundColor="Pink"
            TextColor="White"></Button>
        <Label Text="Previous period" Margin="5" TextColor="DeepPink" FontSize="25"></Label>
        <Label x:Name="lengthcyc" Text="Currently no data is available here, sorry!" FontSize="20"
            Margin="5" TextColor="Black"></Label>
        <Label x:Name="lengthperiod" FontSize="20" Margin="5" TextColor="Black"></Label>
        <Label x:Name="dayslate" FontSize="20" Margin="5" TextColor="Black"></Label>
    </StackLayout>
</ContentPage>

```

HomePage.xaml.cs

```

using App8.Model;
using System;
using System.Collections.Generic;
using Xamarin.Forms;
using Xamarin.Forms.Xaml;
using Microcharts;
using SkiaSharp;

namespace App8
{
    [XamlCompilation(XamlCompilationOptions.Compile)]
    public partial class HomePage : ContentPage
    {
        //Initialises the is so that it can be used anywhere within the page
        public int id;
        public HomePage(int userid1)
        {
            InitializeComponent();
            id = userid1;

            //CurrentT is used to create a model for the tracker
            var database = Querying.Connection();
            var table3 = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
            if (table3 != null)
            {
                int len1 = table3.AvPLen;
                int len2 = table3.AvCycLen - len1;

                var entries = new List<ChartEntry>
                {
                    new ChartEntry(len1) { Color = SKColor.Parse("#FF6384") },
                    new ChartEntry(len2) { Color = SKColor.Parse("#36A2EB") },
                };

                //Used a doughnut chart to make a thin circular diagram that displays proportionally how
                //many days they bleed for compared to the rest of their period
                chartView.Chart = new DonutChart
                {
                    Entries = entries,
                    BackgroundColor = SKColors.Transparent,
                    HoleRadius = 0.9f,
                };
            }

            //Uses functions to display the period tracker to the user
            UpdateCurrentDay();
            CheckIfLate();
            GetCurrentPhase();
            HeaderLabel();
            DayLabel();
            PhaseLabel();
            DueOrLateLabel();
            LogButtonLabel();
        }
    }
}

```

```

    PrevCycleSection();
}

//This function displays the details of the previous cycle to the user
public void PrevCycleSection()
{
    var database = Querying.Connection();
    var last = database.Table<LastT>().FirstOrDefault(u => u.UserId == id);
    if (last != null)
    {
        if (last.Recent1 == DateTime.MinValue)
            //This value is stored if they have no details stored for the previous
            {
                return;
            }
        else
        {
            int diff = (last.Recent2 - last.Recent1).Days;
            lengthcyc.Text = "total cycle length: " + diff + " days";
            lengthperiod.Text = "total period length: " + last.Length + " days";
            if (last.LastDaysLate > 0)
            {
                dayslate.Text = "Days late: " + last.LastDaysLate;
            }
        }
    }
}

//This function is used to display a greeting message to the user
public void HeaderLabel()
{
    //Its uses the time of day
    string timeofday;
    int h = DateTime.Now.Hour;
    if ((h >= 6) && (h < 12))
    {
        timeofday = "morning, ";
    }
    else if (h < 17)
    {
        timeofday = "afternoon, ";
    }
    else if (h < 21)
    {
        timeofday = "evening, ";
    }
    else
    {
        timeofday = "night, ";
    }

    var database = Querying.Connection();
    var fn = database.Table<LogIns>().FirstOrDefault(u => u.UserId == id);
    if (fn != null)

```



```

    {
        //Also uses the stored first name of the user
        Header1.Text = "Good" + timeofday + fn.FirstName + ".";
    }

}

//This function displays to the user what day they are on
public void DayLabel()
{
    var database = Querying.Connection();
    var curr = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
    if (curr != null)
    {
        //It uses the CCycLen stored in the database
        DayHold.Text = "Day " + curr.CCycLen;
    }
}

//This function displays what phase the user is on by using the integer stored in the database
public void PhaseLabel()
{
    var database = Querying.Connection();
    var b = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == id);
    if (b != null)
    {
        int p = b.phase;
        switch (p)
        {
            case 1:
                PhaseHold.Text = "Phase: Menstrating";
                break;
            case 2:
                PhaseHold.Text = "Phase: Follicular";
                break;
            case 3:
                PhaseHold.Text = "Phase: Ovulating";
                break;
            case 4:
                PhaseHold.Text = "Phase: Luteal";
                break;
        }
    }
}

//This function displays to the user how many days they are kare ti starting their period
//If they are not late then it displays how many days there are till their period is due
public void DueOrLateLabel()
{
    var database = Querying.Connection();
    var z = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
    if (z != null)
    {
        if (z.OnPeriod == false)

```

```

    {
        if (z.DaysLate == 0)
        {
            if (z.CCycLen == z.AvCycLen)
            {
                DueOrLate.Text = "Period due today or tomorrow!";
            }
            else
            {
                int diff = z.AvCycLen - z.CCycLen;
                DueOrLate.Text = "Next period due in: " + diff + " days";
            }
        }
        else
        {
            DueOrLate.Text = "Days late: " + z.DaysLate;
        }
    }
}
}

//This function determines the text of the button to log or end their period, depending if the user is
on their period or not
public void LogButtonLabel()
{
    var database = Querying.Connection();
    var a = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
    if (a != null)
    {
        if (a.OnPeriod == true)
        {
            LogPeriod.Text = "Period ended";
        }
        else
        {
            LogPeriod.Text = "Log period";
        }
    }
}

//If the Help Page is seleted in the tool bar the it navigates to the AdvicePage
public void AP(object obj, EventArgs args)
{
    Navigation.PushAsync(new AdvicePage(id));
}

//If the Profile is delected in the tool bar then it navigates to the UserProfile
public void UP(object obj, EventArgs args)
{
    Navigation.PushAsync(new UserProfile(id));
}

//This function determines what day of their cycle the user is on

```

```

public void UpdateCurrentDay()
{
    var database = Querying.Connection();
    var dates1 = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
    if (dates1 != null)
    {
        DateTime start = dates1.StartDate;
        DateTime end = DateTime.Now;
        int diff = (end.Date - start.Date).Days;
        //Gets the difference between the date right now and the startdate of their period, which is
        //stored in the database
        if (diff == 0)
        {
            //If they are the same day then the current day is updated to one and the current period
            //length is updates the same
            dates1.CCycLen = 1;
            database.Update(dates1);
            if (dates1.OnPeriod == true)
            {
                dates1.CPLen = dates1.CCycLen;
                database.Update(dates1);
            }
        }
        else
        {
            //Else the current day is updated to match the difference
            dates1.CCycLen = diff;
            database.Update(dates1);
            if (dates1.OnPeriod == true)
            {
                //If the user is on their period then the current period length is updated to match the
                //current day
                dates1.CPLen = dates1.CCycLen;
                database.Update(dates1);
            }
        }
    }
}

//This function is used to determine if the user is late to starting their period
public void CheckIfLate()
{
    var database = Querying.Connection();
    var late = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
    if (late != null)
    {
        int diff = late.CCycLen - late.AvCycLen;
        //If they are late then the database is updated, if not then its updated to say 0
        if (diff > 0)
        {
            late.DaysLate = diff;
            database.Update(late);
        }
    }
}

```

```

        else
        {
            late.DaysLate = 0;
            database.Update(late);
        }
    }
}

//This function is used to calculate what phase of their period the user is in
public void GetCurrentPhase()
{
    var database = Querying.Connection();
    var current = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
    var helppage = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == id);
    if (current != null && helppage != null)
    {
        int CurrentCyc = current.CCycLen;
        int AvCyc = current.AvCycLen;
        int half = AvCyc / 2;
        //Menstrating
        if (current.OnPeriod == true)
        {
            helppage.phase = 1;
            database.Update(helppage);
        }
        //Follicular phase
        else if (CurrentCyc < half - 1)
        {
            helppage.phase = 2;
            database.Update(helppage);
        }
        //Ovulating
        else if (CurrentCyc < half + 2)
        {
            helppage.phase = 3;
            database.Update(helppage);
        }
        //Luteal
        else
        {
            helppage.phase = 4;
            database.Update(helppage);
        }
    }
}

//This function returns true or false based on if the database says they are on their period or not
public bool GetOnPeriod()
{
    var database = Querying.Connection();
    var c = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
    return c.OnPeriod;
}

//This function is called when the user clicks the log period / end period button

```

```

public async void PeriodLogged(object obj, EventArgs Args)
{
    //Display a warning to check they wanted to press the button
    bool res = await DisplayAlert("WAIT!", "Are you sure you would like to log a period?", "yes", "no");
    if (res == true)
    {
        bool onperiod = GetOnPeriod();
        if (onperiod == true)
        {
            //Updates details stored in the database
            OnIsTrue();
        }
        else
        {
            //Updates details stored in the database
            OnIsFalse();
        }
    }
    else
    {
        return;
    }
}

void OnIsTrue()
{
    var database = Querying.Connection();
    var c = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);

    //1. Calculates and updates AvPLen
    int current = c.CCycLen;
    int average = c.AvPLen;
    int newaverage = (current + average) / 2;
    c.AvPLen = newaverage;
    database.Update(c);

    //2.Stores the current day in CPLen and changes on period to false
    c.CPLen = current;
    database.Update(c);
    c.OnPeriod = false;
    database.Update(c);

    //ReloadHomePage with new stored database data
    Navigation.PushAsync(new HomePage(id));
}

void OnIsFalse()
{
    var database = Querying.Connection();
    var c = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
    var l = database.Table<LastT>().FirstOrDefault(u => u.UserId == id);

    //1.Calculates and updates AvCycLen
    int current = c.CCycLen;
    int average = c.AvCycLen;
    int newaverage = (current + average) / 2;

```

```

        c.AvCycLen = newaverage;
        database.Update(c);

        //2.Moves data into LastT
        l.Recent1 = c.StartDate;
        database.Update(l);
        l.Recent2 = DateTime.Now;
        database.Update(l);
        l.LastDaysLate = c.DaysLate;
        database.Update(l);
        l.Length = c.CPLen;
        database.Update(l);

        //3.Restores data in CurrentT
        c.CCycLen = 1;
        database.Update(c);
        c.DaysLate = 0;
        database.Update(c);
        c.StartDate = DateTime.Now;
        database.Update(c);
        c.OnPeriod = true;
        database.Update(c);

        //ReloadHomePage with new stored database data
        Navigation.PushAsync(new HomePage(id));
    }
}
}

```

Login.xaml

```

<?xml version="1.0" encoding="utf-8" ?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    x:Class="App8.Login">
    <ContentPage.Content>
        <StackLayout BackgroundColor="FloralWhite">
            <Label Text="^ Go back if you don't already have an account!" Margin="20"></Label>
            <Label Text="Welcome back!" Margin="20,130,0,0" FontSize="20" TextColor="Black"></Label>
            <Label Text="log in:" TextColor="HotPink" FontSize="40" Margin="20"/>
            <Entry x:Name="un" Placeholder="Enter username" Margin="20"></Entry>
            <Entry x:Name="pass" Placeholder="Enter password" Margin="20" IsPassword="True"></Entry>
            <Label x:Name="info" HorizontalTextAlignment="Center" TextColor="red"></Label>
            <Button Text="Log in !" Margin="20" TextColor="white" BackgroundColor="pink"
                Clicked="AppHomePage"></Button>
        </StackLayout>
    </ContentPage.Content>
</ContentPage>

```

Login.xaml.cs

```

using App8.Model;
using System;
using Xamarin.Forms;
using Xamarin.Forms.Xaml;

```

```

namespace App8
{
    [XamlCompilation(XamlCompilationOptions.Compile)]
    public partial class LogIn : ContentPage
    {
        public LogIn()
        {
            InitializeComponent();

            //When the user presses the login button it checks their details and then navigates them to the
            appropriate homepage
            public void AppHomePage(object obj, EventArgs args)
            {
                string username1 = un.Text;
                string password1 = pass.Text;

                //Gets the numeric string of the password (used to compare to the numeric string stored in
                database)
                string password2 = GetHashCode(password1);

                //Returns true or false if the user exists and if the password is linked
                bool checker = CheckUserExists(username1, password2);
                if (checker == true)
                {
                    //If they havent filled out the boxes an error message is displayed
                    if (string.IsNullOrEmpty(username1) || string.IsNullOrEmpty(password1))
                    {
                        info.Text = "Please ensure all feilds are filled out";
                    }
                    else
                    {
                        info.Text = " ";

                        int UserId1 = GetUserId(username1);

                        //This returns true or false dependind if they have had their first perios or not
                        bool decider = NavigateToPage(UserId1);

                        if (decider == true)
                        {
                            //If true then it navigates them to the homepage
                            Navigation.PushAsync(new HomePage(UserId1));
                        }
                        else
                        {
                            //If false it navigates them to the temporary homepage
                            Navigation.PushAsync(new FPFFalse(UserId1));
                        }
                    }
                }
            }
            //If the user does not exist then an error message is displayed
            else
            {

```

```
        info.Text = "incorrect details";
    }

}

//This function converts the password to a string of digits to be stored in the database (for security purposes)
public string GetHashCode(string password1)
{
    //This is used every time a password is inserted into the database or the password is retrieved
    int count = 0;
    foreach (char a in password1)
    {
        count = count + a;
    }

    string string2 = Convert.ToString(count);
    foreach (char a in string2)
    {
        count = count + a;
    }

    return Convert.ToString(count);
}

//This function returns true or false depending if the user has had their first period or not
public bool NavigateToPage(int UserId1)
{
    var database = Querying.Connection();
    var user = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == UserId1);
    if(user.FirstPeriod == false)
    {
        return false;
    }
    else
    {
        return true;
    }
}

//This function checks that the user exists and the username and password match
public bool CheckUserExists(string username1, string password2)
{
    var database = Querying.Connection();
    var user = database.Table<LogIns>().FirstOrDefault(u => u.UserName == username1 &&
u.Password == password2);
    if (user != null)
    {
        return true;
    }
    else
    {
        return false;
    }
}
```



```

    }

    //This function uses their username to get the users id to pass into the homepage
    public int GetUserId(string username1)
    {
        var database = Querying.Connection();
        var user = database.Table<LogIns>().FirstOrDefault(u => u.UserName == username1);
        return user.UserId;
    }
}
}

```

MainPage.xaml

```

<?xml version="1.0" encoding="utf-8" ?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    x:Class="App8.MainPage">
    <ContentPage.Content>
        <StackLayout BackgroundColor="FloralWhite">
            <Label Text="Welcome to Pearl" FontSize="35"
                HeightRequest="318" HorizontalTextAlignment="Center"
                VerticalTextAlignment="End" TextColor="HotPink">
            </Label>
            <Button x:Name="login" Text="Log in" TextColor="White"
                Margin="30,8" Clicked="GoToLogInPage"
                BackgroundColor="LightPink"> </Button>
            <Label Text="or" TextColor="Black" FontSize="Medium"
                HorizontalTextAlignment="Center"
                VerticalTextAlignment="Center"/>
            <Button x:Name="signup1" Text="Create an account"
                Margin="30,8" Clicked="GoToSignUpPage"
                TextColor="White" BackgroundColor="LightPink">
            </Button>
        </StackLayout>
    </ContentPage.Content>
</ContentPage>

```

MainPage.xaml.cs

```

using System;
using Xamarin.Forms;

namespace App8
{
    public partial class MainPage : ContentPage
    {
        public MainPage()
        {
            InitializeComponent();
        }

        //This enables the sign up page to be navigated to when the user clicks the option to create an account
        public void GoToSignUpPage(object obj, EventArgs args)
        {

```

```

        Navigation.PushAsync(new SignUp());
    }
    //This enables the log in page to be navigated to when the user clicks the option to login
    public void GoToLoginPage(object obj, EventArgs args)
    {
        Navigation.PushAsync(new Login());
    }
}
}

```

Querying.cs

```

using System;
using SQLite;
using App8.Model;
using System.IO;

namespace App8
{
    public class Querying
    {
        //This provides the connection to the database and the rest of the code
        private static SQLiteConnection database;
        public static SQLiteConnection Connection()
        {
            if (database == null)
            {
                var databasepath =
                Path.Combine(Environment.GetFolderPath(Environment.SpecialFolder.LocalApplicationData),
                    "App8.db3");
                database = new SQLiteConnection(databasepath);
                database.CreateTable<LogIns>();
                database.CreateTable<HelpPage>();
                database.CreateTable<CurrentT>();
                database.CreateTable<LastT>();
            }
            return database;
        }

        public void CreateUsers(LogIns login, HelpPage helppage, CurrentT current, LastT lastt)
        {
            //Creates a new user
            var data = Querying.Connection();
            data.BeginTransaction();
            data.Insert(login);

            helppage.UserId = login.UserId;
            current.UserId = login.UserId;
            lastt.UserId = login.UserId;

            //Inserts their data into the databases
            data.Insert(helppage);
            data.Insert(current);
            data.Insert(lastt);
        }
    }
}

```

```

        data.Commit();
    }
}
}

```

SignUp.xaml

```

<?xml version="1.0" encoding="utf-8" ?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    x:Class="App8.SignUp">
    <ContentPage.Content>
        <StackLayout BackgroundColor="FloralWhite">
            <Label Text="^ Go back if you already have an account!"
                Margin="10"/>
            <Label Text="Create an Account!" HorizontalOptions="Center"
                TextColor="HotPink" FontSize="30" Margin="0,15"/>
            <Entry x:Name="FN" Placeholder="First name" Margin="20"> </Entry>
            <Entry x:Name="UN" Placeholder="Username" Margin="20"> </Entry>
            <Entry x:Name="Birth" Placeholder="Birthday (DD/MM/YYYY)" Margin="20">
            </Entry>
            <Entry x:Name="P" Placeholder="Password" IsPassword="True" Margin="20"> </Entry>
            <Label x:Name="Information" TextColor="Red" HorizontalTextAlignment="Center"> </Label>
            <Button Text="Proceed" Margin="20" BackgroundColor="LightPink"
                Clicked="Buffer"> </Button>
        </StackLayout>
    </ContentPage.Content>
</ContentPage>

```

SignUp.xaml.cs

```

using App8.Model;
using System;
using System.Collections.Generic;
using System.Globalization;
using System.Linq;
using System.Text.RegularExpressions;
using Xamarin.Forms;
using Xamarin.Forms.Xaml;

namespace App8
{
    [XamlCompilation(XamlCompilationOptions.Compile)]
    public partial class SignUp : ContentPage
    {
        //This page gets the users name, username, password and birthday which is then passed into either
        //the BufferPage or the SignUp2 page
        public SignUp()
        {
            InitializeComponent();
        }
    }
}

```

```

public void Buffer(object ob, EventArgs arg)
{
    int userid;
    //The users inputs into the textboxes are retrieved
    string firstName = FN.Text;
    string username = UN.Text;
    string birthday = Birth.Text;
    string password = P.Text;
    int c = 0;

    //First the username is checked against the database to see if the username is already taken
    bool checker = CheckUserExists(username);
    if (checker == true)
    {
        Information.Text = "username already taken !";
    }
    else
    {
        //If the user has entered nothing into any of the textboxes or spaces then it displays an error
        if (string.IsNullOrEmpty(firstName) || string.IsNullOrEmpty(username) ||
            string.IsNullOrEmpty(birthday) || string.IsNullOrEmpty(password))
        {
            Information.Text = "Please make sure all fields are filled out!";
        }
        else
        {
            //Uses regex to check the username only contains upper and lower case letters and numbers
            string un = @"^[a-zA-Z0-9]*$";
            if (Regex.IsMatch(username, un) && username.Length <= 15)
            {
                c++;
            }
            else
            {
                Information.Text = "please input a valid username (must contain only numbers and letters
                and be no longer than 15 characters long)";
            }
            //Uses regex to check the first name only contain upper and lowercase letters
            string fn = @"^[a-zA-Z]*$";
            if (Regex.IsMatch(firstName, fn) && firstName.Length <= 15)
            {
                c++;
            }
            else
            {
                Information.Text = "please input valid a valid name (you can only use letters and it may be
                no longer than 15 letters long)";
            }
            //Uses regex to check that the birthday entry is a valid date
            string b = @"^(0[1-9]|[12][0-9]|3[01])/(0[1-9]|1[0-2])/d{4}$";
            if (Regex.IsMatch(birthday, b))
            {
                c++;
            }
            else

```

```

        {
            Information.Text = "Please enter birthday in the format DD/MM/YYYY";
        }
        //Uses regex to check that the password the user has entered is valid for security reasons
        string pass = @"^(?=.*[a-zA-Z])(?=.*\d)(?=.*[*&$@!%?])";
        if (Regex.IsMatch(password, pass) && password.Length > 7)
        {
            c++;
        }
        else
        {
            Information.Text = "Password must letters, a number, a special character and be 8 or
more characters long";
        }
    }
    //If all four entries are valid then the data is passed into the BufferPage and it is navigated to
    if (c == 4)
    {
        Information.Text = "";
        string password1 = CreateHash(password);
        userid = CreateUserId();
        DateTime birthday2 = DateTime.ParseExact(birthday, "dd/MM/yyyy",
CultureInfo.InvariantCulture);
        Navigation.PushAsync(new BufferPage(username, firstName, birthday2, password1, userid));
    }
}
//This function creates a hash for the password entered by the user to ensure that their actual
password is not stored
public string CreateHash(string password)
{
    int count = 0;
    foreach (char a in password)
    {
        count = count + a;
    }

    string string2 = Convert.ToString(count);
    foreach (char a in string2)
    {
        count = count + a;
    }
    return Convert.ToString(count);
}

//This function returns true or false to indicate if the user exists or not
public bool CheckUserExists(string username)
{
    var database = Querying.Connection();
    var user = database.Table<LogIns>().FirstOrDefault(u => u.UserName == username);
    if (user != null)
    {
        return true;
    }
    else

```

```

    {
        return false;
    }
}

//This function randomises a number to be used as an id and keeps re randomising if the id is
already taken
public int CreateUserId()
{
    var database = Querying.Connection();
    List<int> intlist1 = database.Table<LogIns>().Select(u => u.UserId).ToList();
    int[] intlist = intlist1.ToArray();

    //merge sorts the array of current ids already taken
    MergeSort(intlist, 0, intlist.Length - 1);

    //Keeps rerandomising until a number is found that isnt used in the database already
    Random random = new Random();
    int a;
    do
    {
        a = random.Next(1, 901);
    }
    //Uses binary search to check that its not contained in the list already
    while (!BinarySearch(intlist, a));
    return a;
}

//The binary search
static bool BinarySearch(int[] intlist, int a)
{
    int min = 0;
    int max = intlist.Length - 1;
    while (min <= max)
    {
        int mid = (min + max) / 2;
        if (intlist[mid] == a)
        {
            return true;
        }
        else if (a < intlist[mid])
        {
            max = mid - 1;
        }
        else
        {
            min = mid + 1;
        }
    }
    return false;
}

//Merge sort algorithm
static void MergeSort(int[] intlist, int l, int r)
{
    int mid;

```

```

    if (r > l)
    {
        mid = (r + l) / 2;
        MergeSort(intlist, l, mid);
        MergeSort(intlist, (mid + 1), r);
        MergeMethod(intlist, l, (mid + 1), r);
    }
}

static void MergeMethod(int[] intlist, int l, int mid, int r)
{
    int[] temp = new int[901];
    int c, lEnd, numelements, temp1;
    lEnd = mid - 1;
    temp1 = l;
    numelements = (r - l + 1);
    while ((mid <= r) && (l <= lEnd) )
    {
        if (intlist[l] <= intlist[mid])
        {
            temp[temp1++] = intlist[l++];
        }
        else
        {
            temp[temp1++] = intlist[mid++];
        }
    }
    while (l <= lEnd)
    {
        temp[temp1++] = intlist[l++];
    }
    while (mid <= r)
    {
        temp[temp1] = intlist[mid++];
    }
    for (c = 0; c < numelements; c++)
    {
        intlist[r] = temp[r];
        r--;
    }
}
}
}

```

SignUp2.xaml

```

<?xml version="1.0" encoding="utf-8" ?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    x:Class="App8.SignUp2">
    <ContentPage.Content>
        <StackLayout BackgroundColor="FloralWhite">

            <Label Text="Continue creating account!" FontSize="30"
                Margin="10" HorizontalTextAlignment="Center"
                TextColor="HotPink" FontAttributes="Bold"></Label>

```

```

<Label Text="We just need a couple more details please, then we will set up your account!"
      FontAttributes="Italic" Margin="5"
      TextColor="Black"></Label>
<Button BackgroundColor="FloralWhite"></Button>
<Label Text="How long does your cycle normally last?"
      FontSize="15" HorizontalTextAlignment="Center"
      TextColor="DeepPink"></Label>
<Entry x:Name="ACL" Placeholder="Cycle Length" Margin="10"></Entry>
<Label Text="How long does your period normally last?"
      FontSize="15" HorizontalTextAlignment="Center"
      TextColor="DeepPink"></Label>
<Entry x:Name="APL" Placeholder="Period Length" Margin="10"></Entry>
<Label Text="What day are you currently on?" FontSize="15"
      HorizontalTextAlignment="Center" TextColor="DeepPink">
</Label>
<Entry x:Name="CCL" Placeholder="Current Day" Margin="10"></Entry>
<Label Text="are you on any birth control?" FontSize="15"
      HorizontalTextAlignment="Center" TextColor="DeepPink">
</Label>
<Entry x:Name="bc" Placeholder="Yes or No (or leave blank if you wish)" Margin="10">
</Entry>
<Label x:Name="Info" HorizontalTextAlignment="Center" TextColor="Red"></Label>
<Button Text="Create account !" Margin="10" BackgroundColor="Pink"
Clicked="AppHomePage">
</Button>
</StackLayout>
</ContentPage.Content>
</ContentPage>

```

SignUp2.xaml.cs

```

using System;
using Xamarin.Forms;
using Xamarin.Forms.Xaml;
using App8.Model;

namespace App8
{
    [XamlCompilation(XamlCompilationOptions.Compile)]
    public partial class SignUp2 : ContentPage
    {
        //Initialises variables to be used anywhere in the page
        public string username2;
        public string firstName2;
        public DateTime birthday2;
        public string password2;
        public int userid2;
        public bool period1;
        public DateTime startdate;
        public bool onperiod;
        public int Late;

        public SignUp2 (string username1, string firstName1, DateTime birthday1, string password1, int
userid1, bool period)
        {
            //Assigns the passed in data to be used within the page

```



```

        InitializeComponent ();
        username2 = username1;
        firstName2 = firstName1;
        birthday2 = birthday1;
        password2 = password1;
        userid2 = userid1;
        period1 = period;
    }
    //This navigates the user to the HomePage if they completely set up their account
    public void AppHomePage(object obj, EventArgs args)
    {
        int CurrentPeriod;
        bool Bc;
        //This gets the users inputs in the entrys on the page
        string AverageCycle1 = ACL.Text;
        string AveragePeriod1 = APL.Text;
        string CurrentDay1 = CCL.Text;
        string birthcon = bc.Text;
        //Checks that the entrys are integer values
        bool a = int.TryParse(AverageCycle1, out int res);
        bool b = int.TryParse(AveragePeriod1, out int resu);
        bool c = int.TryParse(CurrentDay1, out int resv);
        if(a == true && b == true && c == true)
        {
            int AverageCycle = Convert.ToInt32(AverageCycle1);
            int AveragePeriod = Convert.ToInt32(AveragePeriod1);
            int CurrentDay = Convert.ToInt32(CurrentDay1);
            //This ensures that all the integers entered are appropriate sizes in
            //For example, that the person has not said that their average period
            //their average cycle length as this is not possible
            if (AverageCycle > (AveragePeriod + 9) && 120 > AverageCycle &&
25 > AveragePeriod)
            {
                //assigns true if the user says they are on birth control, and
                false if they say no or do not specify
                if(birthcon == "yes" || birthcon == "Yes" || birthcon == "Y" ||
birthcon == "y")
                {
                    Bc = true;
                }
                else if (birthcon == "no" || birthcon == "No" || birthcon ==
"N" || birthcon == "n")
                {
                    Bc = false;
                }
                else
                {
                    Bc = false;
                }
                Info.Text = "";
                //Sees if the user is on their period right now
                if(CurrentDay <= AveragePeriod)
                {

```

```

        CurrentPeriod = CurrentDay;
        onperiod = true;
    }
    else
    {
        CurrentPeriod = 0;
        onperiod = false;
    }
    //Sees if they are currently late
    if (CurrentDay > AverageCycle)
    {
        Late = CurrentDay - AverageCycle;
    }
    //Calculates the day their cycle began
    if (CurrentDay > 0)
    {
        DateTime datetime = DateTime.Now;
        startdate = datetime.AddDays(-(CurrentDay));
    }
    else
    {
        startdate = DateTime.Now;
    }

    //Then checks to see if an account has already been made for
    them (from FPPage logging their first period)
    var database = Querying.Connection();
    var FromFPFalsePage =
    database.Table<LogIns>().FirstOrDefault(u => u.UserId == userid2);
    //If they have then their account is deleted
    if (FromFPFalsePage != null)
    {
        database.Delete(FromFPFalsePage);
    }

    //Then a new user is created using the details passed into the
    page and the ones calculated on this page
    AddNewUserSignUp(firstName2, username2, birthday2, password2, userid2, Bc,
    AverageCycle, CurrentDay, AveragePeriod, startdate, onperiod, Late);
    var getnewid = database.Table<LogIns>().FirstOrDefault(u =>
    u.UserName == username2 && u.Password == password2);
    int newid = getnewid.UserId;
    //they are navigated to the home page
    Navigation.PushAsync(new HomePage(newid));
    }

    //error message displayed if integers entered are wrong in relation to
    eachother
    else
    {
        Info.Text = "Ensure that you have entered your details
    correctly";
    }
    }

    //If they are not integers then a message is displayed
    else

```

```

        {
            Info.Text = "Ensure all entries are a whole number!";
        }

        //This function adds a new user to the database
        void AddNewUserSignUp(string firstName2, string username2, DateTime birthday2, string
password2,
                                                                    int userid2, bool bc, int
AverageCycle, int CurrentDay, int AveragePeriod,
                                                                    DateTime startdate, bool
onperiod, int Late)
        {
            var AddUserLI = new LogIns
            {
                UserId = userid2,
                UserName = username2,
                FirstName = firstName2,
                Password = password2,
                Birthday = birthday2,
            };
            Querying.Connection().Insert(AddUserLI);

            var AddUserHP = new HelpPage
            {
                UserId = AddUserLI.UserId,
                Id = AddUserLI.UserId,
                FirstPeriod = true,
                BC = bc,
            };

            var AddUserCT = new CurrentT
            {
                UserId = AddUserLI.UserId,
                Id = AddUserLI.UserId,
                AvCycLen = AverageCycle,
                CCycLen = CurrentDay,
                AvPLen = AveragePeriod,
                CPLen = AveragePeriod,
                DaysLate = Late,
                StartDate = startdate,
                OnPeriod = onperiod,
            };

            var AddUserLT = new LastT
            {
                UserId = AddUserLI.UserId,
                Id = AddUserLI.UserId,
                Length = 0,
                Recent1 = DateTime.MinValue,
                Recent2 = DateTime.MinValue,
                LastDaysLate = 0,
            };

            Querying.Connection().Insert(AddUserHP);
            Querying.Connection().Insert(AddUserCT);

```

```

        Querying.Connection().Insert(AddUserLT);
    }
}
}
}

```

UserProfile.xaml

```

<?xml version="1.0" encoding="utf-8" ?>
<ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
    xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
    x:Class="App8.UserProfile"
    Title="Profile">
    <ContentPage.ToolbarItems>
        <ToolbarItem Text="Tracker"
            Priority="0"
            Order="Primary"
            Clicked="T"/>
        <ToolbarItem Text="Help Page"
            Priority="1"
            Order="Primary"
            Clicked="AP"/>
    </ContentPage.ToolbarItems>
    <StackLayout>
        <Label x:Name="UsersProfile" Margin="20" FontSize="35" TextColor="HotPink"
            HorizontalTextAlignment="Center" Text="Personal Details"></Label>
        <Label x:Name="UN" Margin="15" FontSize="20"></Label>
        <Label x:Name="Pass" Margin="15" FontSize="20"></Label>
        <Label x:Name="FN" Margin="15" FontSize="20"></Label>
        <Label x:Name="Age" Margin="15" FontSize="20"></Label>
        <Label x:Name="Bc" Margin="15" FontSize="20"></Label>
        <Button Text="Edit details" Clicked="UpdateDeatils" Margin="40" BackgroundColor="LightPink"
            TextColor="White"></Button>
        <Button Clicked="Back" Text="Log out" Margin="10" BackgroundColor="LightPink"
            TextColor="White"></Button>
        <Button Text="Delete account" Clicked="Deleteacc" Margin="10" BackgroundColor="LightPink"
            TextColor="White"></Button>
    </StackLayout>
</ContentPage>

```

UserProfile.xaml.cs

```

using App8.Model;
using System;
using Xamarin.Forms;
using Xamarin.Forms.Xaml;

namespace App8
{
    [XamlCompilation(XamlCompilationOptions.Compile)]
    public partial class UserProfile : ContentPage
    {
        //Variables that can be accessed anywhere in page
        public int id1;
        public string birthcon;
    }
}

```

```

public UserProfile(int userid1)
{
    InitializeComponent();
    int id = userid1;
    id1 = userid1;

    //Uses the userid to access the tables in the database
    var database = Querying.Connection();
    var LI = database.Table<LogIns>().FirstOrDefault(u => u.UserId == id);
    var HP = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == id);
    //These display the information to the user about the data stored:
    if (LI != null)
    {
        UN.Text = "Username: " + LI.UserName;
        Pass.Text = "Password: ***** ";
        FN.Text = "Name: " + LI.FirstName;
        DateTime dateTime = LI.Birthday;
        Age.Text = "Age: " + Convert.ToString(GetAge(dateTime));
    }
    if (HP != null)
    {
        bool Bool = HP.BC;
        if(Bool == true)
        {
            Bc.Text = "On birth control: yes";
        }
        else
        {
            Bc.Text = "On birth control: no";
        }
    }
}

//Uses the toolbar to navigate to the HomePage
public void T(object obj, EventArgs args)
{
    Navigation.PushAsync(new HomePage(id1));
}

//Uses the toolbar to navigate to the AdvicePage
public void AP(object obj, EventArgs args)
{
    Navigation.PushAsync(new AdvicePage(id1));
}

//If the user pushes the update details button then they are navigated to the EditDetails page
public void UpdateDeatils(object obj, EventArgs Args)
{
    Navigation.PushAsync(new EditDetails(id1, birthcon));
}

//This function works out how old the user is based on their birthday stored in the database and the
date now
public int GetAge(DateTime dateTime)
{
    DateTime a = DateTime.Now;

```

```

    int age = a.Year - dateTime.Year;
    if (dateTime.Month > a.Month)
    {
        age--;
    }
    return age;
}

//If the user presses the button to log out then an alert is displayed asking f they actually want to
public async void Back(object obj, EventArgs Args)
{
    bool res = await DisplayAlert("", "Are you sure you would like to log out?", "yes", "no");
    //if they select yes then the Goback function is called
    if (res == true)
    {
        Goback();
    }
    //if they select no then they remain logged in
    else
    {
        return;
    }
}

//This function deletes the users account if they wish to delete it
public async void Deleteacc(object obj, EventArgs Args)
{
    bool res = await DisplayAlert("WAIT", "Are you sure you would like to PERMANENTLY delete your
account", "yes", "no");
    if (res == true)
    {
        var database = Querying.Connection();
        var dellog = database.Table<LogIns>().FirstOrDefault(u => u.UserId == id1);
        var delhelp = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == id1);
        var delcurr = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id1);
        var dellast = database.Table<LastT>().FirstOrDefault(u => u.UserId == id1);
        database.Delete(dellog);
        database.Delete(delhelp);
        database.Delete(delcurr);
        database.Delete(dellast);
        Goback();
    }
    else
    {
        return;
    }
}

//Navigates back to the MainPage
void Goback()
{
    Navigation.PushAsync(new MainPage());
}
}
}

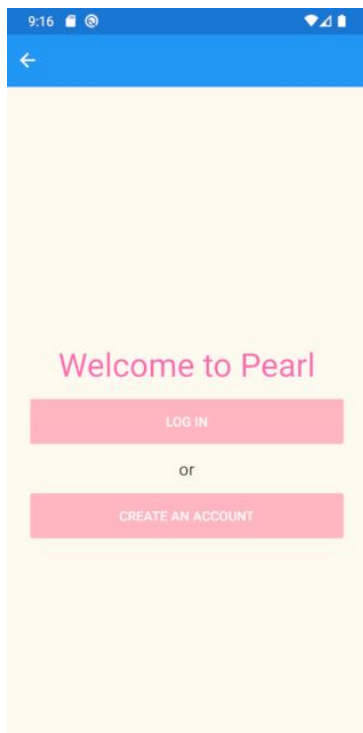
```

Testing Document

Testing the WelcomePage

Objectives being tested for

- 12. Navigation between all the pages works
- 12.1 . Navigation in log in and sign up system works



This is the first page that loads up when the app is run. It asks the user if they would like to create an account or log into an existing account.

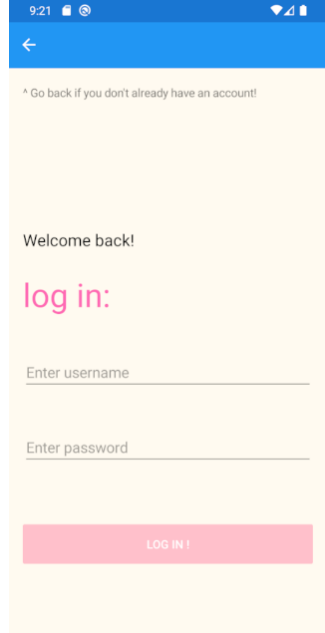
- The log in button should direct them to the log in page (LogIn)
- The create an account button should direct them to the sign up / create an account page (SignUp).

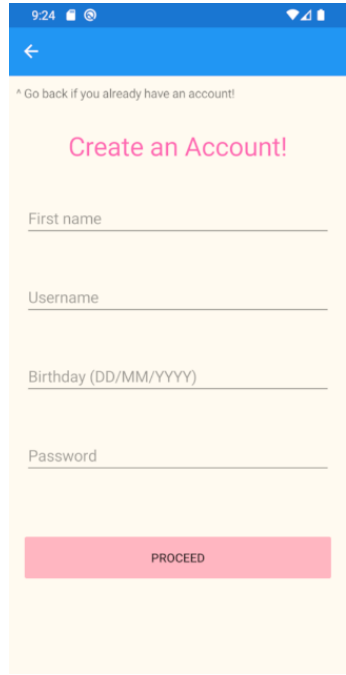
Test case 1

Test type: typical

Actual result:

Test Data: Press the log in button

Expected result: Should navigate to the LogIn page	 Test Result = PASS
Reason: The user wishes to log into an existing account and they pressed the button to be able to do this	

Test case 2	
Test type: typical	Actual result:  Test Result = PASS
Test Data: Press the create an account button	
Expected result: Should navigate to the SignUp page (says create an account across the top)	
Reason: The user wants to make an account so should guide them to the necessary page to do this	

These two tests show that the welcome page's navigation works correctly.

Testing the SignUp page

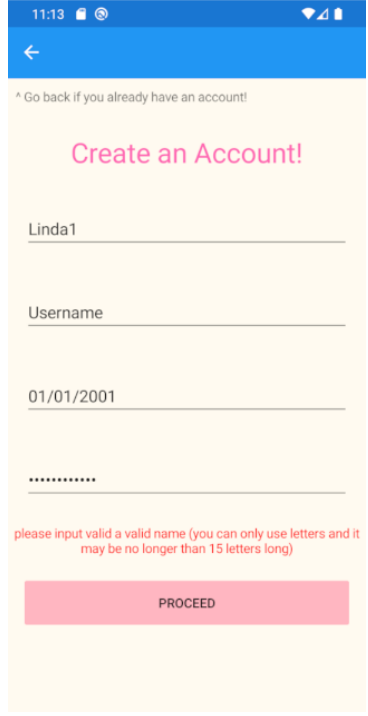
Objectives being tested for

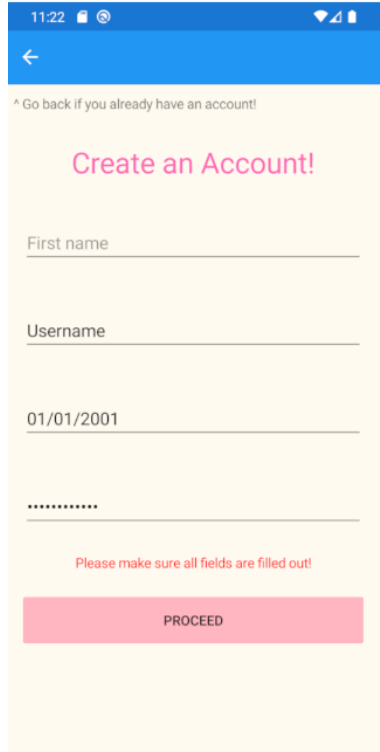
2. Use of a sign up and log in system
 - 2.1. Data must be checked and validated when the user signs up (e.g. username cannot have been used before, password must meet requirements for password)
 - 2.3. When an account is created, they must be added to the database

To test the sign up page, I will be using a mix of erroneous, typical and extreme testing.

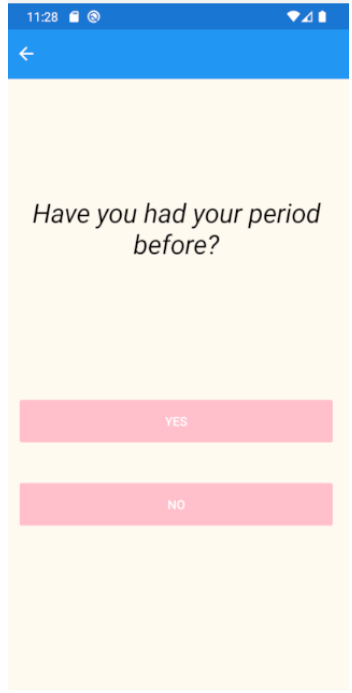
- **First name entry** : can't be blank or contain whitespace, must only contain upper or lower case letters, must be between 1 and 15 characters long.
- **Username entry** : can't be blank or contain whitespace, must only contain upper or lower case letters or numbers and must be between 1 and 15 characters long.
- **Birthday entry**: can't be blank or contain whitespace, must be in the format DD/MM/YYYY. DD must be in the range 1-31, MM must be in the range 1-12, YYYY must be a valid year.
- **Password entry** : can't be blank or contain whitespace, must contain lower or uppercase letters, be 8 or more characters long, must contain at least one special character and at least one number.

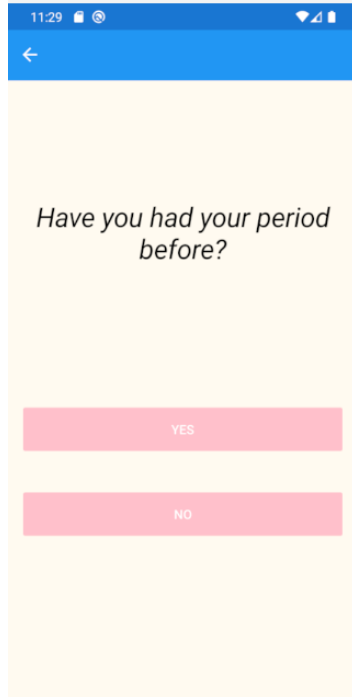
Test case 1	
Test type: erroneous	Actual result:
Test Data: First name entry "Linda1"	
Expected result: "please input a valid name (you can only use letters and it may be no longer than 15 letters long)"	
Reason: Contains a number	

	 Test Result = PASS
--	---

Test case 2	
Test type: erroneous	Actual result:  Test Result = PASS
Test Data: First name entry ""	
Expected result: "Please make sure all fields are filled out!"	
Reason: Is blank	

Test case 3	
Test type: typical	Actual result:

	
Test Data: First name entry "Linda"	
Expected result: the first name should be accepted and the app should move on to the next page	
Reason: It is 1-15 characters long and only contains upper or lower case letters	Test Result = PASS

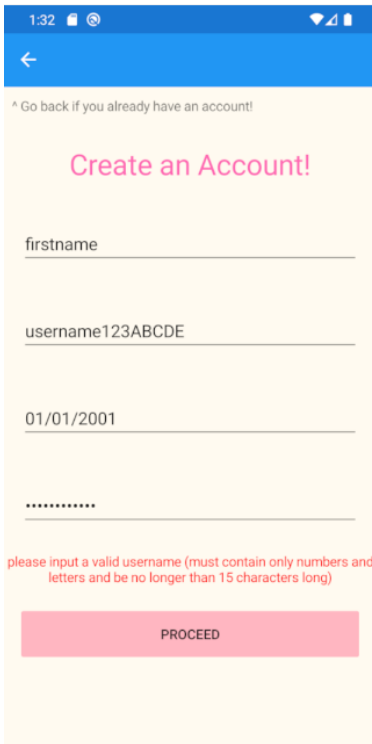
Test case 4	
Test type: Extreme	Actual result: 
Test Data: First name entry: "a"	
Expected result: the first name should be accepted and the app should move on to the next page	
Reason: It is one character long so should be allowed	Test Result = PASS

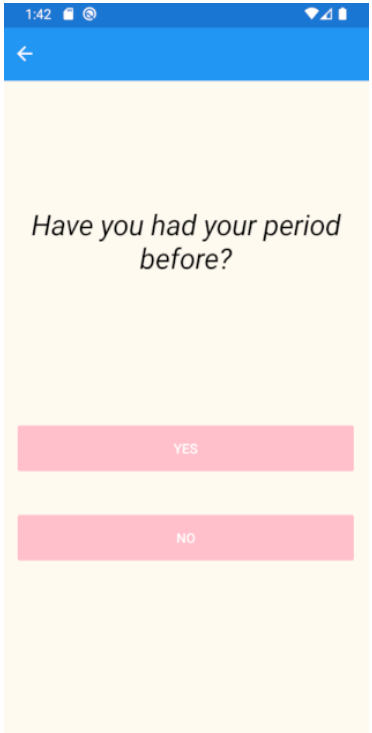
Test case 5	
Test type: erroneous	Actual result:

	
Test Data: username entry ""	
Expected result: "Please make sure all fields are filled out"	
Reason: contains blank space	
Test Result = PASS	

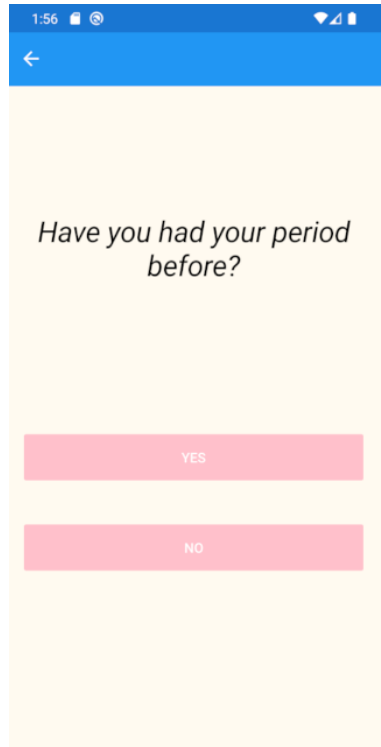
Test case 6	
Test type: erroneous	Actual result:
Test Data: username entry "User#@"	
Expected result: "please input a valid username (must contain only numbers and letters and be no longer than 15 characters long)"	
Reason: contains special character	
Test Result = PASS	

Test case 7

Test type: erroneous	Actual result: 
Test Data: username entry “username1234ABCDE”	
Expected result: “please input a valid username (must contain only numbers and letters and be no longer than 15 characters long)”	
Reason: longer than 15 characters long	
Test Result = PASS	

Test case 8	
Test type: extreme	Actual result: 
Test Data: username entry “username1234ABC”	
Expected result: the username should be accepted and the app should move on to the next page	
Reason: It is 15 characters long which should be accepted	
Test Result = PASS	

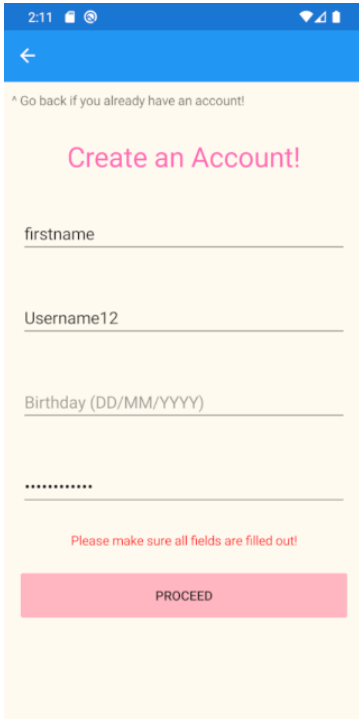
Test case 9

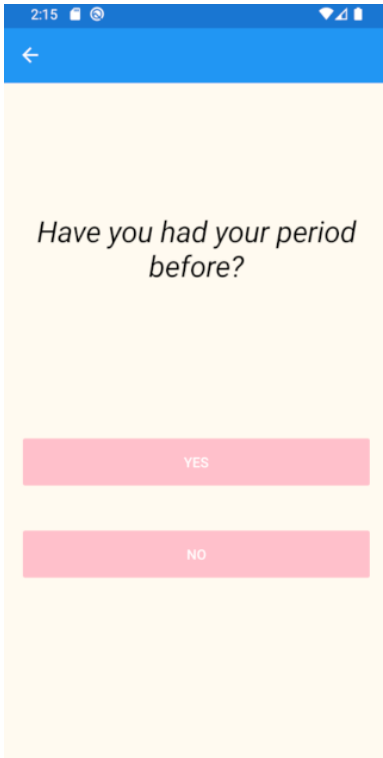
Test type: typical	Actual result: 
Test Data: username entry: "Username12"	
Expected result: the username should be accepted and the app should move on to the next page	
Reason: It is between 1 and 15 characters and contains valid characters	
Test Result = PASS	

Test case 10	
Test type: error	Actual result: 
Test Data: birthday entry “33/02/2001”	
Expected result: : “Please enter birthday in the format DD/MM/YYYY”	
Reason: DD is not from 01-31	
Test Result = PASS	

<u>Test case 11</u>	
Test type: erroneous	Actual result: 
Test Data: birthday entry "13/00/2001"	
Expected result: "Please enter birthday in the format DD/MM/YYYY"	
Reason: MM is not from 01-12	
Test Result = PASS	

Test case 12	
Test type: error	Actual result:  Test Result = PASS
Test Data: birthday entry “/07-2007”	
Expected result: “Please enter birthday in the format DD/MM/YYYY”	
Reason: it is not in the correct format	

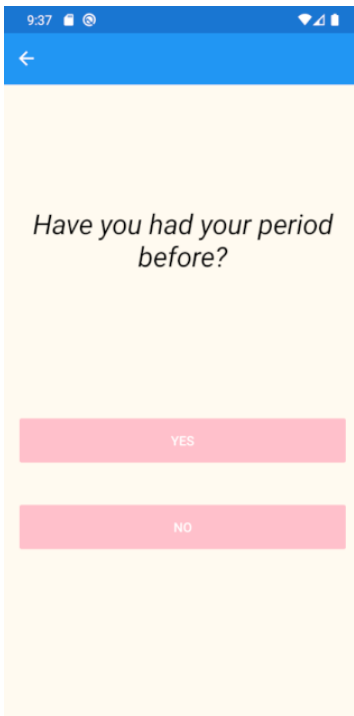
Test case 13	
Test type: erroneous	Actual result: 
Test Data: birthday entry ""	
Expected result: "Please make sure all fields are filled out!"	
Reason: There is blank space	
Test Result = PASS	

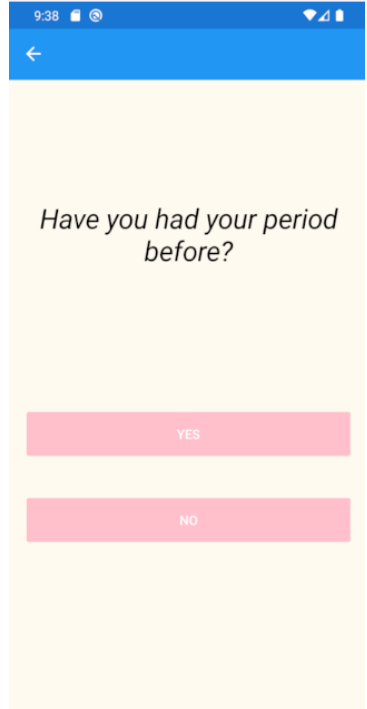
<u>Test case 14</u>	
Test type: extreme	Actual result: 
Test Data: birthday entry “31/07/2015”	
Expected result: the birthday should be accepted and the app should move on to the next page	
Reason: DD is on the boundary 31	
Test Result = PASS	

Test case 15	
Test type: erroneous	Actual result: 
Test Data: password entry "Yay1!"	
Expected result: "Password must letters, a number, a special character and be 8 or more characters long"	
Reason: The password is to short in length	
Test Result = PASS	

Test case 16	
Test type: erroneous	Actual result: 
Test Data: password entry "Password*"	
Expected result: "Password must letters, a number, a special character and be 8 or more characters long"	
Reason: no numbers	
Test Result = PASS	

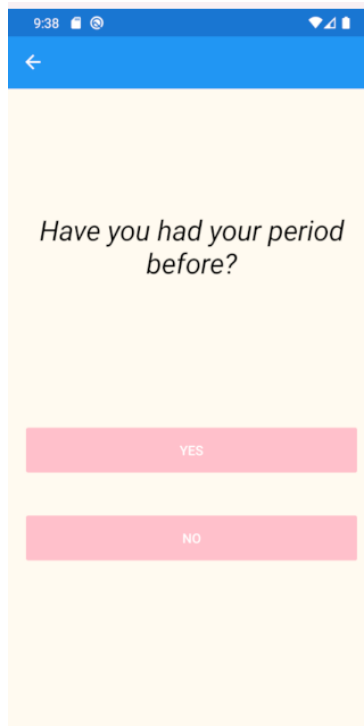
<u>Test case 17</u>	
Test type: erroneous	Actual result: 
Test Data: password entry ""	
Expected result: "Please make sure all fields are filled out"	
Reason: is blank space	
Test Result = PASS	

<u>Test case 18</u>	
Test type: extreme	Actual result: 
Test Data: password entry “123@yaYy”	
Expected result: the password entry is accepted and it moves onto the next page	
Reason: it is exactly 8 characters long which is the bound	
Test Result = PASS	

Test case 19	
Test type: typical	Actual result:  Test Result = PASS
Test Data: password entry "Hello1234!"	
Expected result: the password entry is accepted and it moves onto the next page	
Reason: Meets all the requirements	

Testing the BufferPage

Objectives being tested for
12. Navigation between all the pages works 12.1. Navigation in log in and sign up system works 12.3. Navigation for a user who has not experienced their first period works

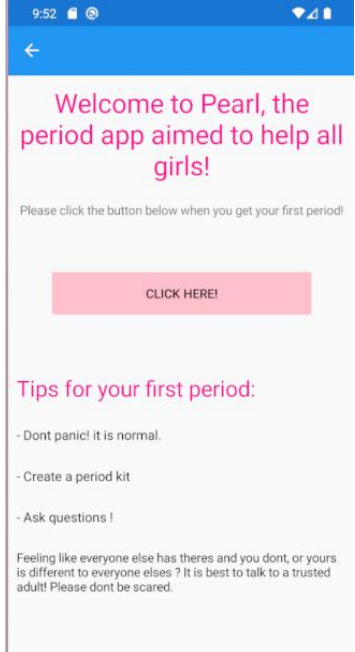


This page asks the user if they have got their first period yet, is they have then they press the yes button, if not they press the no button.

- The yes button takes them to the SignUp2 page where they are asked to input more data
- The no button takes them to a temporary home page, the FPPFalse page, it also then inputs their input data into the database

Test case 1	
Test type: typical	Actual result: Test Result = PASS
Test Data: the yes button on BufferPage is clicked	
Expected result: should direct the user to the SignUp2 page	
Reason: they have their first period so further data is needed to create their account	

Test case 2			
Test type: typical	Test Data: the no button on the BufferPage is clicked	Expected result: should direct the user to the FPPage	Reason: They do not have their period yet so a

		and add them to the database	temporary user (not a full account) is created for them
Actual result: 1.data placed in database  Test Result = PASS		2.FPFalse page is navigated to 	

Testing the hash code in SignUp

Objectives being tested for
2. Implement a hash code for the password
d. Hash code works

I have used a hash code for the password in my program to ensure more security for the user.

Test1: I input the password "Password123!". Here is a trace table of the hash code that was created:

password	string2	password1	a	count
Password123!	null	null	Null	0
			P	80
			a	177
			s	292
			s	407
			w	526
			o	637
			r	751
			d	851
			1	900
			2	950

			3	1001
			!	1034
	1034			
			1	1083
			0	1131
			3	1182
			4	1234
		1234		

Test2: I input the password "18*Dogs!". Here is a trace table of the hash code that was created:

password	string2	password1	a	count
18*Dogs!.	null	null	Null	0
			1	49
			8	105
			*	147
			D	215
			o	326
			g	429
			s	544
			!	577
	577			
			5	630
			7	685
			7	740
		740		

These show that the hash code is working correctly and that different password produce different hash codes.

Testing the SignUp2 Page – data validation and database insertion

Objectives being tested for
1.3. Query to the database including inserts, updates, reads and deletes
2. Use of a sign up and log in system
2.1. Data must be checked and validated when the user signs up (e.g. username cannot have been used before, password must meet requirements for password)
2.3. When an account is created, they must be added to the database
12. Navigation between all the pages works
12.1. Navigation in log in and sign up system works

Data validation

11:11

←

Continue creating account!

We just need a couple more details please, then we will set up your account!

How long does your cycle normally last?

Cycle Length

How long does your period normally last?

Period Length

What day are you currently on?

Current Day

are you on any birth control?

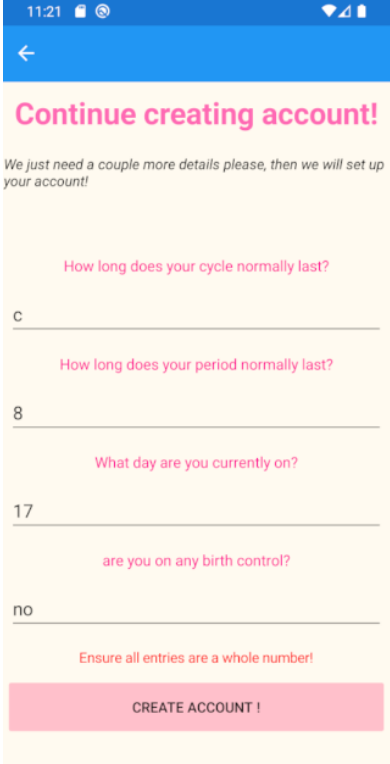
Yes or No (or leave blank if you wish)

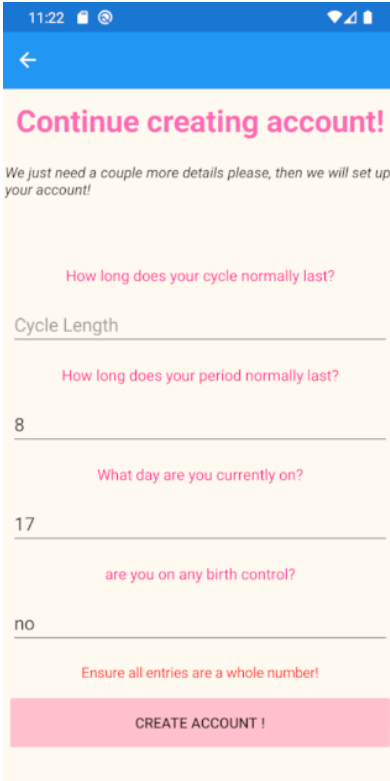
CREATE ACCOUNT !

- **Cycle Length** – can't be blank or white space. Must be a reasonable integer size (smaller than 120).
- **Period Length** – cant be blank or white space, must be an integer less than cycle length and also less than 25.
- **Current Day** – cant be blank or white space, must be an integer that's less than or equal to the cycle length.
- **Birth control** - Validation for the birth control entry does not need to be tested because if the user does not enter in anything random, then it is saved as false in the database because the user does not need to provide this information with the app if they do not wish too.

Initially I will only test data that should be erroneous. Data that is typical or boundary will be tested with in the database insertion section.

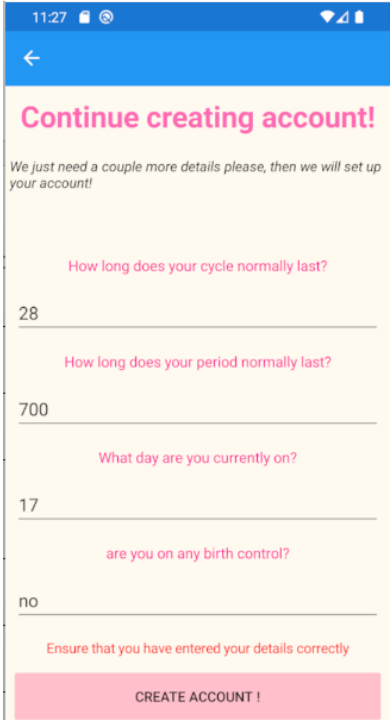
Test case 1	
Test type: erroneous	Actual result:
Test Data: Average cycle length entry: "c"	
Expected result: error message "ensure all entries are a whole number"	

	
Reason: not an integer	

Test case 2	
Test type: erroneous	Actual result:
Test Data: Average cycle length entry: ""	
Expected result: error message "ensure all entries are a whole number"	
Reason: is blank	

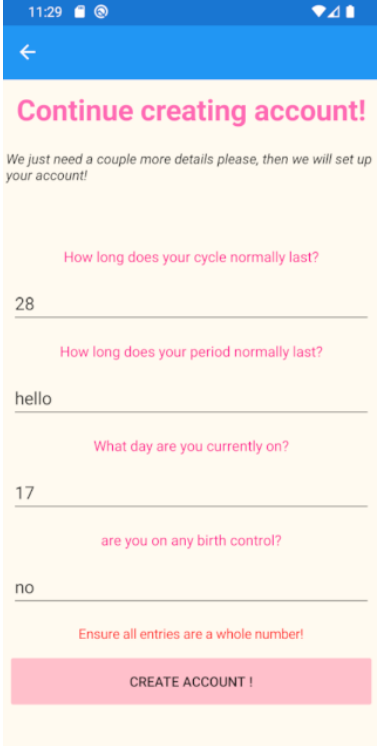
Test case 3

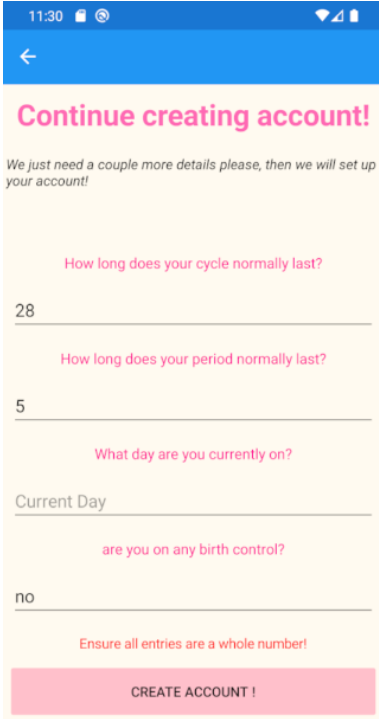
Test type: erroneous	Actual result: 
Test Data: Average cycle length entry: "130"	
Expected result: error message "ensure that you have entered your details correctly"	
Reason: greater than 120	

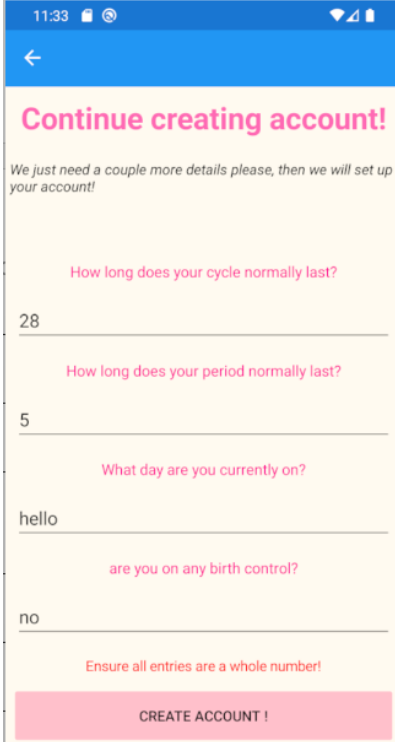
Test case 4	
Test type: erroneous	Actual result: 
Test Data: Average period length entry: "700"	
Expected result: error message "ensure that you have entered your details correctly"	
Reason: Average period length must be smaller than average cycle length	

Test case 5

Test type: erroneous	Actual result: 
Test Data: Average period length entry: "27"	
Expected result: error message "ensure that you have entered your details correctly"	
Reason: Must be smaller than 25	

Test case 6	
Test type: erroneous	Actual result: 
Test Data: Average period length entry: "hello"	
Expected result: error message "ensure all entries are a whole number"	
Reason: not an integer	

Test case 7	
Test type: erroneous	Actual result: 
Test Data: Current day entry: ""	
Expected result: : error message "ensure all entries are a whole number"	
Reason: is blank	

Test case 8	
Test type: erroneous	Actual result: 
Test Data: Current day entry: "hello"	
Expected result: error message "ensure all entries are a whole number"	
Reason: not a valid integer	

Database insertion

For this test of the SignUp2 page I will complete 3 tests.

- The first will be a lower boundary test of all the data being inserted to ensure that gets accepted and I will also look at the data being stored in the database to ensure this is correct as well.
- The second will be an upper bound test of all the data being inserted to ensure this also gets accepted and I will again look at the data being stored in the database to ensure this is correct.
- The third will be a typical test for the data validation and also a look to see the database is being stored correctly.

Test case 1 – lower boundary test

Data to be input:

First name – Lowerbound
 Username – lowerboundtest
 Birthday – 01/01/2001
 Password – Hello1234!
 First period - yes

Average cycle length – 5
 Average period length – 1
 Current day – 1
 On birth control – no

Proof of data inputs:

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Database

The image displays three screenshots of a mobile application interface, likely a menstrual tracker, showing different user interface elements and data.

Screenshot 1 (Left): Tracker Screen

- Header: 3:11, Signal, Wi-Fi, and Battery icons. Navigation: Back arrow, Tracker, HELP PAGE, PROFILE.
- Greeting: Goodafternoon, Lowerbound.
- Progress Indicator: A circular progress bar with a red segment and the text "Day 1".
- Phase: Menstrating
- Button: PERIOD ENDED
- Section: Previous period
- Status: Currently no data is available here, sorry!

Screenshot 2 (Middle): Help Page

- Header: 3:11, Signal, Wi-Fi, and Battery icons. Navigation: Back arrow, Help Page, TRACKER, PROFILE.
- Greeting: Hi, Lowerbound
- Text Box 1: You are in the menstrual phase of your cycle ! This is the first out of 4 phases. You may experience symptoms such as cramps, blating, irritability, tiredness, lower back pain. These are all completely normal but if you have any concerns please speak to a doctor!
- Text Box 2: If you are experiencing anything abnormal from your regular period, consider speaking to someone as this could be a big indicator for a health issue!
- Text Box 3: Thank you for using Pearl!

Screenshot 3 (Right): Profile Screen

- Header: 3:11, Signal, Wi-Fi, and Battery icons. Navigation: Back arrow, Profile, TRACKER, HELP PAGE.
- Section: Personal Details
- Fields: Username: lowerboundtest, Password: *****
- Fields: Name: Lowerbound, Age: 24
- Field: On birth control: no
- Buttons: EDIT DETAILS, LOG OUT, DELETE ACCOUNT

TEST RESULT = PASS

112

Test case 2 – upper boundary test

Data to be input:

First name – Upperbound

Username – upperboundtest

Birthday – 01/01/2001

Password – Hello1234!

First period – yes

Average cycle length – 119

Average period length – 24

Current day – 119

On birth control – no

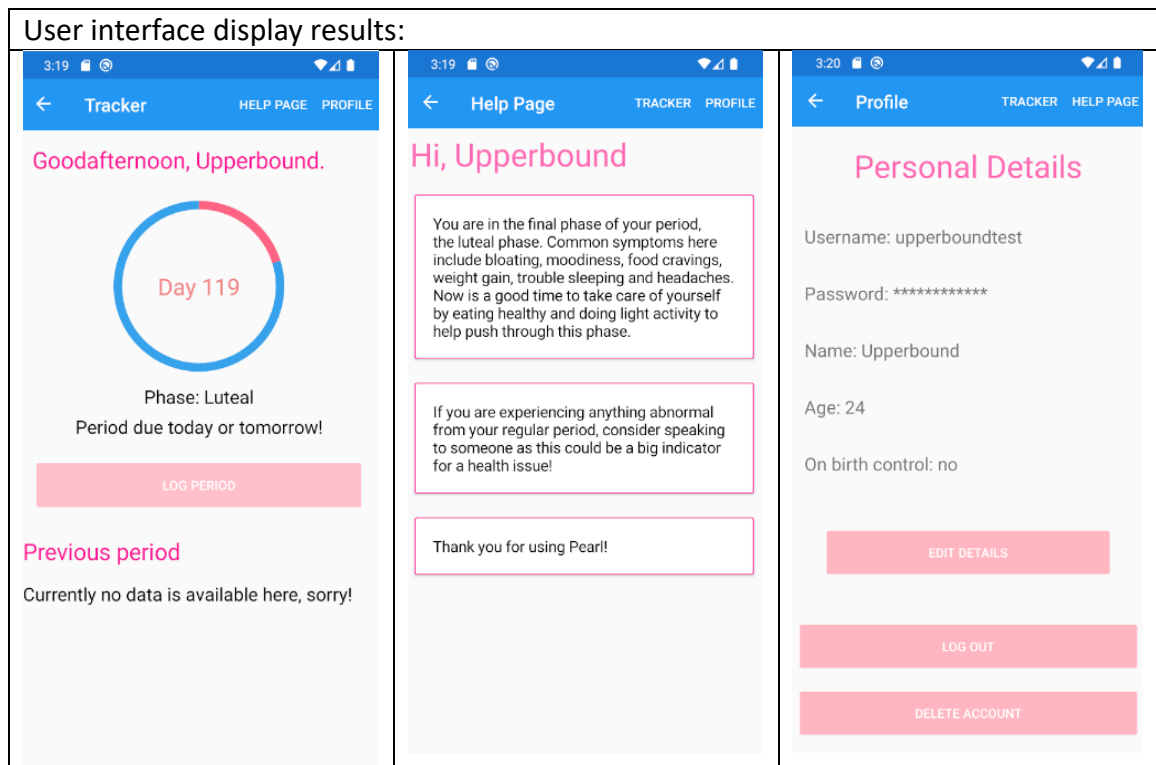
Proof of data inputs:

Three screenshots of the app's account creation process:

- Screenshot 1:** "Create an Account!" screen. Fields: Upperbound, upperboundtest, 01/01/2001, and a password field. A "PROCEED" button is at the bottom.
- Screenshot 2:** Confirmation screen: "Have you had your period before?". Buttons: YES, NO.
- Screenshot 3:** "Continue creating account!" screen. Fields: 119, 24, 119, no. A "CREATE ACCOUNT !" button is at the bottom.

Database

AddUserLI {App8.Model.LogIns} - Birthday {System.DateTime} - FirstName View "Upperbound" - Password View "894" - UserId 87 - UserName View "upperboundtest" - Non-public members	AddUserHP {App8.Model.HelpPage} - BC false - FirstPeriod true - Id 88 - UserId 87 - phase 0 - Non-public members
AddUserCT {App8.Model.CurrentT} - AvCycLen 119 - AvPLen 24 - CCycLen 119 - CPLen 24 - DaysLate 0 - Id 76 - OnPeriod false - StartDate {System.DateTime} - UserId 87 - Non-public members	AddUserLT {App8.Model.LastT} - Id 70 - LastDaysLate 0 - Length 0 - Recent1 {System.DateTime} - Recent2 {System.DateTime} - UserId 87 - Non-public members



TEST RESULT = PASS

The results from this show that the correct data is being input into the database and the data validation works as well.

Test case 3 - typical test

Data to be input:

First name – typical
 Username – typicaltest1
 Birthday – 01/01/2001
 Password – Hello1234!
 First period – yes

Average cycle length – 30
 Average period length – 6
 Current day – 8
 On birth control – yes

Proof of data inputs:

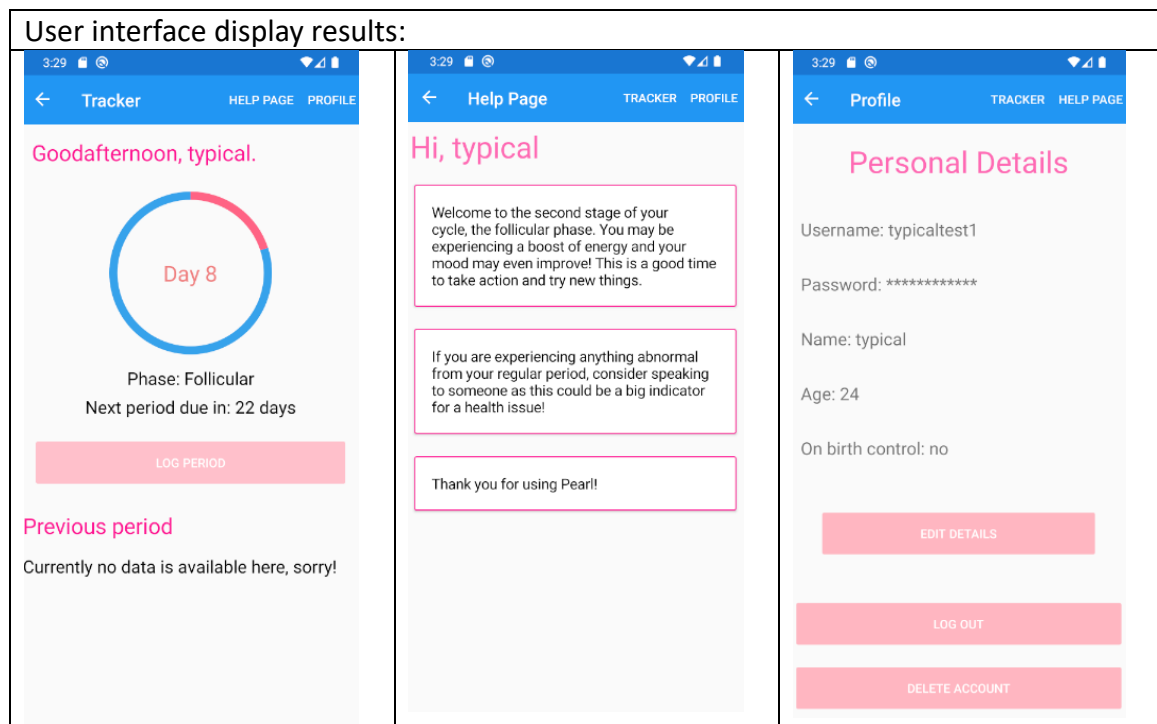
The first screenshot shows the 'Create an Account!' screen with the following inputs: 'typical' for the first name, 'typicaltest1' for the second name, '01/01/2001' for the birthday, and a password field with dots. A 'PROCEED' button is at the bottom.

The second screenshot shows a question: 'Have you had your period before?' with 'YES' and 'NO' buttons. A mouse cursor is pointing at the 'YES' button.

The third screenshot shows the 'Continue creating account!' screen with the following inputs: '30' for 'How long does your cycle normally last?', '6' for 'How long does your period normally last?', '8' for 'What day are you currently on?', and 'no' for 'are you on any birth control?'. A 'CREATE ACCOUNT !' button is at the bottom.

Database

AddUserLI {App8.Model.LogIns} - Birthday (System.DateTime) - FirstName View "typical" - Password View "894" - UserId 88 - UserName View "typicaltest1" - Non-public members	AddUserHP {App8.Model.HelpPage} - BC false - FirstPeriod true - Id 89 - UserId 88 - phase 0 - Non-public members
AddUserCT {App8.Model.CurrentT} - AvCycLen 30 - AvPLen 6 - CCycLen 8 - CPLen 6 - DaysLate 0 - Id 77 - OnPeriod false - StartDate (System.DateTime) - UserId 88 - Non-public members	AddUserLT {App8.Model.LastT} - Id 71 - LastDaysLate 0 - Length 0 - Recent1 (System.DateTime) - Recent2 (System.DateTime) - UserId 88 - Non-public members



TEST RESULT = PASS

The results from this show that the correct data is being input into the database and the data validation works as well.

Testing the FPFalse Page – button navigation

Objectives being tested for
1.3. Query to the database including inserts, updates, reads and deletes
9. Users are able to use a simplified version of the app if they have never had a period
9.1. Have the app ask the user if they have had their first period when they create an account
9.2. If they have not, the app is directed to a temporary homepage
9.3. It also creates a temporary account for them
9.4. Have a button that leads the user to finish creating their account
12. Navigation between all the pages works
12.1. Navigation in log in and sign up system works
12.2. Navigation for a user who has not experienced their first period works

1. I will create a user and select no for first period

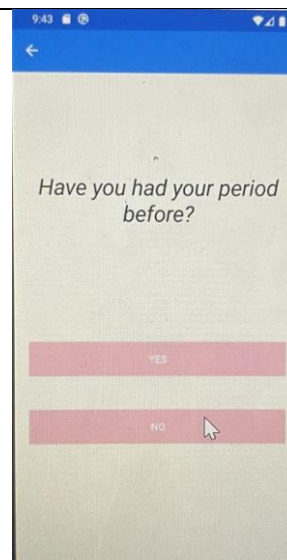
- This should guide the user to the FPFalse page and stored their data into the database

2. I will then log a first period for them

- a. Their details from their account should be passed into the SignUp2 page.
- b. The SignUp2 page is navigated to.

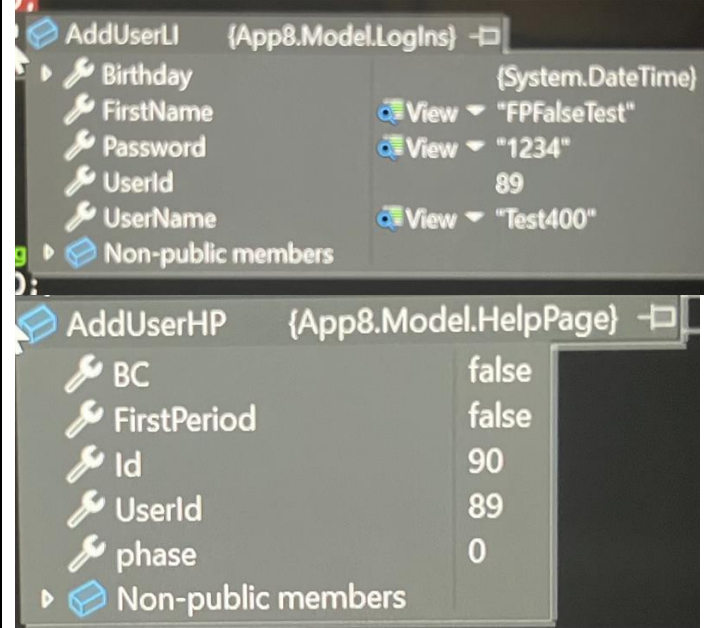
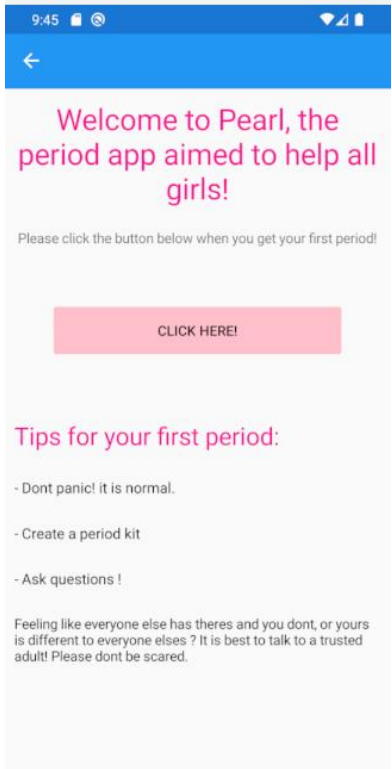
3. I will then enter the rest of their details in

- a. Their details should now be deleted.
- b. This should be restored into the database

1. Creating the user**Evidence of creating user**

Expected results: The data should be stored in the database the navigated to the FPFalse page

Actual results:

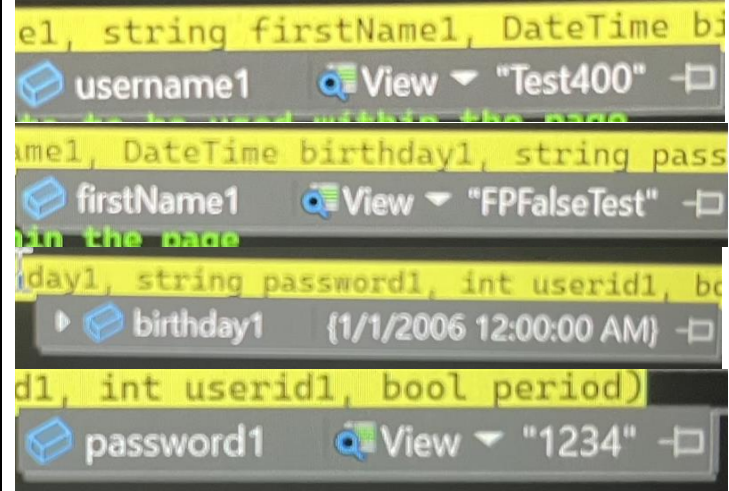
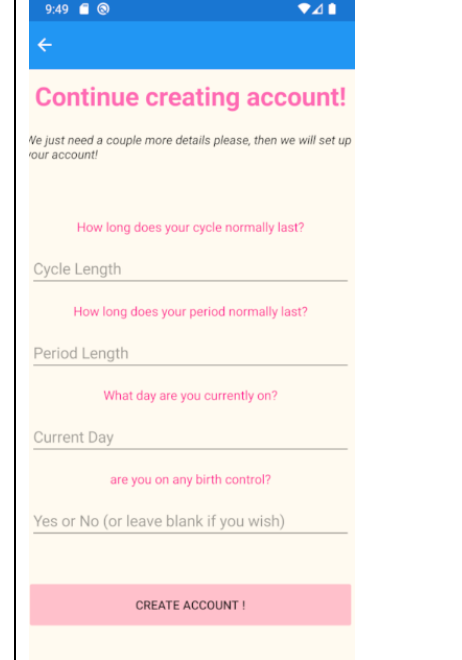
 This shows the data has been stored	This shows the page has been navigated to  Test Result = PASS
--	---

2. Logging a first period for them

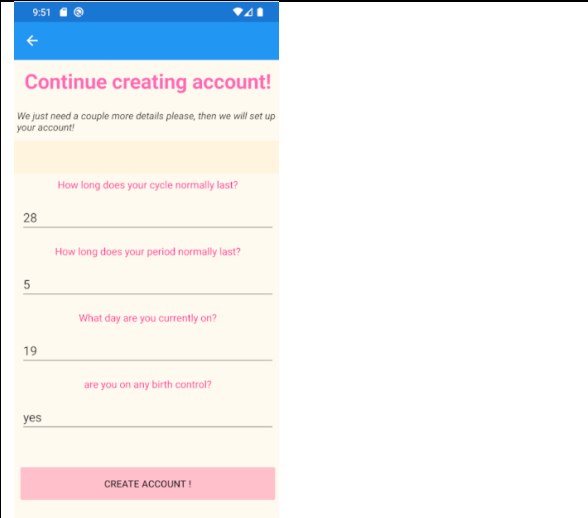
I pressed the "click here!" button

- Their details from their account should be passed into the SignUp2 page.
- The SignUp2 page should be navigated to.

Results:	
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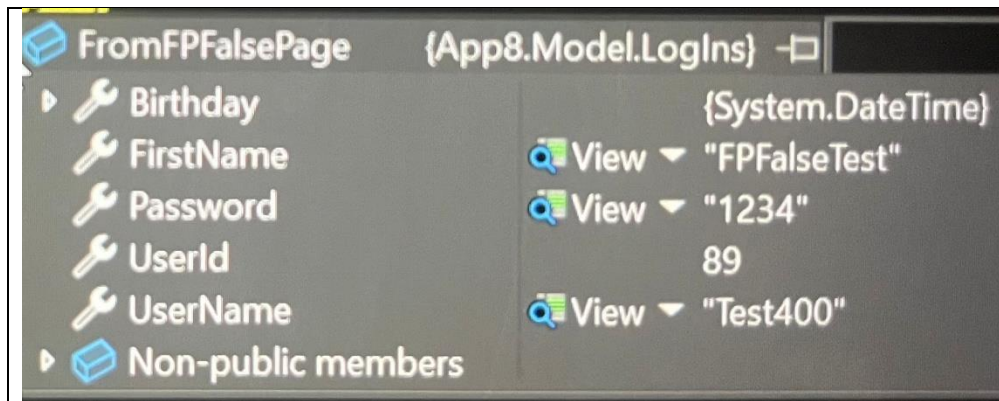
 el, string firstName1, DateTime b username1 View "Test400" ame1, DateTime birthday1, string pass in the page day1, string password1, int userid1, b birthday1 {1/1/2006 12:00:00 AM} d1, int userid1, bool period) password1 View "1234"	 Continue creating account! We just need a couple more details please, then we will set up your account! How long does your cycle normally last? Cycle Length How long does your period normally last? Period Length What day are you currently on? Current Day are you on any birth control? Yes or No (or leave blank if you wish) CREATE ACCOUNT !
This shows that the data is passed into the SignUp2 page correctly The Signup2 page is navigated to Test Result = PASS	

3. Entering the rest of details

Evidence of inputs	 Continue creating account! We just need a couple more details please, then we will set up your account! How long does your cycle normally last? 28 How long does your period normally last? 5 What day are you currently on? 19 are you on any birth control? yes CREATE ACCOUNT !
--------------------	--

Expected results: The current data should be deleted
Then the data should be re stored

Actual results:



^ This is the data that gets deleted

Below is the data that gets stored:

Test Result = PASS	

These results show that the user is being created correctly, the navigation works and the recreation when they log a period works correctly.

Testing Login page

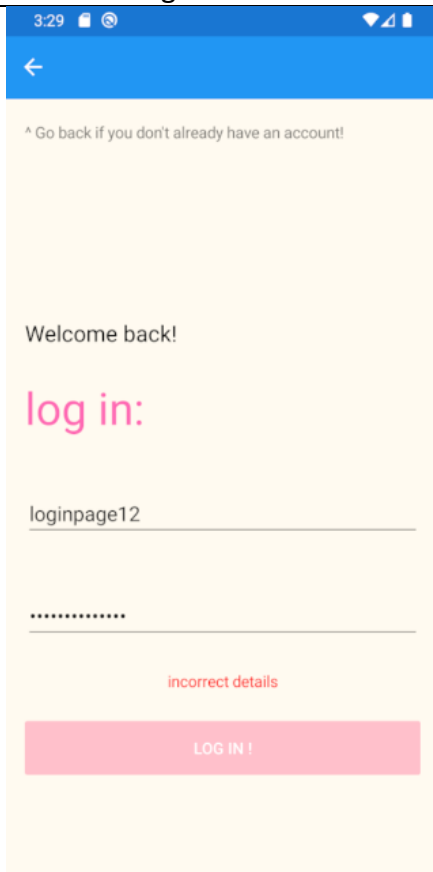
Objectives being tested for

2. Use of a sign up and log in system
 - 2.4. Log in only allowed once been checked against the database to see if the user exists

I will complete three individual tests to check that the LogIn page is working correctly

1. Entering details of an account that does not exist
 - a. An error message should be displayed
2. Entering the details of an account that does correct
 - a. The HomePage should be navigated to
3. Entering the username of an account that does exist but with the wrong password
 - a. An error message should be displayed

1. Entering details of an account that does not exist

Expected result:	Error message that states the details entered are incorrect
Actual result:	 The screenshot shows a mobile app interface for logging in. At the top, there's a blue header with a back arrow. Below it, a yellow background contains the text 'Welcome back!' and 'log in:'. There are two input fields: the first contains 'loginpage12' and the second is masked with dots. A red error message 'incorrect details' is displayed below the password field. At the bottom, there is a pink 'LOG IN !' button.
Test Result = PASS	

2. Entering the details of a user that does exist

Proof a user was created:

3:37 3:38

←

^ Go back if you already have an account!

Create an Account!

Test

test800

01/02/2003

PROCEED

Continue creating account!

We just need a couple more details please, then we will set up your account!

How long does your cycle normally last?

30

How long does your period normally last?

6

What day are you currently on?

17

are you on any birth control?

yes

CREATE ACCOUNT !

Expected results: once the login button is pressed, the log in should be accepted and the page should navigate to the HomePage

Data input:

3:39

←

^ Go back if you don't already have an account!

Welcome back!

log in:

test800

LOG IN !

Result:

3:39

← Tracker HELP PAGE PROFILE

Goodafternoon, Test.

Day 17

Phase: Luteal

Next period due in: 13 days

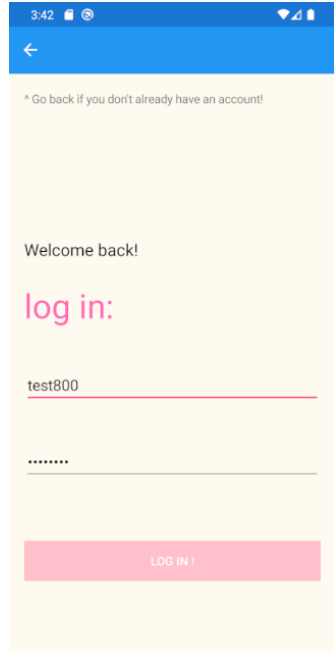
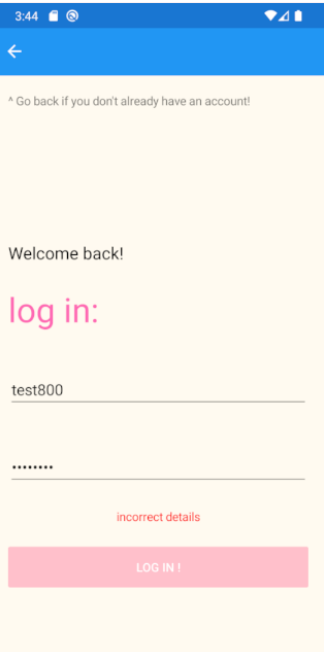
LOG PERIOD

Previous period

Currently no data is available here, sorry!

Test Result = PASS

3. Entering the username of an account that does exist but with the wrong password

Expected result: an error should be displayed that shows that the details entered are incorrect		
Data input: 	You can see from this screen shot that the password entered is clearly incorrect due to the length of the password entered	Result: 
Test Result = PASS		

Testing the merge sort and binary search

Objectives being tested for
1. Implement a merge sort and binary search - Use of a merge sort to order the user ids that have been used - Use of a binary search to check a randomly generated user id doesn't exist in the sorted list

- As there are lots of items of data being sorted using the merge sort I will adapt my code to run in a normal console application.
- This was I can create a smaller array of data to sort.
- It will function the same as my code
- Here you can see they are doing the same thing:

Code in my app	Console application code
<pre>public int CreateUserId() { var database = Querying.Connection(); List<int> intlist1 = database.Table<LogIns>().Select(u => u.UserId).ToList(); int[] intlist = intlist1.ToArray();</pre>	<pre>static void Main(string[] args) { int[] intlist = { 217, 92, 31, 67, 107, 104}; MergeSort(intlist, 0, intlist.Length -1); for(int i = 0; i < intlist.Length; i++) {</pre>

<pre> MergeSort(intlist, 0, intlist.Length - 1); Random random = new Random(); int a; do { a = random.Next(1, 901); } while (!BinarySearch(intlist, a)); return a; } static bool BinarySearch(int[] intlist, int a) { int min = 0; int max = intlist.Length - 1; while (min <= max) { int mid = (min + max) / 2; if (intlist[mid] == a) { return true; } else if (a < intlist[mid]) { max = mid - 1; } else { min = mid + 1; } } return false; } static void MergeSort(int[] intlist, int l, int r) { int mid; if (r > l) { mid = (r + l) / 2; MergeSort(intlist, l, mid); MergeSort(intlist, (mid + 1), r); MergeMethod(intlist, l, (mid + 1), r); } } static void MergeMethod(int[] intlist, int l, int mid, int r) </pre>	<pre> Console.WriteLine(intlist[i] + " "); } Random random = new Random(); int a; do { a = random.Next(1, 901); } while (!BinarySearch(intlist,a)); Console.WriteLine("id created: " + a); } static bool BinarySearch(int[] intlist, int a) { int min = 0; int max = intlist.Length - 1; while(min<= max) { int mid = (min+max)/2; if(intlist[mid] == a) { return true; } else if (a < intlist[mid]) { max= mid - 1; } else { min = mid + 1; } } return false; } static void MergeSort(int[] intlist, int l, int r) { int mid; if(r > l) { mid = (r+l)/2; MergeSort(intlist, l, mid); MergeSort(intlist, (mid + 1), r); MergeMethod(intlist, l, (mid+1), r); } } static void MergeMethod(int[] intlist, int l, int mid, int r) </pre>
---	---

<pre> { int[] temp = new int[901]; int c, lEnd, numelements, temp1; lEnd = mid - 1; temp1 = l; numelements = (r - l + 1); while ((l <= lEnd) && (mid <= r)) { if (intlist[l] <= intlist[mid]) { temp[temp1++] = intlist[l++]; } else { temp[temp1++] = intlist[mid++]; } } while (l <= lEnd) { temp[temp1++] = intlist[l++]; } while (mid <= r) { temp[temp1] = intlist[mid++]; } for (c = 0; c < numelements; c++) { intlist[r] = temp[r]; r--; } } </pre>	<pre> { int[] temp = new int[901]; int c, lEnd, numelements, temp1; lEnd = mid - 1; temp1 = l; numelements = (r - l + 1); while((l<=lEnd) && (mid<=r)) { if (intlist[l] <= intlist[mid]) { temp[temp1++] = intlist[l++]; } else { temp[temp1++] = intlist[mid++]; } } while (l <= lEnd) { temp[temp1++] = intlist[l++]; } while (mid <= r) { temp[temp1] = intlist[mid++]; } for(c = 0; c<numelements; c++) { intlist[r] = temp[r]; r--; } } </pre>
---	---

- The difference is that the code in my app uses an integer array determined by the data stored in the database
- I can preset an array in the console application for simplicity and to save the trace table being extremely long

Trace table for Merge sort:

			Int[] intlist						Int[] Temp					
mid	r	l	0	1	2	3	4	5	0	1	2	3	4	5
			217	92	31	67	107	104	0	0	0	0	0	0
0	5	0												
2														
0	2													
1														
0	1													
	0													

	1														
		1													
		0													
1															
2									92						
		1								217					
				217											
	0			92											
	-														
	1														
									0	0					
0	1	0													
1	2														
0		2													
1		0													
2															
3									31						
		1								92					
		2									217				
					217										
					92										
	0														
				31											
	-														
	1														
									0	0	0				
1	2	0													
2	5														
0		3													
4															
0	4														
3															
0	3														
3	4														
0		4													
3		3													
4															
		4										67			
5													107		
	3														
												0	0		
	2														
3	4	3													
4	5														
0		5													
4		3													
5															
												67			
6													104		
		5												107	

								107							
	3							104							
	2														
												0	0	0	
4	5	3													
2		0													
3															
		1							31						
4										67					
		2									92				
5												104			
6													107		
		3												217	
								217							
	4														
								107							
	3														
							104								
	2														
						92									
	1														
					67										
	0														
			31												
	-														
	1														
									0	0	0	0	0	0	
2	5	0													

This shows that the merge sort does correctly sort the list of data. It can now be placed searched using a binary search.

Testing binary search

To test the binary search, instead of using the random function I modified the code to set a to be the value of a number within the list to check that it finds it correctly. If the number was found then the program would create a new random number, however in my example a is not being re assigned.

It will display a message to say if the value was found of not.

The modified code:

```
static void Main(string[] args)
{
    int[] intlist = { 217, 92, 31, 67, 107, 104 };

    MergeSort(intlist, 0, intlist.Length - 1);

    for (int i = 0; i < intlist.Length; i++)
    {
        Console.WriteLine(intlist[i] + " ");
    }
    Console.WriteLine("enter value to search for");
    int a = Convert.ToInt32(Console.ReadLine());
    bool b = BinarySearch(intlist, a);
    if (b == true)
    {
        Console.WriteLine("The value " + a + " was found in list");
    }
    else
    {
        Console.WriteLine("The value " + a + " was not found");
    }
}

static bool BinarySearch(int[] intlist, int a)
{
    int min = 0;
    int max = intlist.Length - 1;
    while (min <= max)
    {
        int mid = (min + max) / 2;
        if (intlist[mid] == a)
        {
            return true;
        }
        else if (a < intlist[mid])
        {
            max = mid - 1;
        }
        else
        {
            min = mid + 1;
        }
    }
    return false;
}
```

<u>Value of a</u>	<u>Expected result</u>	<u>Actual result</u>	<u>Test result</u>
67	"Was found in the list"	31 67 92 104 107 217 enter value to search for 67 The value 67 was found in list	PASS
104	"Was found in the list"	31 67 92 104 107 217 enter value to search for 104 The value 104 was found in list	PASS
306	"Was not found"	31 67 92 104 107 217 enter value to search for 306 The value 306 was not found	PASS
9	"Was not found"	31 67 92 104 107 217 enter value to search for 9 The value 9 was not found	PASS
217	"Was found in the list"	31 67 92 104 107 217 enter value to search for 217 The value 217 was found in list	PASS

Testing functionality of the tracker

Objectives being tested for
3.The system must predict when the user will get their next period

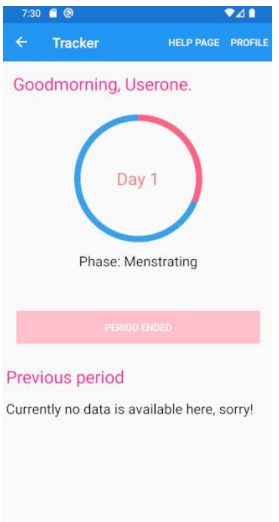
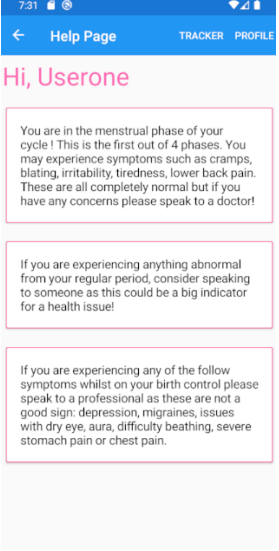
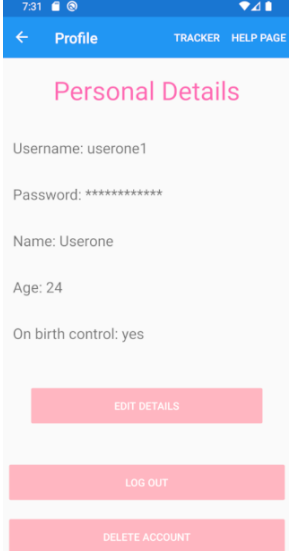
3.1. Uses data stored in database to make a prediction of when their next 3.2. period will be 3.3. When a full cycle has passed, the system must use the data in the database and the data from the cycle that's just ended to update its predictions
4. Displays data about the user's current cycle 4.1. Must display a model to show their cycle in some way 4.2. Must display what day the user is on and their current phase 4.3. Must display if the user is late to start their period
5. Displays data about the user's previous cycle 5.1. Uses data stored in database to display to user how long their cycle and period lasted, and if they were, how many days late they were (if they were)
6. Use of a help page to display advice to the user 6.1. Use the user's birthday to work out their age and display appropriate information (age specific) 6.2. Display appropriate advice for if they are on birth control or not 6.3. Display information based upon what phase of their period they are in currently
3. Symptom tracker a. User can view common symptoms they may experience currently in their cycle
4. Mood tracker a. User can view common moods they may experience currently in their cycle

Day 1 - 03/03/2025

I began by creating my users by following my testing plan

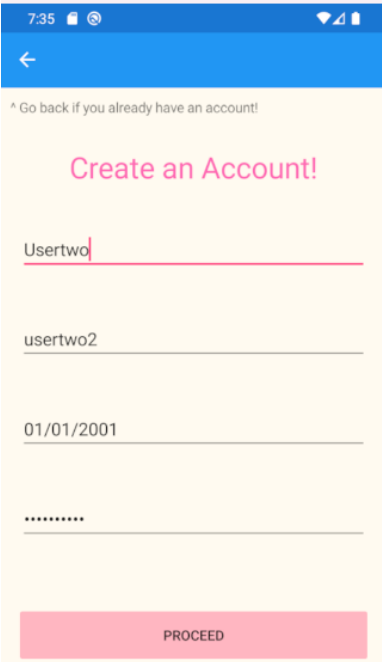
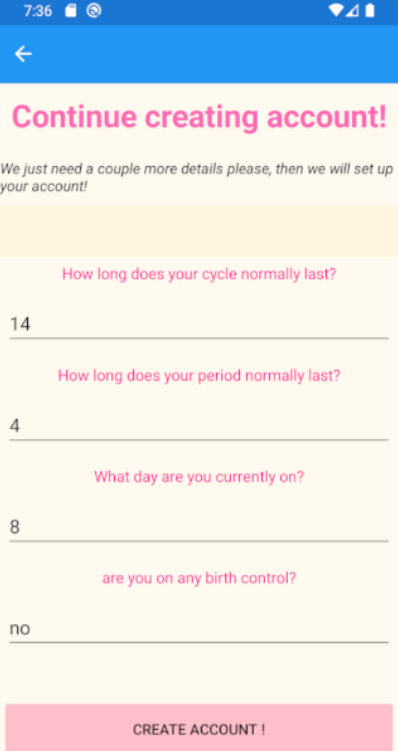
User 1:

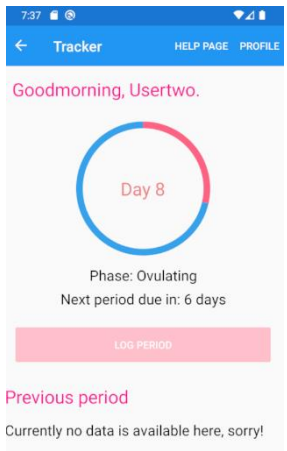
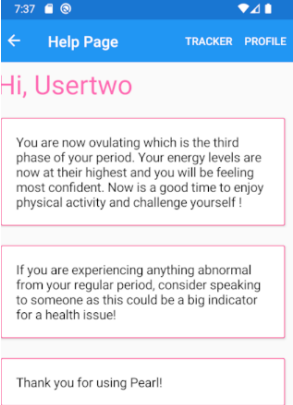
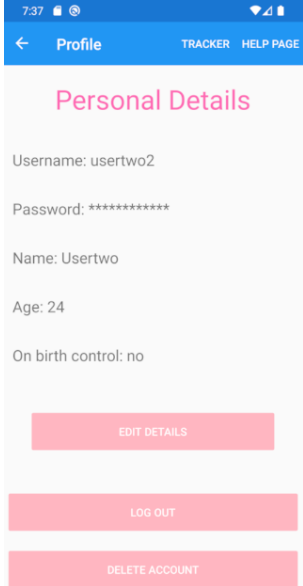
Inputs: First name: Userone Username: userone1 Password: Hello1234! Birth control: true First period: true Average cycle length: 16 Average period length: 5 Current day: 1	Evidence of inputs: 	
--	--	---

Expected outputs: Current day = 1 Phase = menstruating Previous period = n/a	Actual result: 		
Test Result = PASS			

Day 1 – 03/03/2025

User 2:

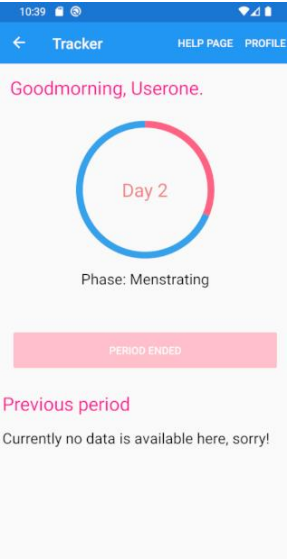
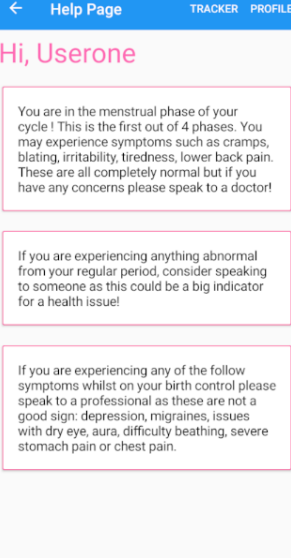
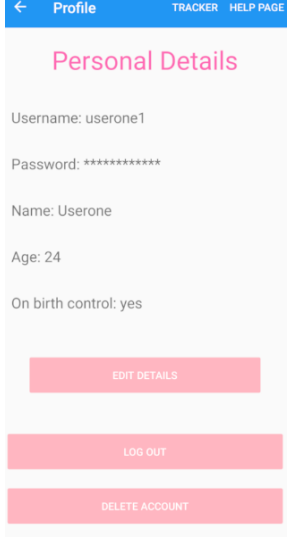
Inputs: First name: Usertwo Username: usertwo2 Password: Hello1234! Birth control: false First period: true Average cycle length: 14 Average period length: 4 Current day: 8	Evidence of inputs: 	
---	---	--

Expected outputs: Current day = 8 Phase = Ovulation Due in = 6 days Previous period = n/a	Actual result: 		
Test Result = PASS			

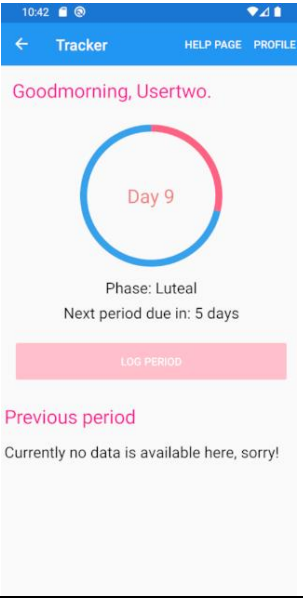
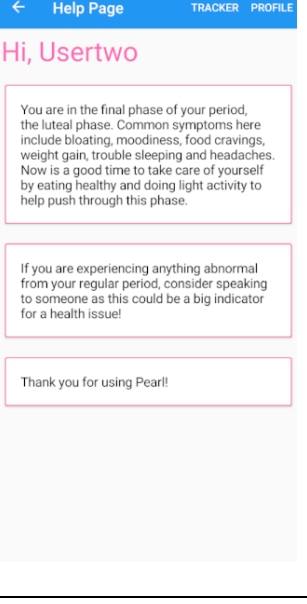
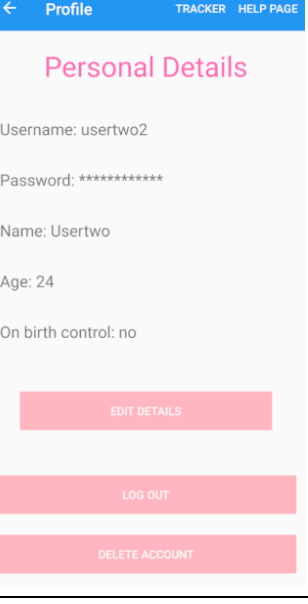
The results from day 1 suggest that the data is being input into the database correctly, and the functions to display the data to the users screen works correctly.

Day 2 – 04/03/25

User 1:

Expected outputs: Current day = 2 Phase = menstruating Previous period = N/A	Actual result: 		
Test Result = PASS			

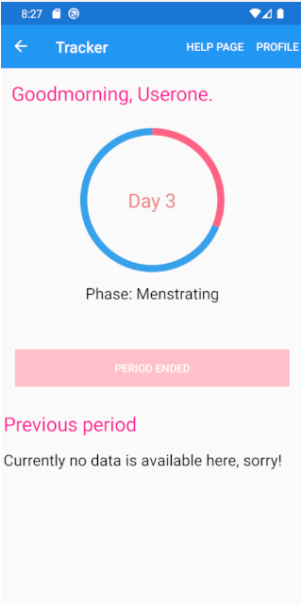
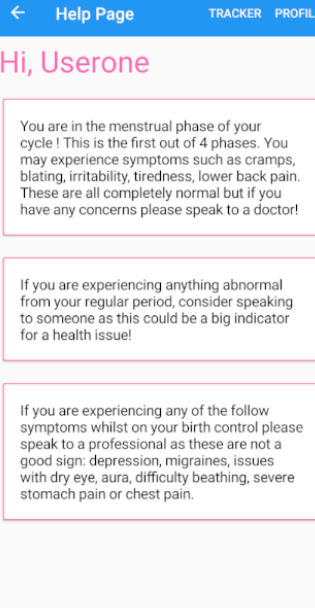
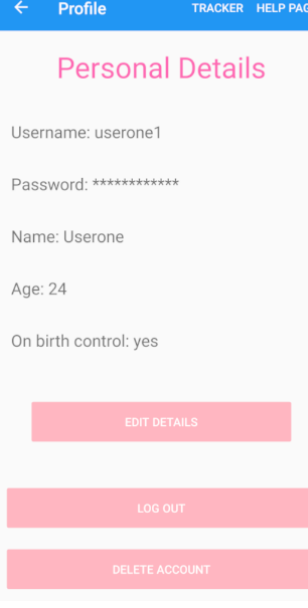
User 2:

Expected outputs: Current day = 9 Phase = Luteal Due in = 5 days Previous period = N/A	Actual result: 		
Test Result = PASS			

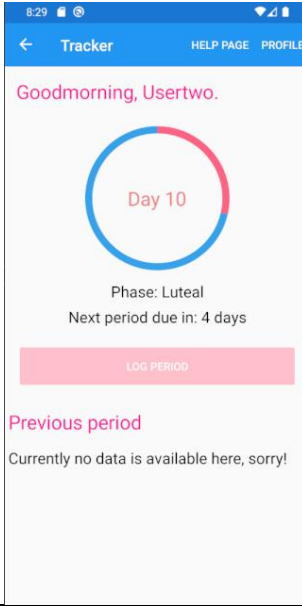
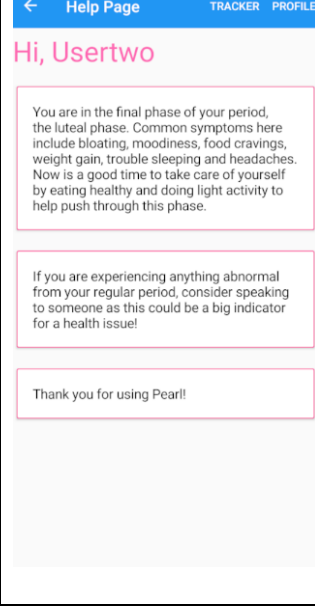
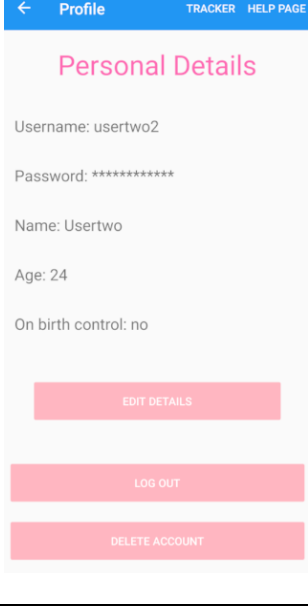
Results from day 2 suggest that the get current day function and the due in function work correctly.

Day 3 – 05/03/2025

User 1:

Expected outputs: Current day = 3 Phase = menstruating Previous period = N/A	Actual result: 		
Test Result = PASS			

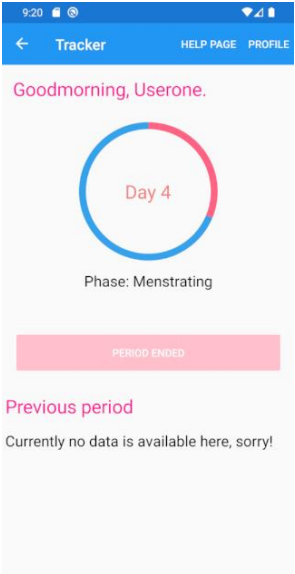
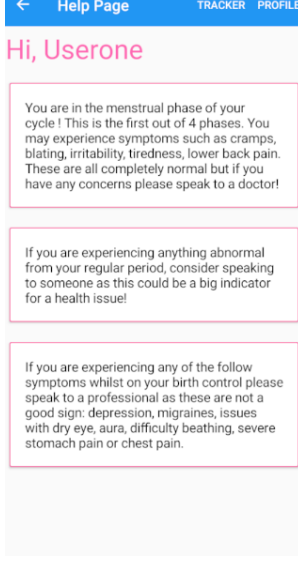
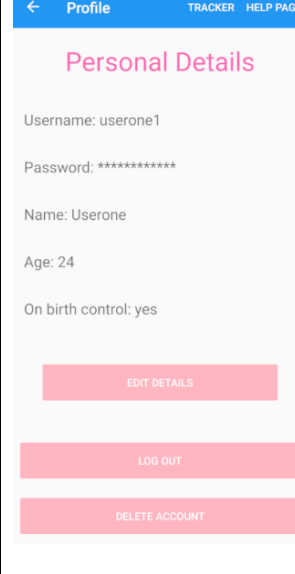
User 2:

Expected outputs: Current day = 10 Phase = Luteal Due in = 4 days Previous period = N/A	Actual result: 		
Test Result = PASS			

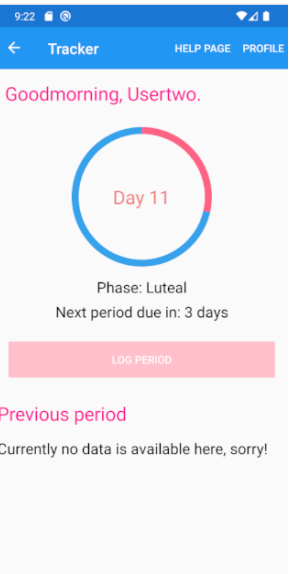
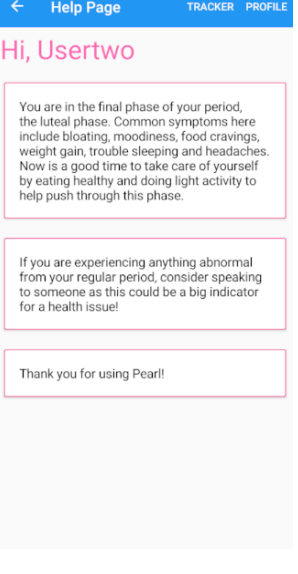
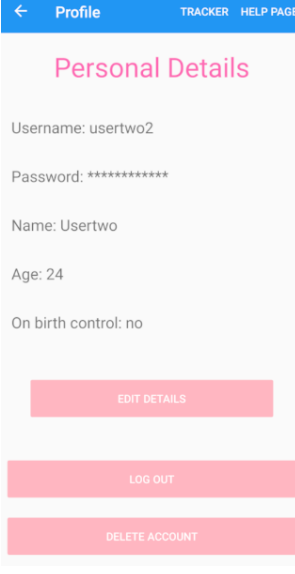
The results from day 3 show that the current day, due in and phase functions are all still working correctly.

Day4 – 06/03/2025

User 1:

Expected outputs: Current day = 4 Phase = menstruating Previous period = N/A	Actual result: 		
Test Result = PASS			

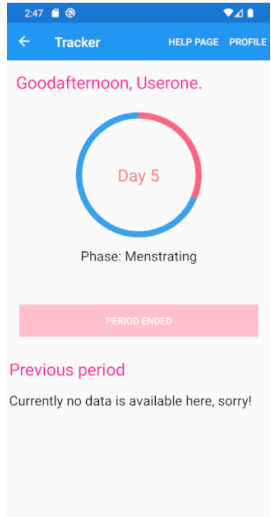
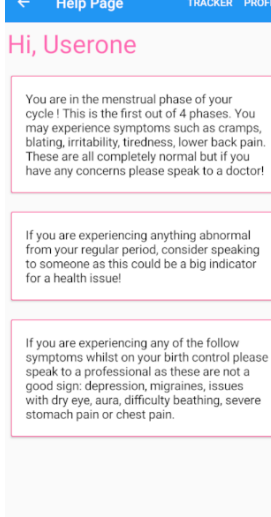
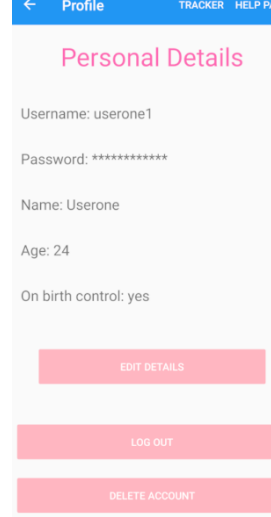
User 2:

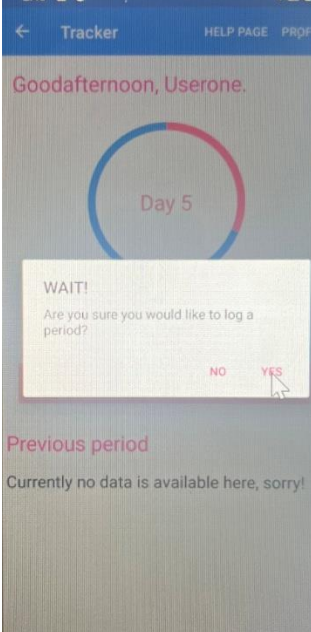
Expected outputs: Current day = 11 Phase = Luteal Due in = 3 days Previous period = N/A	Actual result: 		
Test Result = PASS			

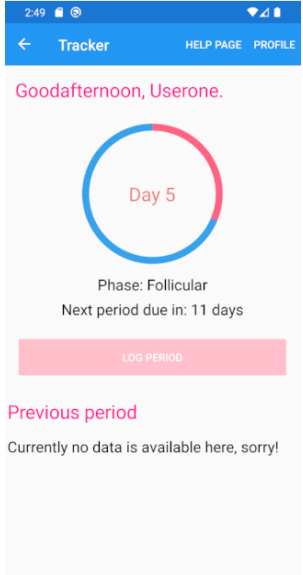
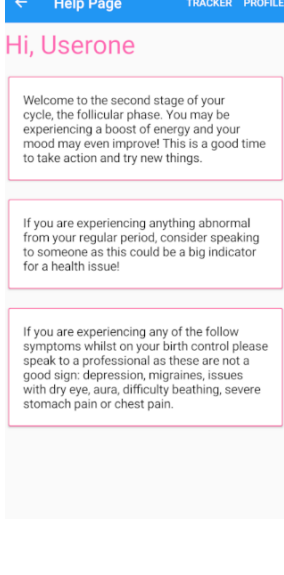
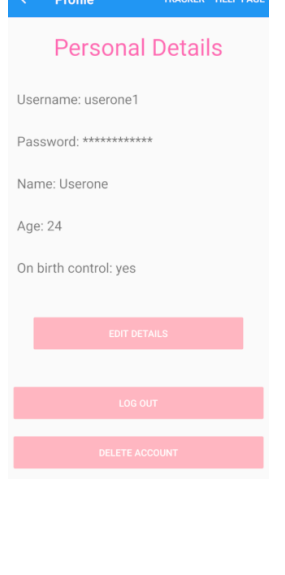
Day 4 shows that the homepage functions are still working correctly.

Day 5 – 07/03/2025

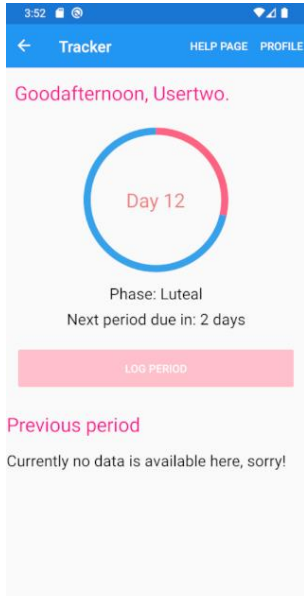
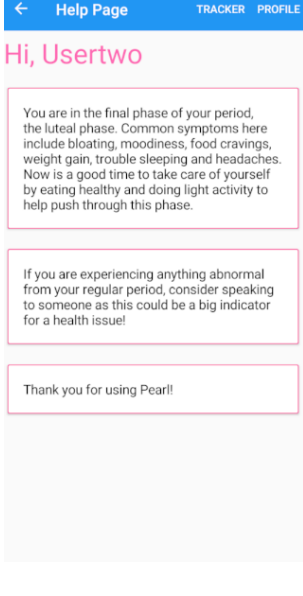
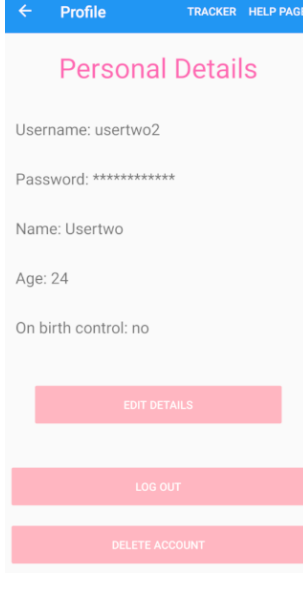
User 1:

Expected outputs: Current day = 5 Phase = menstruating Previous period = N/A	Actual result: 		
Test Result = PASS			

The end period button is then pressed and yes is selected. The result is displayed in the next table.	
---	--

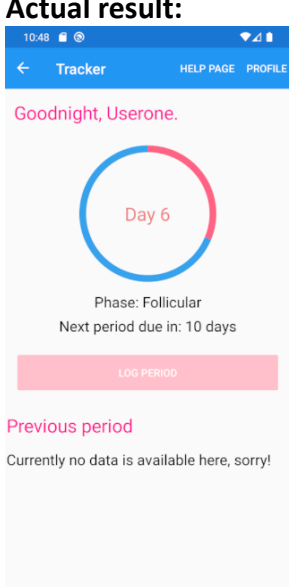
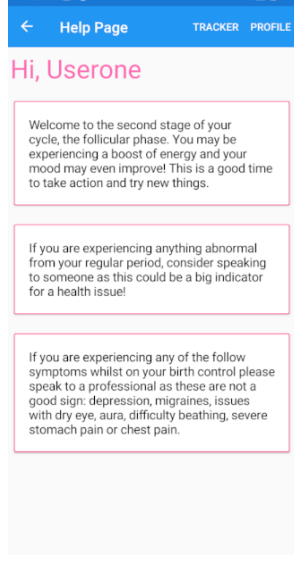
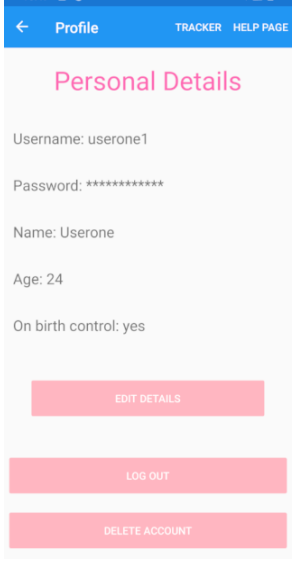
Expected outputs: Current day = 5 Phase = follicular Due in = 11 days Previous period = N/A	Actual result: 		
Test Result = PASS			

User 2:

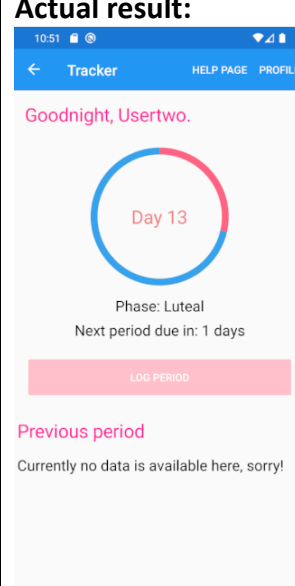
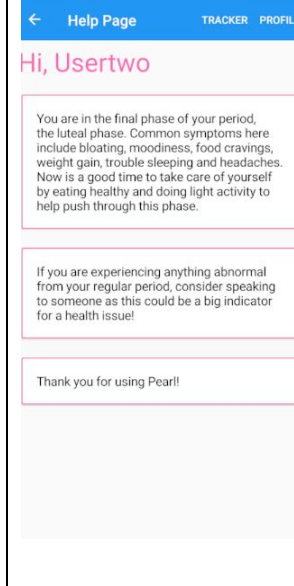
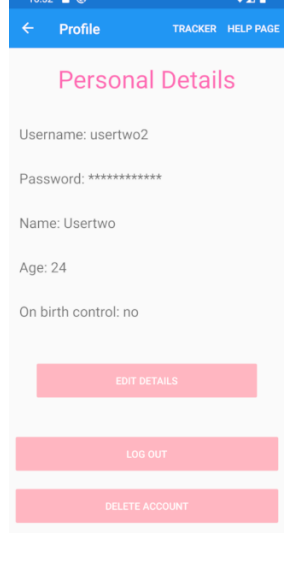
Expected outputs: Current day = 12 Phase = Luteal Due in = 2 days Previous period = N/A	Actual result: 		
Test Result = PASS			

Day 6 – 08/03/2025

User 1:

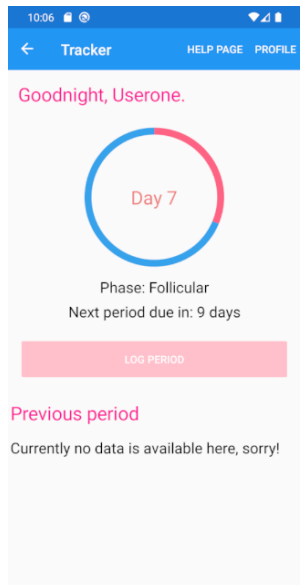
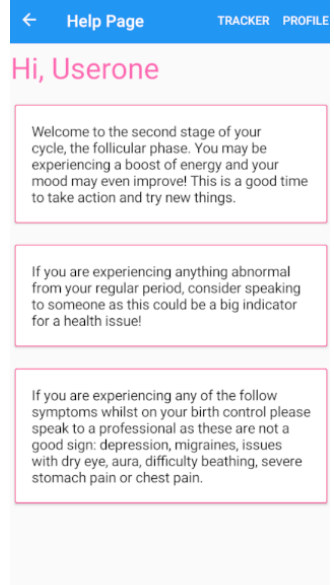
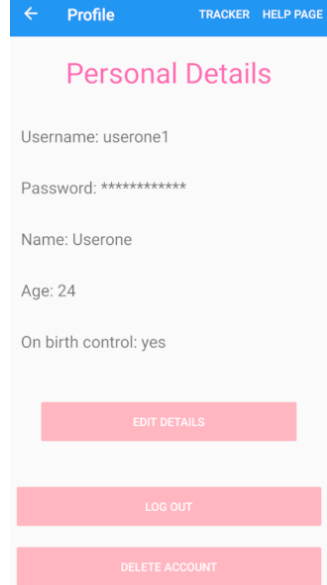
Expected outputs: Current day = 6 Phase = follicular Due in = 10 days Previous period = N/A	Actual result: 		
Test Result = PASS			

User 2:

Expected outputs: Current day = 13 Phase = Luteal Due in = due today or tomorrow Previous period = N/A	Actual result: 		
Test Result = PASS			

Day 7 – 09/03/2025

User 1:

Expected outputs: Current day = 7 Phase = ovulation Due in = 9 days Previous period = N/A	Actual result: 		
Test Result = FAIL			

This shows that the phase calculator isn't working correctly. The program should take days 7,8 and 9 as ovulation days for user one.

The code used:

```

var database = Querying.Connection();
var current = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
var helppage = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == id);
if (current != null && helppage != null)
{
    int CurrentCyc = current.CCycLen;
    int AvCyc = current.AvCycLen;
    int half = AvCyc / 2;
    int lower = half - 1;
    int higher = half + 1;
    //Menstrating
    if (current.OnPeriod == true)
    {
        helppage.phase = 1;
        database.Update(helppage);
    }
    //Follicular phase
    else if (lower >= CurrentCyc)
    {
        helppage.phase = 2;
        database.Update(helppage);
    }
}

```



```

    }
    //Ovulating
    //This is the part of the if statement that should run for userone
    else if (higher >= CurrentCyc)
    {
        helppage.phase = 3;
        database.Update(helppage);
    }
    //Luteal
    else
    {
        helppage.phase = 4;
        database.Update(helppage);
    }
}

```

Trace table of what actually happens:

CurrentCyc	AvCyc	half	lower	higher	Helppage.phase
7	16				
		8	7	9	
					2

This shows that the code goes wrong when it is going through the if statement because Helppage.phase should take the value 3, not 2. Therefore, I got rid of the higher and lower variables and modified the code to fix this.

Modified code:

```

var database = Querying.Connection();
var current = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
var helppage = database.Table<HelpPage>().FirstOrDefault(u => u.UserId == id);
if (current != null && helppage != null)
{
    int CurrentCyc = current.CCycLen;
    int AvCyc = current.AvCycLen;
    int half = AvCyc / 2;
    //Menstrating
    if (current.OnPeriod == true)
    {
        helppage.phase = 1;
        database.Update(helppage);
    }
    //Follicular phase
    else if (CurrentCyc < half - 1)
    {
        helppage.phase = 2;
        database.Update(helppage);
    }
}

```

```

//Ovulating
else if (CurrentCyc < half + 2)
{
    helppage.phase = 3;
    database.Update(helppage);
}
//Luteal
else
{
    helppage.phase = 4;
    database.Update(helppage);
}
}

```

I then reran the program to ensure it worked now

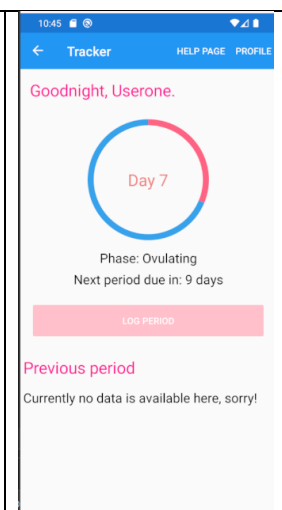
Expected outputs:

Current day = 7

Phase = ovulation

Due in = 9 days

Previous period = N/A



Test Result = **PASS**

Test 2

User 2:

Expected outputs:

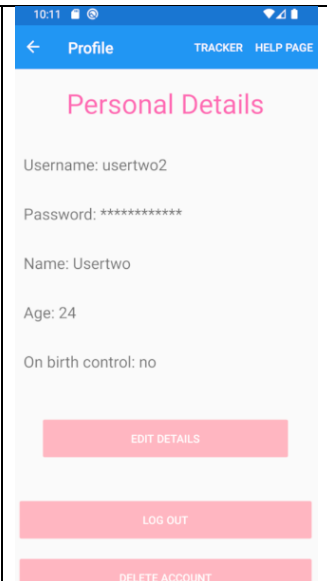
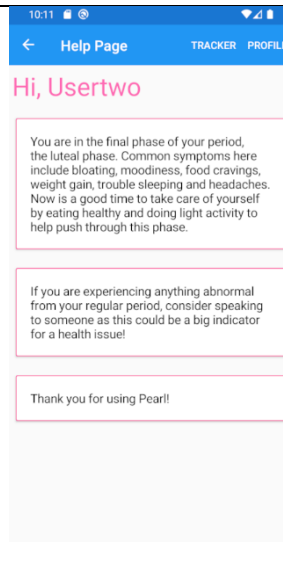
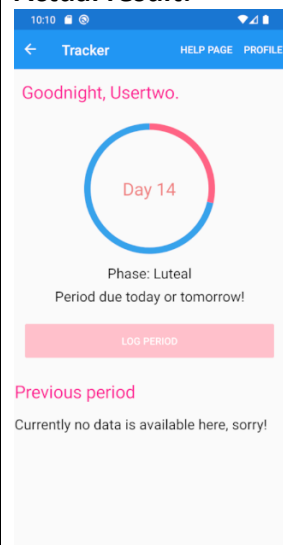
Current day = 14

Phase = Luteal

Due in = today or tomorrow

Previous period = N/A

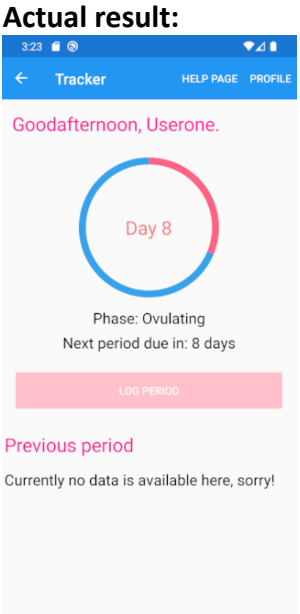
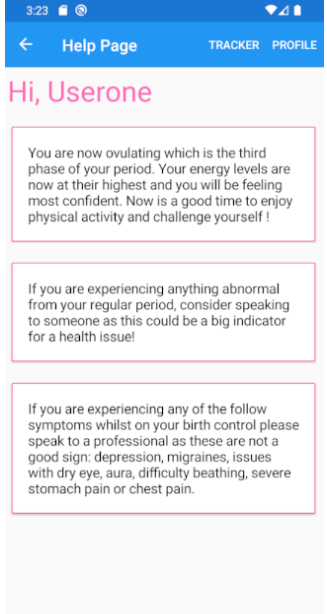
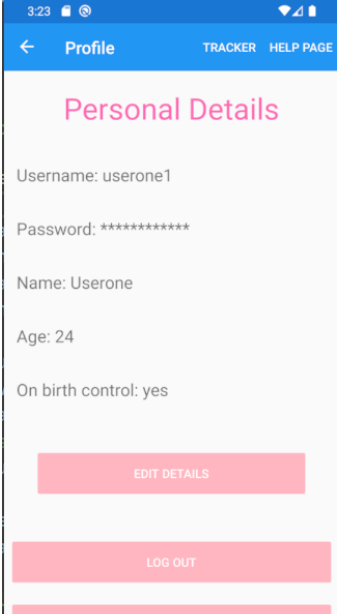
Actual result:



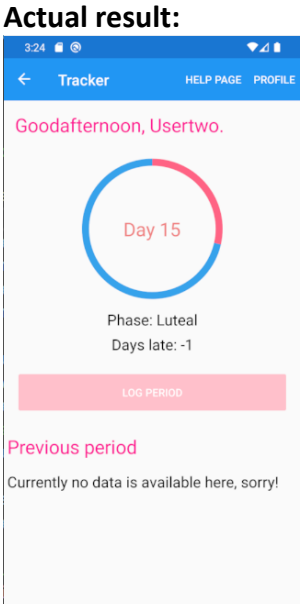
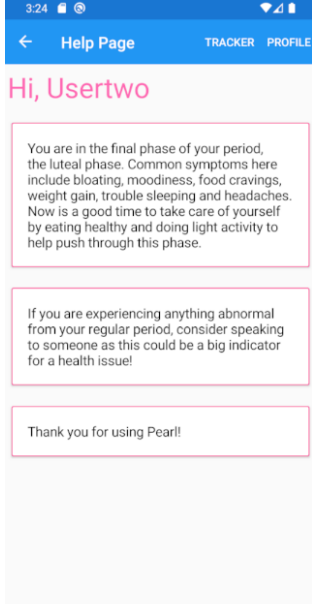
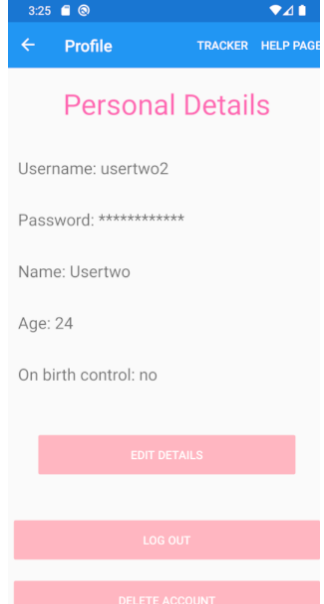
Test Result = **PASS**

Day 8 – 10/03/2025

User 1:

Expected outputs: Current day = 8 Phase = ovulation Due in = 8 days Previous period = N/A	Actual result: 		
Test Result = PASS			

User 2:

Expected outputs: Current day = 15 Phase = Luteal Late = 1 day Previous period = N/A	Actual result: 		
Test Result = FAIL			

This test shows that the days late is being calculated wrong. It should display 1 but it displays -1 instead

The code being used:

```
//This function is used to determine if the user is late to starting their period
public void CheckIfLate()
{
    var database = Querying.Connection();
    var late = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
    if (late != null)
    {
        int diff = late.AvCycLen - late.CCycLen;
        //If they are late then the database is updated, if not then its updated to say 0
        if (diff < 0)
        {
            late.DaysLate = diff;
            database.Update(late);
        }
        else
        {
            late.DaysLate = 0;
            database.Update(late);
        }
    }
}
```

Trace table of what is happening:

late.AvCycLen	late.CCycLen	diff	late.DaysLate
14	15		
		-1	
			-1

This shows that the value of difference is being calculated wrong. I changed this and I also changed how

The modified code:

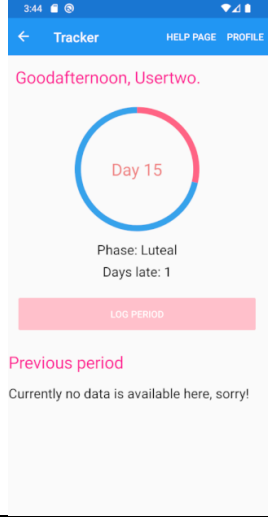
```
//This function is used to determine if the user is late to starting their period
public void CheckIfLate()
{
    var database = Querying.Connection();
    var late = database.Table<CurrentT>().FirstOrDefault(u => u.UserId == id);
    if (late != null)
    {
        int diff = late.CCycLen - late.AvCycLen;
        //If they are late then the database is updated, if not then its updated to say 0
    }
}
```

```

    if (diff > 0)
    {
        late.DaysLate = diff;
        database.Update(late);
    }
    else
    {
        late.DaysLate = 0;
        database.Update(late);
    }
}
}

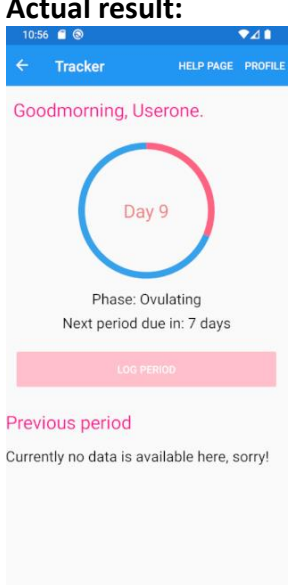
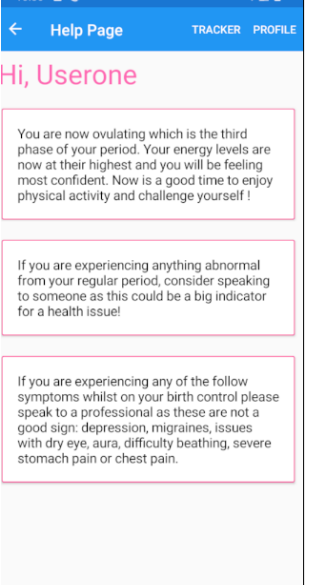
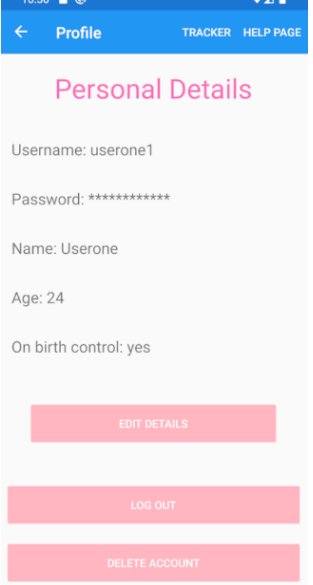
```

I then reran the program to ensure it worked now

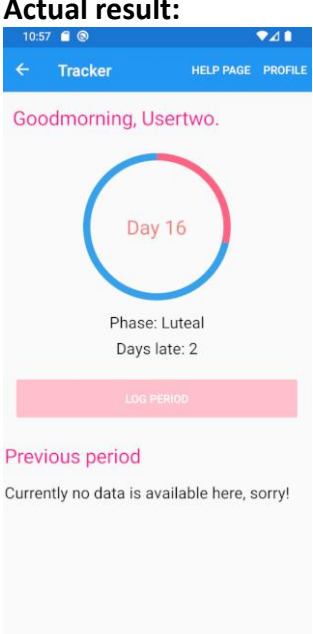
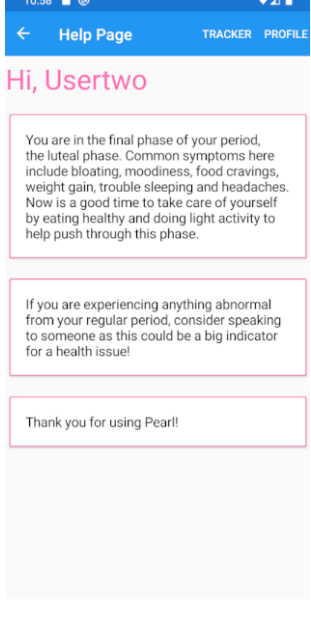
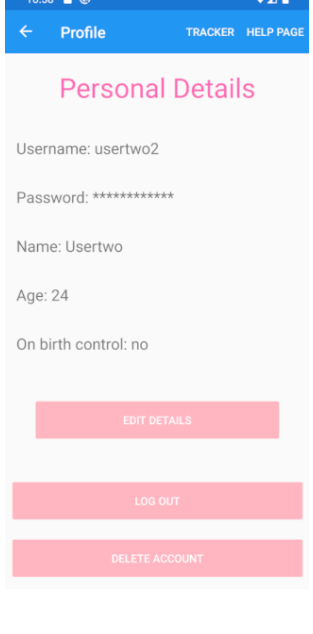
Expected outputs: Current day = 15 Phase = Luteal Late = 1 day Previous period = N/A	
Test Result = PASS	Test 2

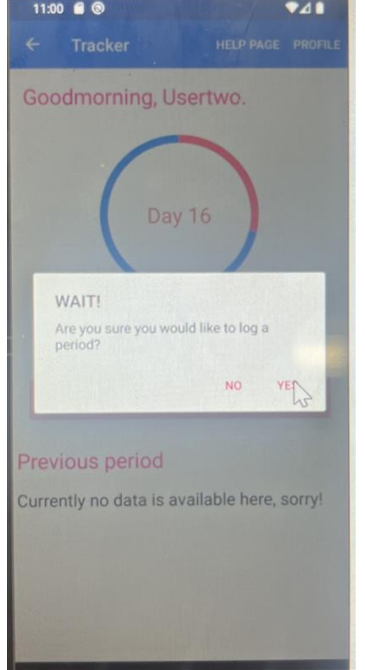
Day 9 – 11/03/2025

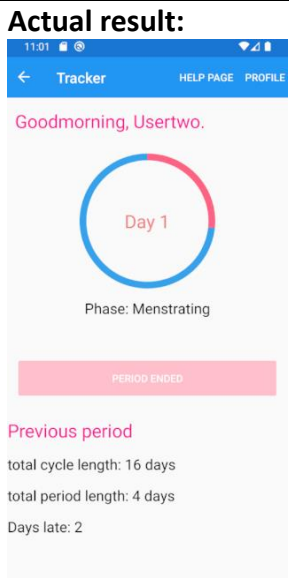
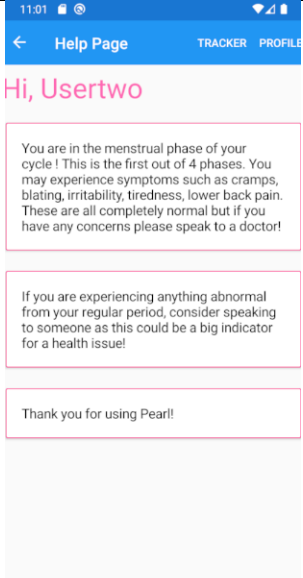
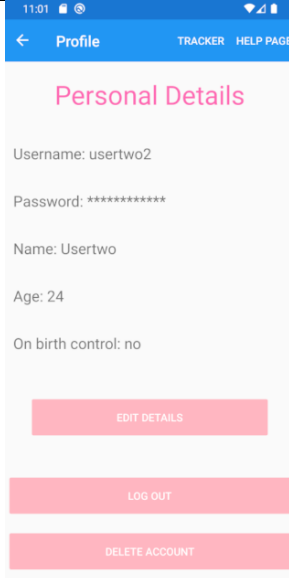
User 1:

Expected outputs: Current day = 9 Phase = ovulation Due in = 7 days Previous period = N/A	Actual result: 		
Test Result = PASS			

User 2:

Expected outputs: Current day = 16 Phase = Luteal Late = 2 days Previous period = N/A	Actual result: 		
Test Result = PASS			

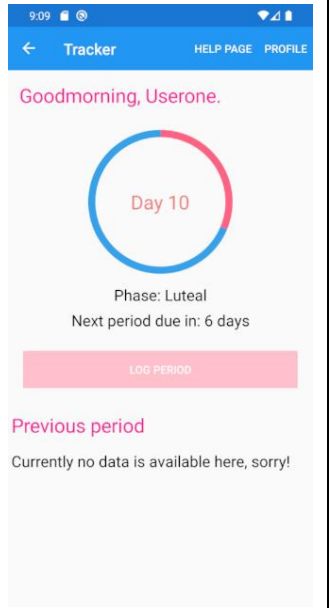
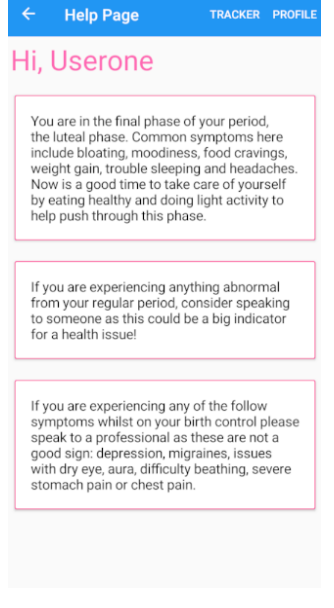
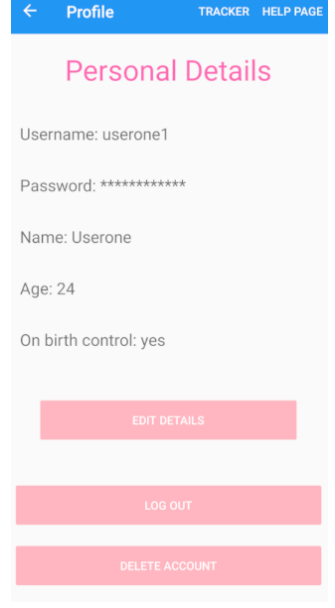
• I then pressed the Log period button to indicate that the user has ended their period. • I confirmed the warning message by selecting yes	
--	---

Expected outputs: Current day = 1 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days	Actual result: 		
Test Result = PASS			

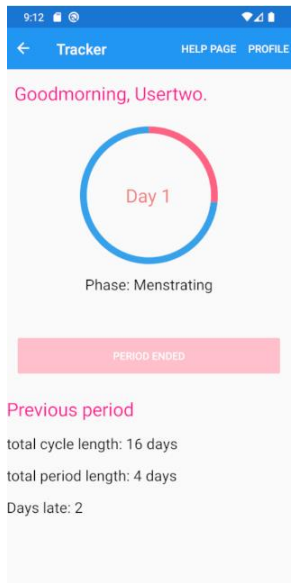
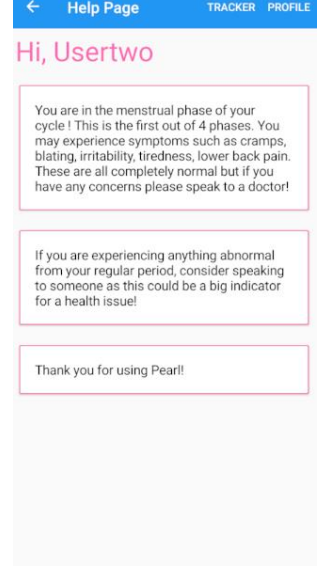
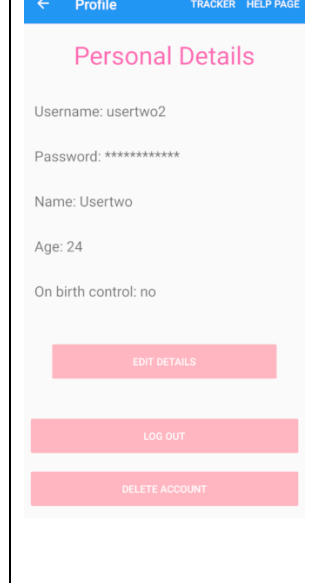
Day 9 confirms that the Log period button is working correctly and displays the correct data to the user still.

Day 10 – 12/03/2025

User 1:

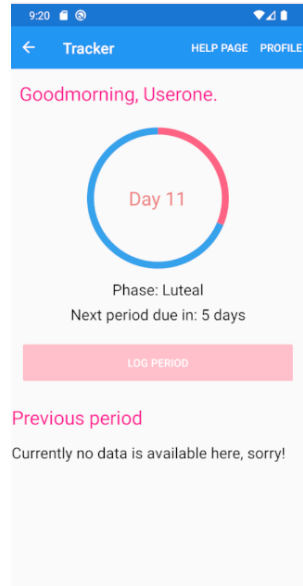
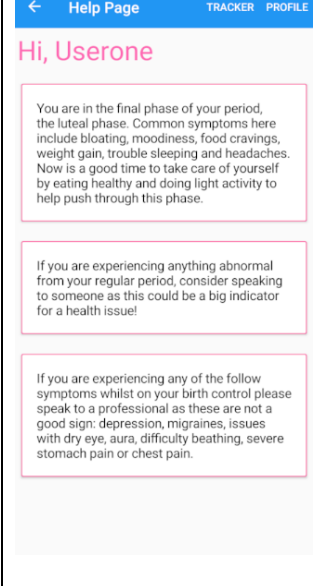
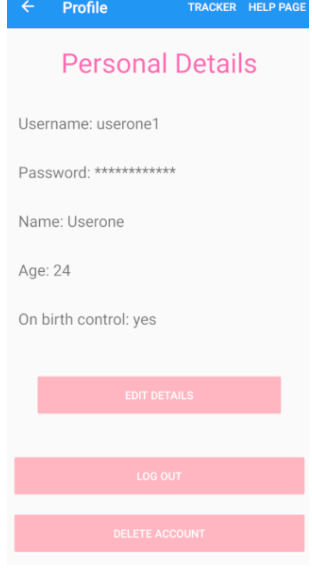
Expected outputs: Current day = 10 Phase = luteal Due in = 6 days Previous period = N/A	Actual result: 		
Test Result = PASS			

User 2:

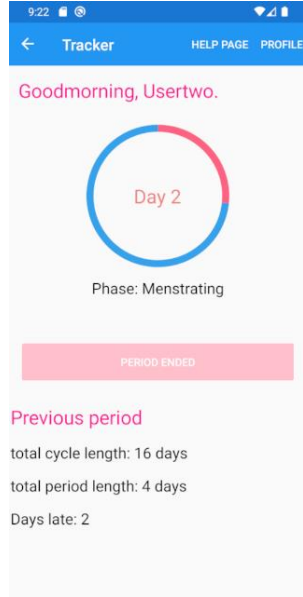
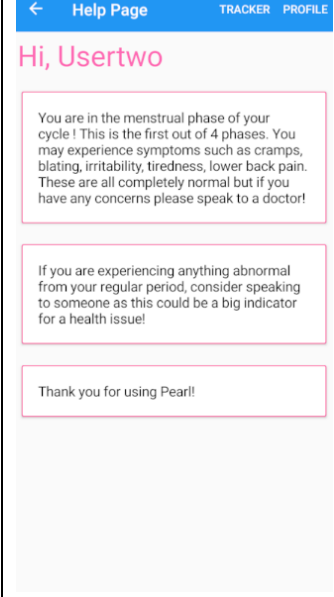
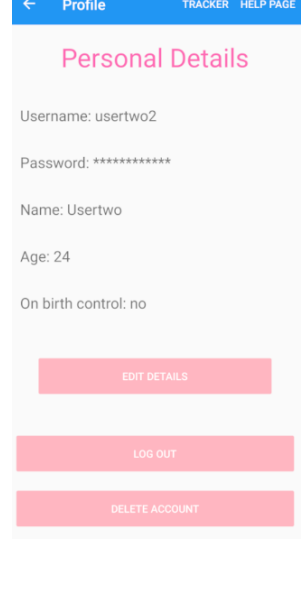
Expected outputs: Current day = 1 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days	Actual result: 		
Test Result = PASS			

Day 11 - 13/03/2025

User 1:

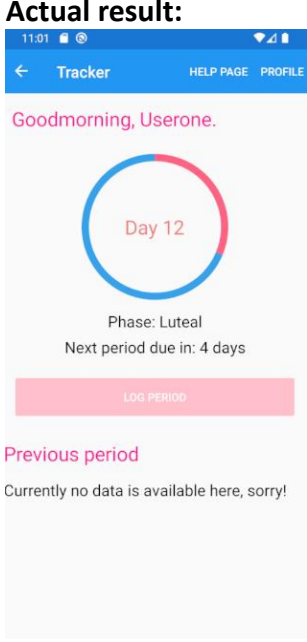
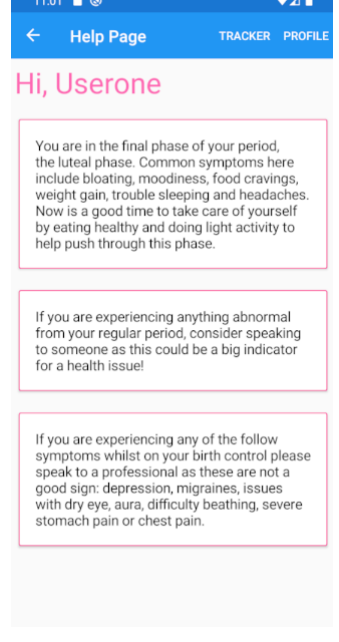
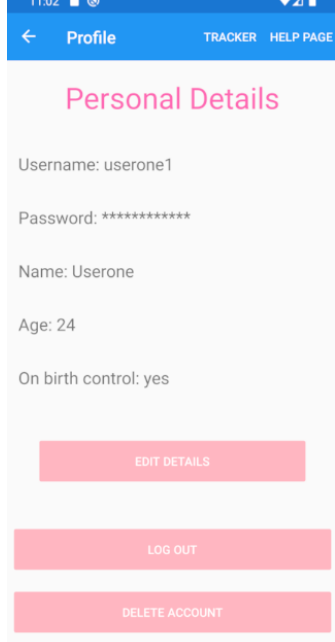
Expected outputs: Current day = 11 Phase = luteal Due in = 5 days Previous period = N/A	Actual result: 		
Test Result = PASS			

User 2:

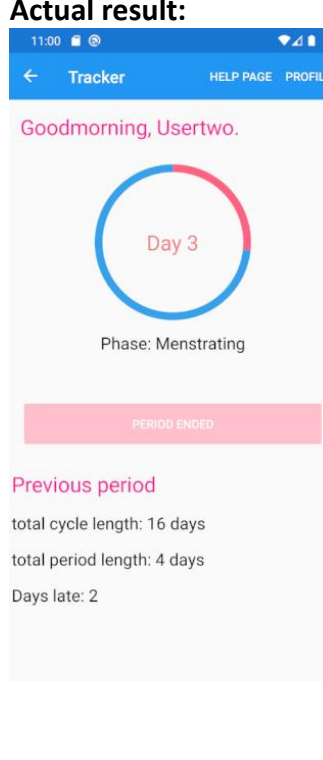
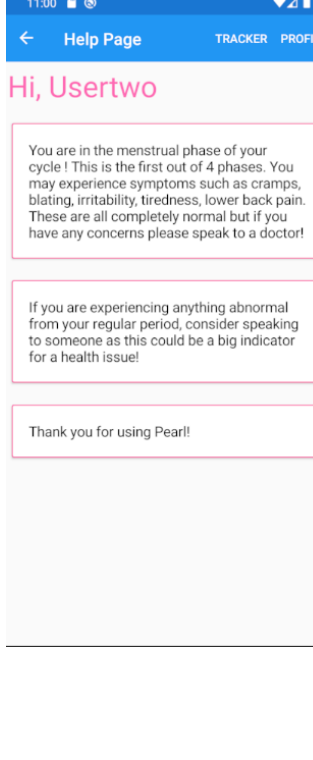
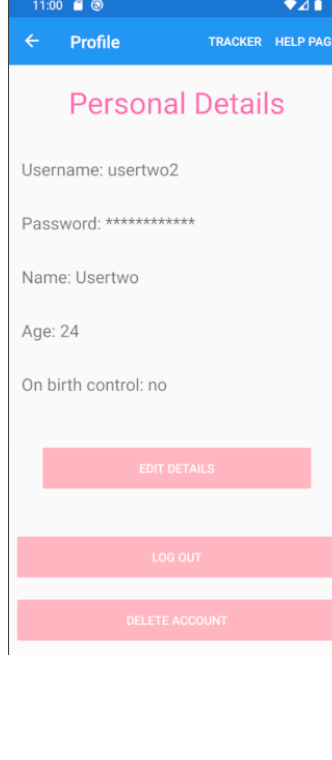
Expected outputs: Current day = 2 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days	Actual result: 		
Test Result = PASS			

Day 12 - 14/03/2025

User 1:

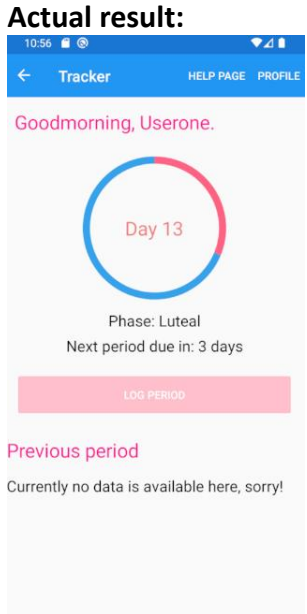
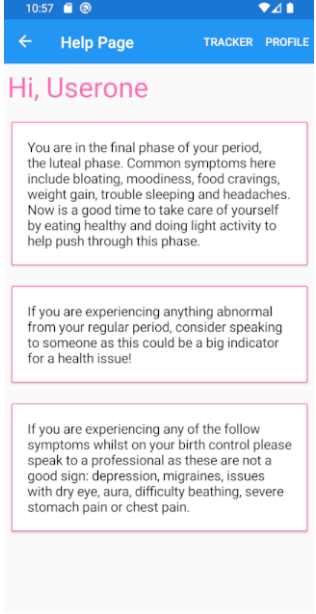
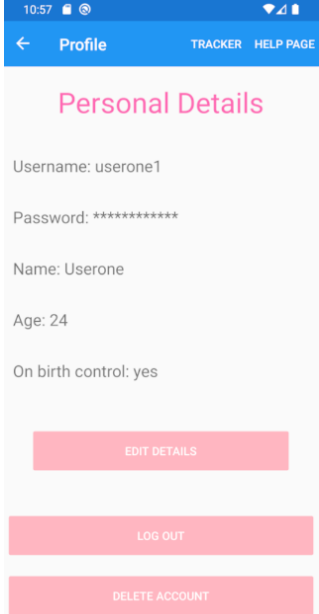
Expected outputs: Current day = 12 Phase = luteal Due in = 4 days Previous period = N/A	Actual result: 		
Test Result = PASS			

User 2:

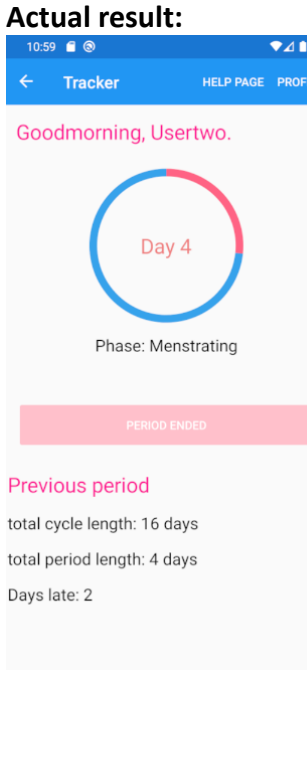
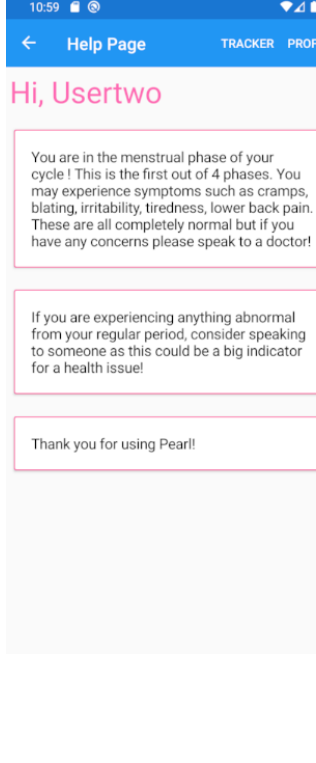
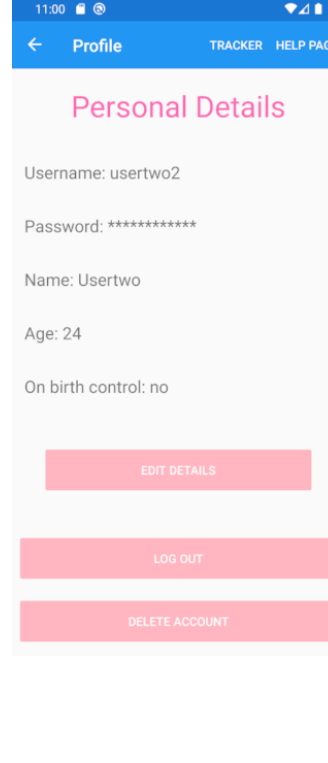
Expected outputs: Current day = 3 Phase = menstruating Previous period: Cycle length = 16 days Period length = 4 days Late by= 2 days	Actual result: 		
Test Result = PASS			

Day 13 - 15/03/2025

User 1:

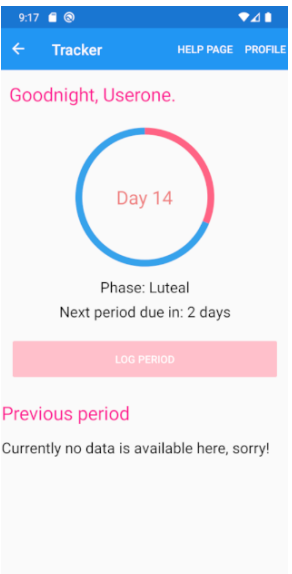
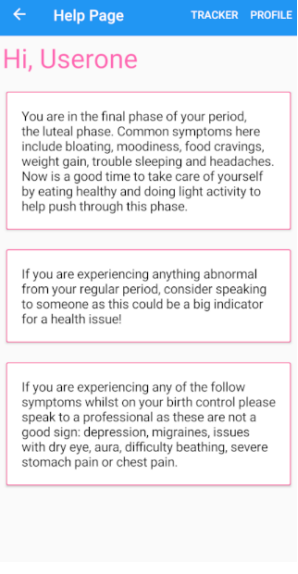
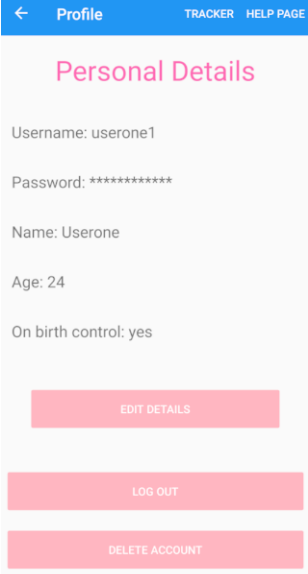
Expected outputs: Current day = 13 Phase = luteal Due in = 3 days Previous period = N/A	Actual result: 		
Test Result = PASS			

User 2:

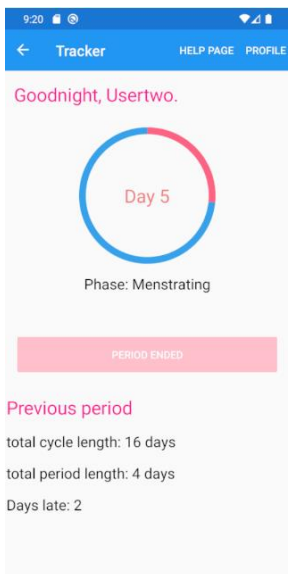
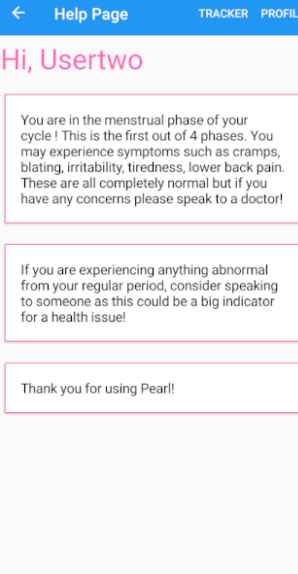
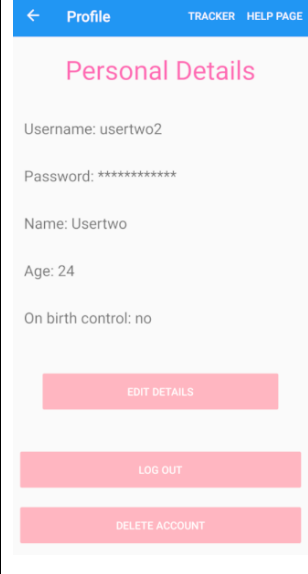
Expected outputs: Current day = 4 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days	Actual result: 		
Test Result = PASS			

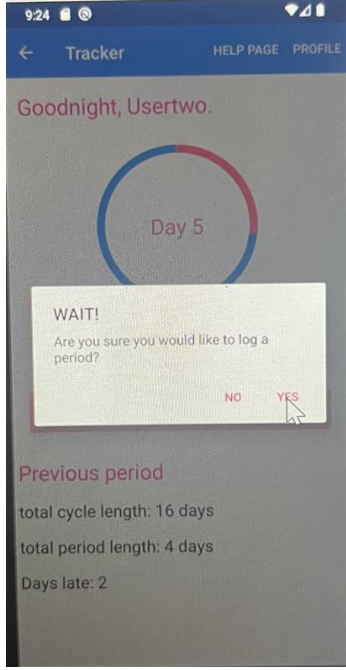
Day 14 - 16/03/2025

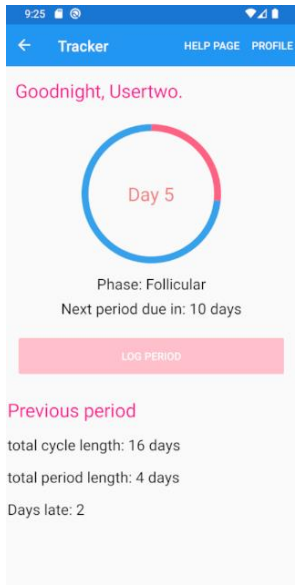
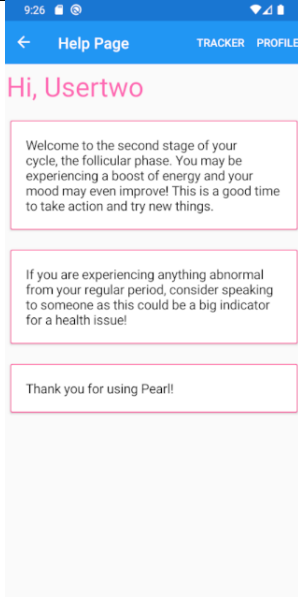
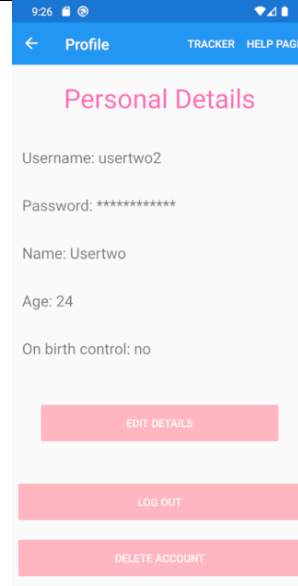
User 1:

Expected outputs: Current day = 14 Phase = luteal Due in = 2 days Previous period = N/A	Actual result: 		
Test Result = PASS			

User 2:

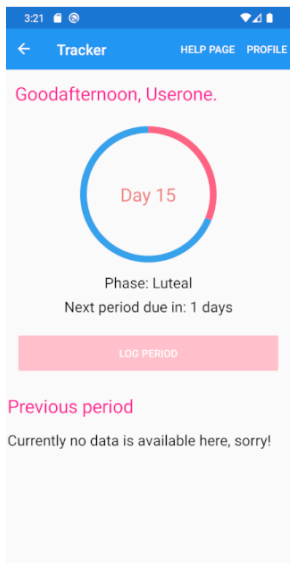
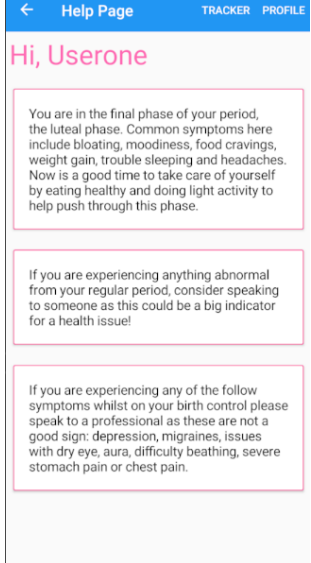
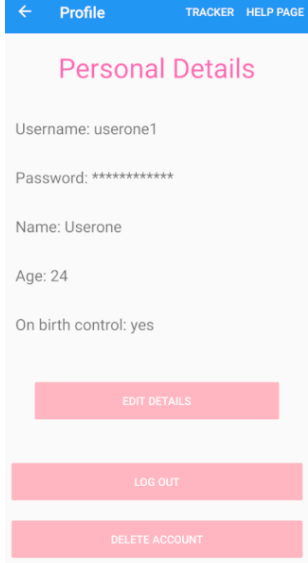
Expected outputs: Current day = 5 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days	Actual result: 		
Test Result = PASS			

- I then selected the period ended button to confirm that the users period has ended - I selected yes	
--	--

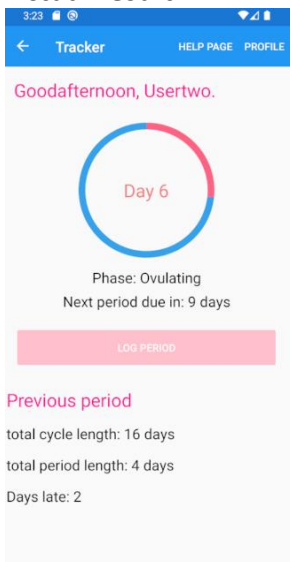
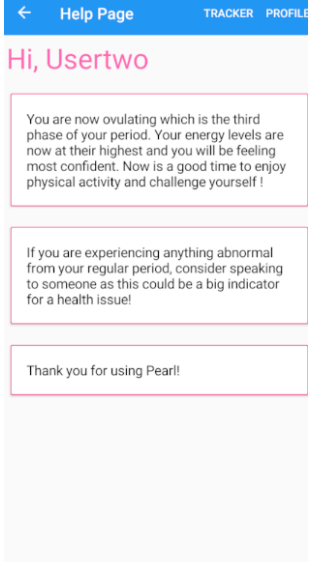
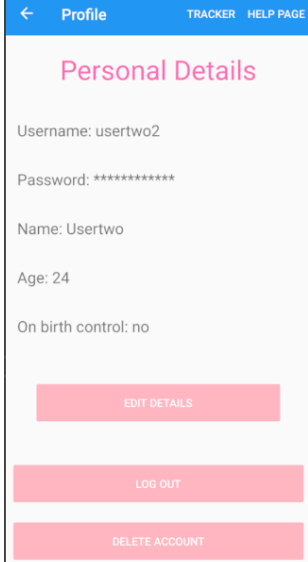
Expected outputs: Current day = 5 Phase = follicular Due in = 10 days <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days	Actual result: 		
Test Result = PASS			

Day 15 - 17/03/2025

User 1:

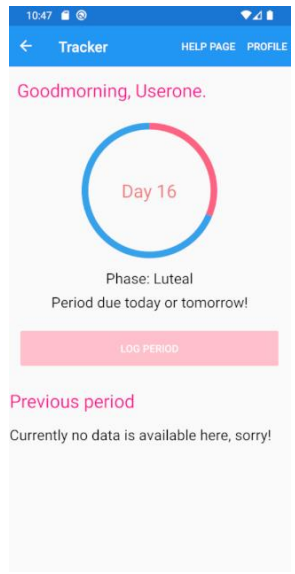
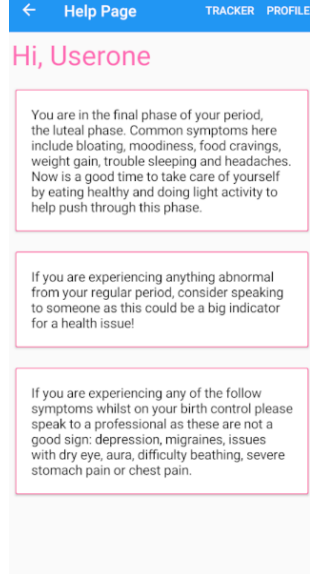
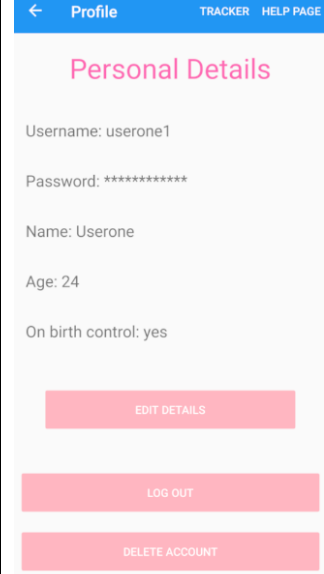
Expected outputs: Current day = 15 Phase = luteal Due in = 1 days Previous period = N/A	Actual result: 		
Test Result = PASS			

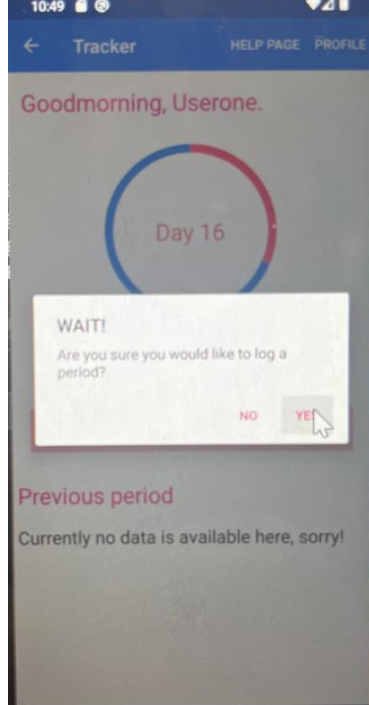
User 2:

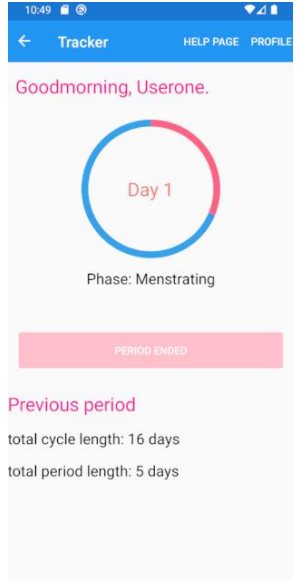
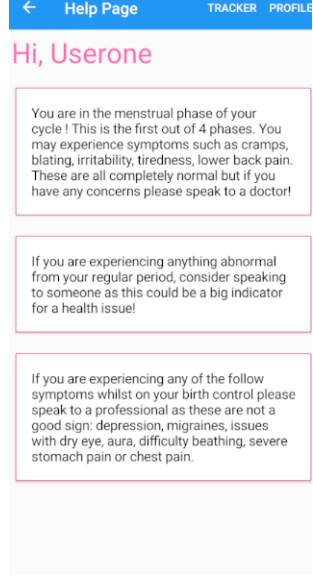
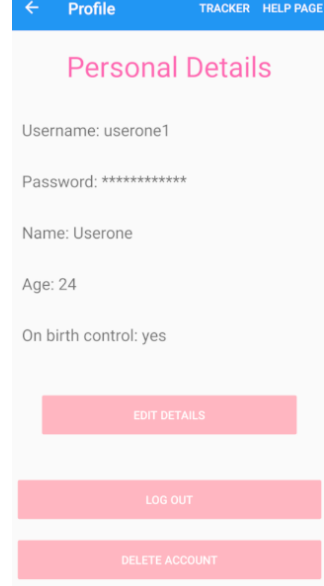
Expected outputs: Current day = 6 Phase = ovulating Due in = 9 days <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days	Actual result: 		
Test Result = PASS			

Day 16 - 18/03/2025

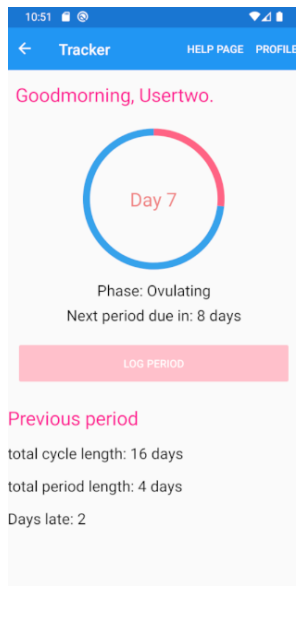
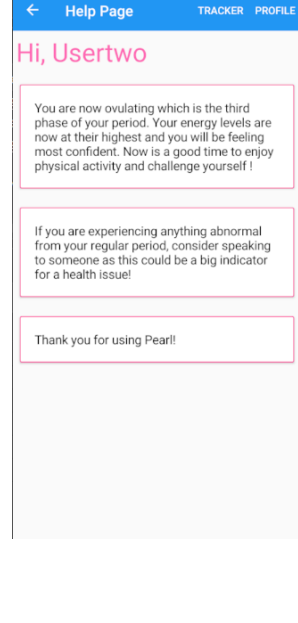
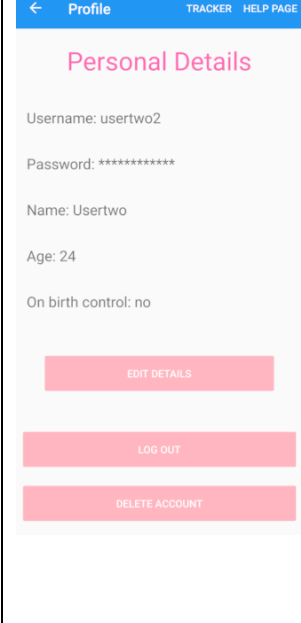
User 1:

Expected outputs: Current day = 16 Phase = luteal Due in = today or tomorrow Previous period = N/A	Actual result: 		
Test Result = PASS			

I then pressed the log period button and pressed yes	
--	--

Expected outputs: Current day = 1 Phase = menstruating <u>Previous period:</u> Cycle length = 16 days Period length = 5 days	Actual result: 		
Test Result = PASS			

User 2:

Expected outputs: Current day = 8 Phase = ovulating Due in = 8 days <u>Previous period:</u> Cycle length = 16 days Period length = 4 days Late by= 2 days	Actual result: 		
Test Result = PASS			

The 16 day testing plan testing for the functionality of my tracker shows that the main pages are working correctly and is displaying the correct information to the user.

Trace tables of HomePage functions

Objectives being tested for
4.Displays data about the user's current cycle

- 4.1. Must display a model to show their cycle in some way
- 4.2. Must display what day the user is on and their current phase
- 4.3. Must display if the user is late to start their period

In this section I will tracing the processes of a few functions in my HomePage.

I will create a user with the following inputs the follow some of the functions on the HomePage to display the correct data to the user:

• First name = hp • Username = HomePageTest90 • Birthday = 02/02/2002 • Password = !Password19	• Cycle length = 25 • Period length = 7 • Current day = 12 • BirthControl = no
---	---

Evidence of Inputs:

UpdateCurrentDay() function

Expected result:

- dates1.CCycLen = 12

start	end	diff	dates1.CCycLen
3/6/2025 11:15:07AM			
	3/6/2025 11:17:56 AM		
		12	

			12
--	--	--	----

Test Result = **PASS**

CheckIfLate() function

Expected result:

Late.DaysLate = 0

late.CCycLen	late.AvCycLen	diff	late.DaysLate
12	25		
		-13	
			0

Test Result = **PASS**

GetCurrentPhase() function

Expected result:

helppage.phase = 3

CurrentCyc	AvCyc	half	helppage.phase
0	0	0	0
12			
	25		
		12	
			3

Test Result = **PASS**

DueOrLateLabel() function

z.OnPeriod	z.DaysLate	z.CCycLen	z.AvCycLen	diff	DueOrLate.Text
false					
	0				
		12	25		
				13	
					"Next period due in: 13 days"

Test Result = **PASS**

Testing the Advice Page

Objectives being tested for
6.Use of a help page to display advice to the user 6.1.Use the user's birthday to work out their age and display appropriate information (age specific) 6.2.Display appropriate advice for if they are on birth control or not

6.3.Display information based upon what phase of their period they are in currently

The advice page displays information to the user based upon data they have input and data stored in the database. Here I will test that the correct information is displayed to each user.

Test case	Data stored in database	Expected result	Actual result	Pass or Fail
1	phase = 1	Information displayed regarding menstruation	You are in the menstrual phase of your cycle ! This is the first out of 4 phases. You may experience symptoms such as cramps, bloating, irritability, tiredness, lower back pain. These are all completely normal but if you have any concerns please speak to a doctor!	PASS
2	phase = 2	Information displayed regarding the follicular phase	Welcome to the second stage of your cycle, the follicular phase. You may be experiencing a boost of energy and your mood may even improve! This is a good time to take action and try new things.	PASS
3	phase = 3	Information displayed regarding ovulation	You are now ovulating which is the third phase of your period. Your energy levels are now at their highest and you will be feeling most confident. Now is a good time to enjoy physical activity and challenge yourself !	PASS
4	phase = 4	Information displayed regarding the luteal phase	You are in the final phase of your period, the luteal phase. Common symptoms here include bloating, moodiness, food cravings, weight gain, trouble sleeping and headaches. Now is a good time to take care of yourself by eating healthy and doing light activity to help push through this phase.	PASS
5	Birthday = 08/04/2011 (age 13)	Advice about irregular periods	If you are experiencing ongoing irregular periods, extremely bad cramps, very heavy bleeding, consider speaking to a professional for help!	PASS
6	Birthday = 27/07/2002 (age 22)	Advice about if anything abnormal happens	If you are experiencing anything abnormal from your regular period, consider speaking to someone as this could be a big indicator for a health issue!	PASS
7	Birthday = 19/05/1979 (age 45)	Advice about menopause	Lucky you ! your periods may be coming to an end yay ! Symptoms of menopause include irregular periods or a general change in your cycle, hot flushes, palpitations, difficulty sleeping, brain fog, changes in mood, dry and itchy skin and migraines. If you are concerned with any of your symptoms please seek professional help.	PASS
8	BC = true	Symptoms to watch out for	If you are experiencing any of the follow symptoms whilst on your birth control please speak to a professional as these are not a good sign: depression, migraines, issues with dry eye, aura, difficulty breathing, severe stomach pain or chest pain.	PASS

9	BC - false	A thank you message is displayed		PASS
---	------------	----------------------------------	--	------

Testing the User Profile

Objectives being tested for
7.Users are able to view their personal details 7.1.Have user be able to view their username, first name and password stored on the database
10.User able to delete account 10.1.If they press the button to delete the account, a warning is displayed to check they definitely want to 10.2.All their data is permanently deleted from the database
11.Able to log out without reloading the app 11.1The user is given the option to log out via a button

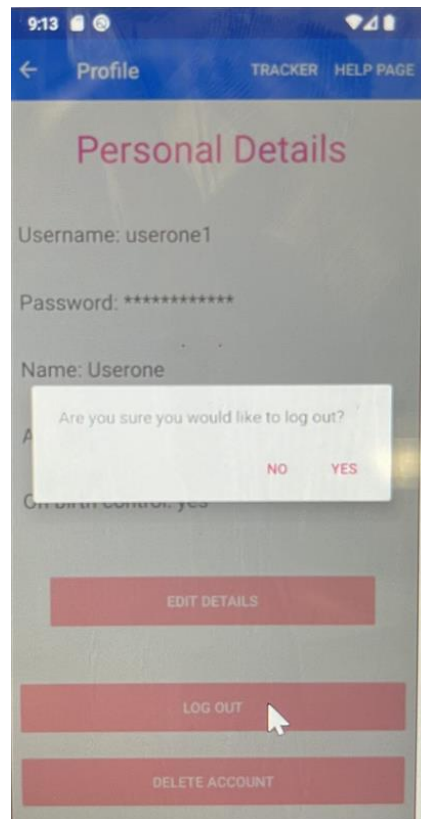
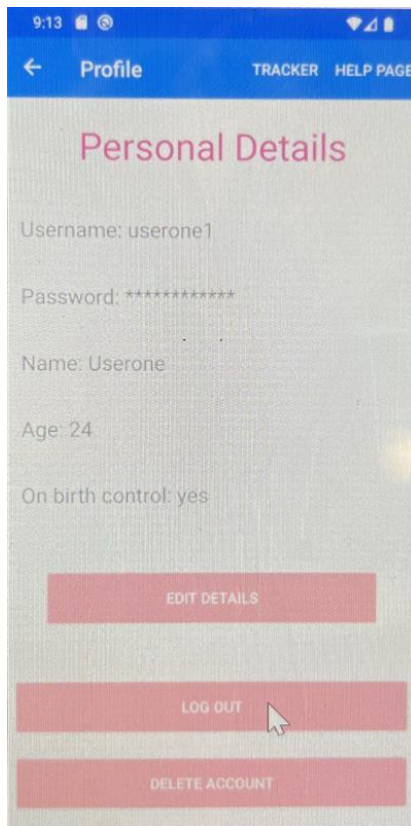
Log out button

- If the log out button is selected, a warning message should appear to say “are you sure you would like to log out”.
- If the no button is then pressed the screen should be returned to the UserProfile
- If yes is pressed then the screen should return to the WelcomePage

Test 1 – checking the warning displays:

The log out button:

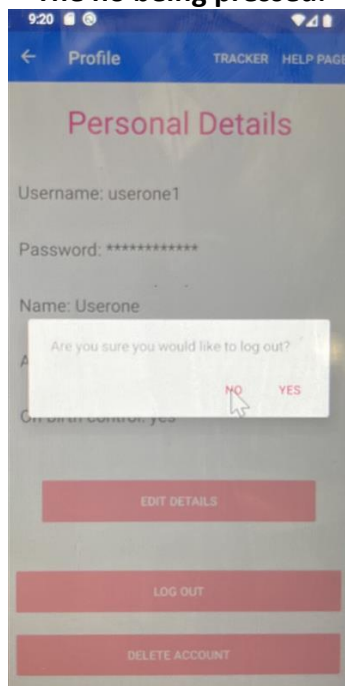
The screen displayed when it is pressed:



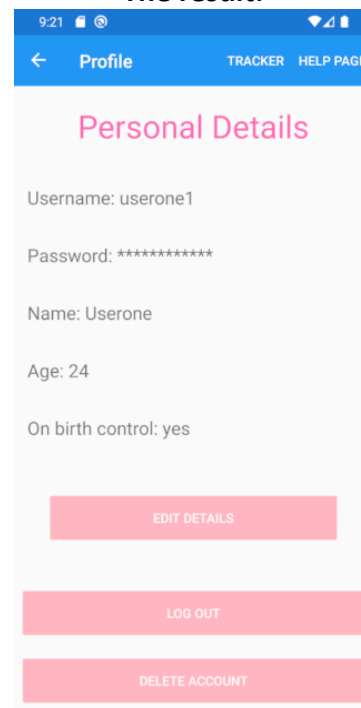
Test result = **PASS**

Test 2 – checking that when no is selected the screen returns to the user profile

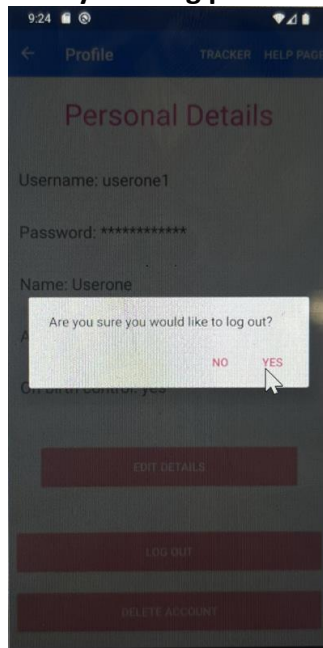
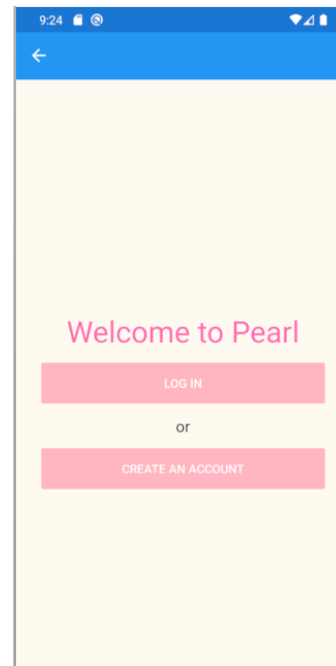
The no being pressed:



The result:

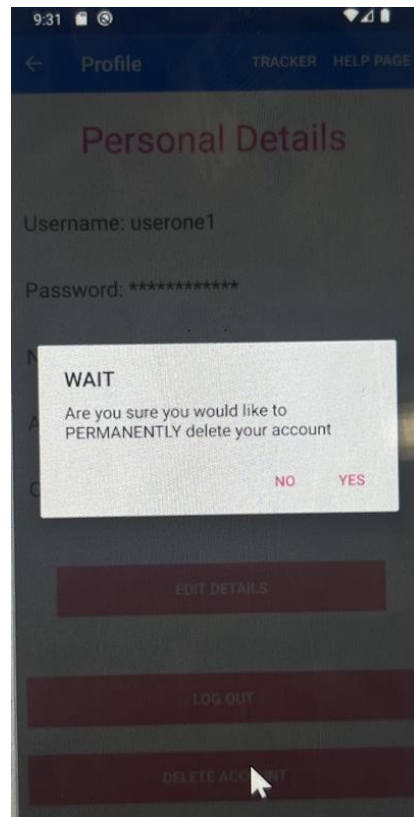
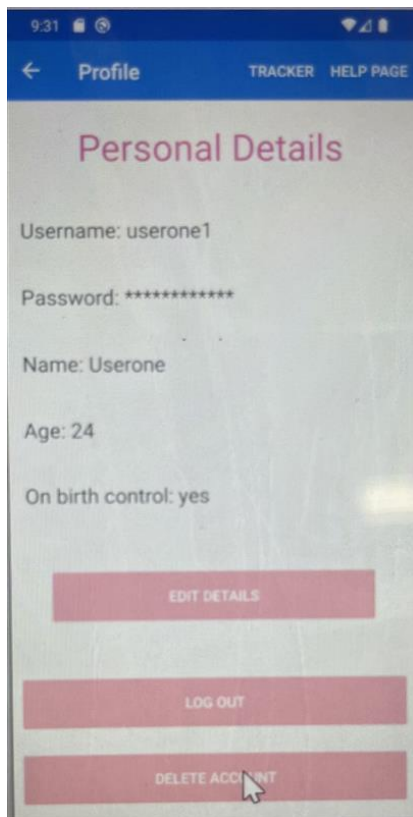


Test result = **PASS**

Test 3 – checking it logs the user out when yes is pressed**The yes being pressed:****The result:****Test result = PASS****Delete button**

- If the delete button is selected, a warning message should appear to say “are you sure you would like to delete your account”.
- If the no button is then pressed the screen should be returned to the UserProfile
- If yes is pressed then the screen should return to the WelcomePage and delete their details from the database

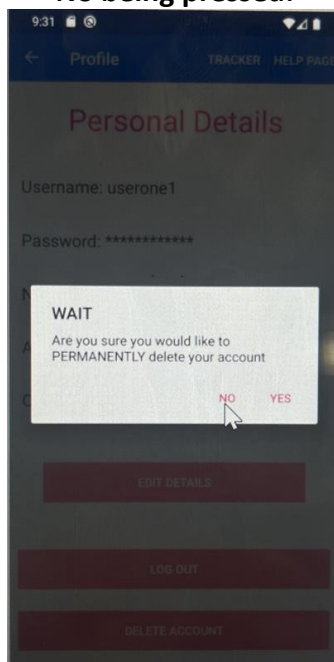
Test 1 – checking the warning displays:**The delete button:****The screen displayed when it is pressed:**



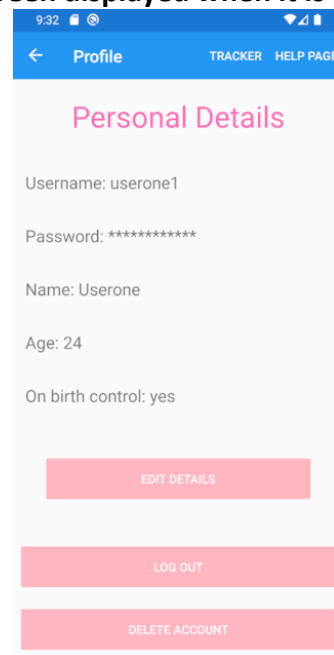
Test result = **PASS**

Test 2 – checking no returns it to the user profile:

No being pressed:

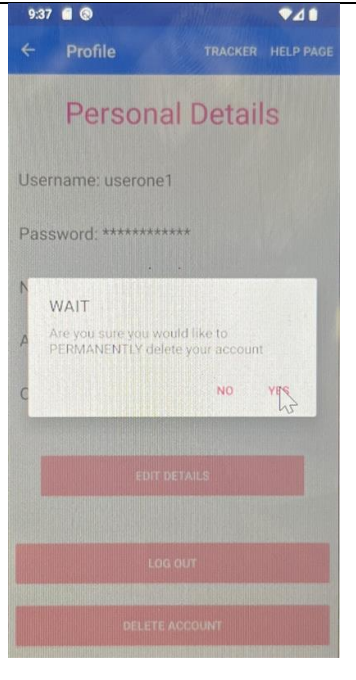
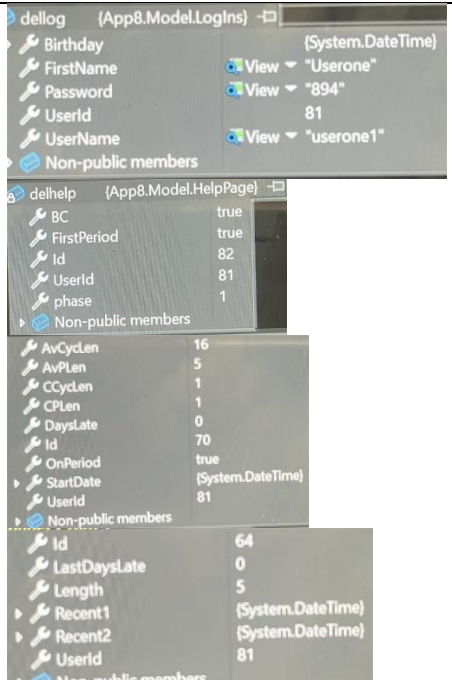
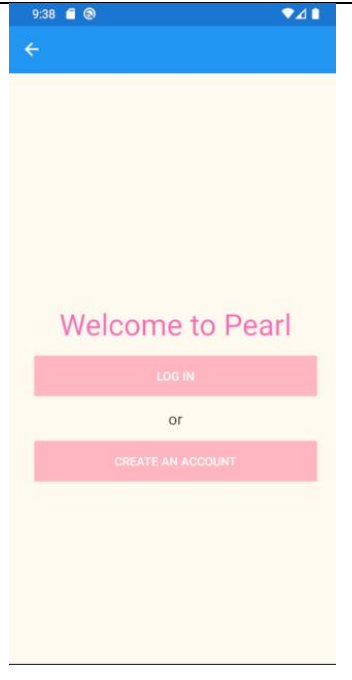


The screen displayed when it is pressed:



Test result = **PASS**

Test 3 – checking that yes causes data to be deleted from database and WelcomePage to appear:

Yes being pressed:	Data deleted from database:	Navigation:
		

Test result = PASS

Update User profile – EditDetails Page

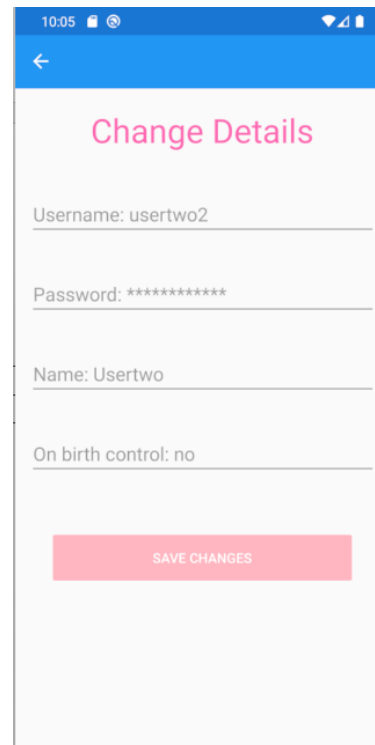
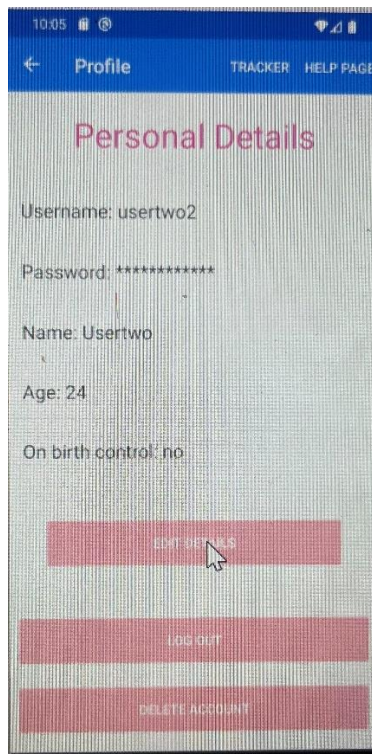
Objectives being tested for

- 8.Users are able to change their personal details
 - 8.1.Have user be able to change their username, first name and password that's stored on the database

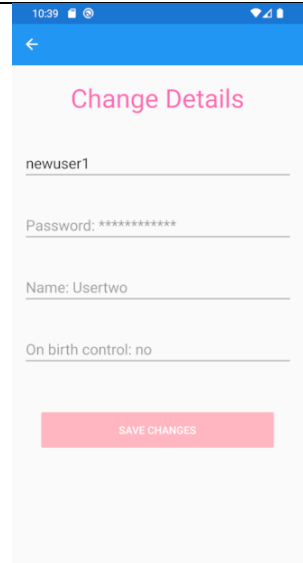
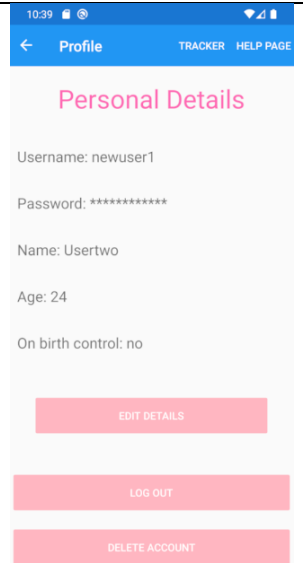
Test 1 – Navigation to Edit details:

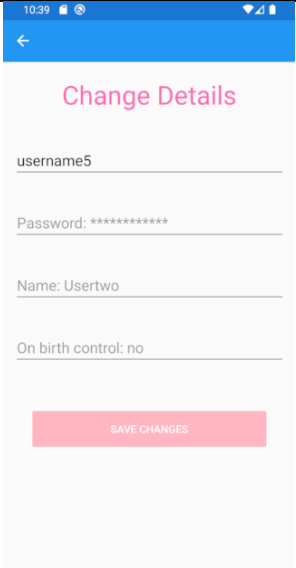
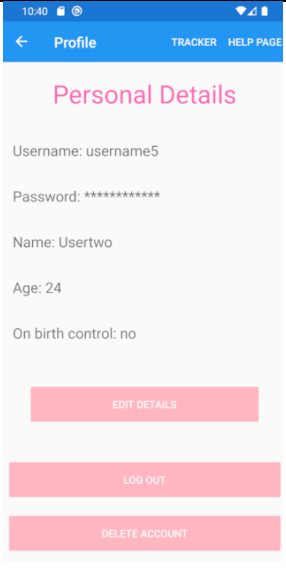
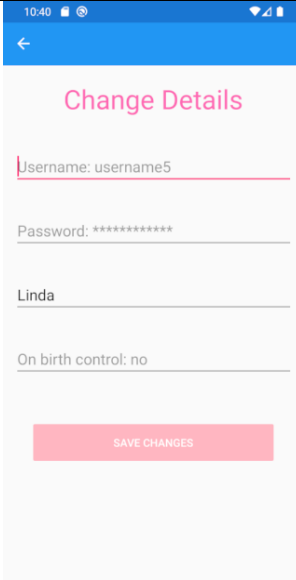
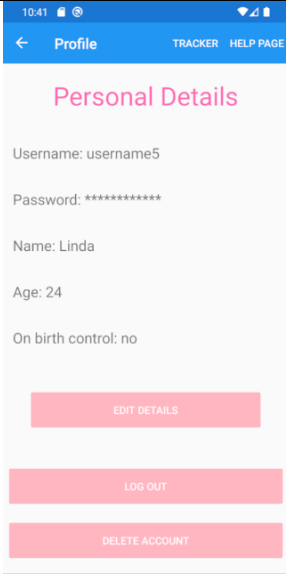
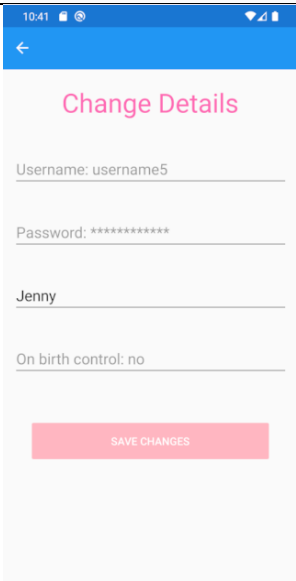
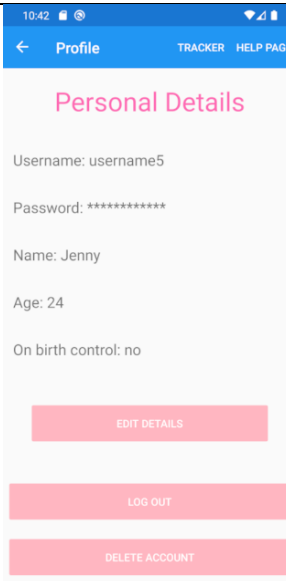
Button being pressed:

Result:



Test Result = **PASS**

Test case	Data to enter	Proof entry	Expected result	Actual result	Pass or Fail
1	Username = newuser1		When update button is pressed, username should update		PASS

2	Username = username5		When update button is pressed, username should update		PASS
3	Name = Linda		When update button is pressed, name should update		PASS
4	Name = Jenny		When update button is pressed, name should update		PASS

Evaluation

Objectives Overview

	Core Objectives
1	Connect to a database
1.1	Have a database coded into my project
	This objective has been met. I programmed a database containing four tables (Logins, HelpPage, CurrentT and LastT) into my app.
1.2	Have a connection string connecting this to the rest of the code
	This objective has been met. A connection string in a sperate page named Querying.cs has been created and allows the database to be accessed in any page anywhere in my code when called.
1.3	Query to the database including inserts, updates, reads and deletes
	This objective has been met. • Insert statements are included when creating a user in the BufferPage.xaml.cs or SignUp2.xaml.cs. • Updates are used multiple times throughout the code, especially in the HomePage.xaml.cs and EditDetails.xaml.cs pages when a user's cycle resets or they change their information stored in the database. • Reads are used all throughout the code as well to gather data stored in the databased to determine what is displayed to the user. This is used for example to determine whether a user exists when signing up or logging in or to display al the information on the HomePage or to determine the advice given to the user on the HelpPage. • Deletes are used when a new user is created from the FPFfalse page and when a user clicks the button to delete their account on the UserProfile
2	Use of a sign up and log in system
2.1	Data must be checked and validated when the user signs up (e.g. username cannot have been used before, password must meet requirements for password)
	This objective has been met. Multiple checks are performed on the data when a user creates an account to ensure that the entered data is valid, and if necessary, unique. These all work correctly and appropriately get users to re-enter information if necessary. • All the entries are checked to see that they aren't blank or contain whitespace

	• The username is checked to see that it doesn't already exist, only consists of letters and numbers and is an appropriate length • The first name is checked to ensure that it only contains letters and is an appropriate length • The birthday is checked to ensure that a valid date is entered in the format dd/mm/yyyy • The password is checked to ensure it contains letters, numbers, special characters and is an appropriate length • The birth control entry is checked to ensure that the user enters yes or no or another variation of these. However, if this is left blank then the Boolean is set to false as this is the only entry that allows for this • The average cycle length is checked to ensure it is a valid integer of an appropriate size • The average period length is checked to ensure it is a valid integer of an appropriate size • The current cycle length is checked to ensure it is a valid integer of an appropriate size
2.2	Must be easy for the user to understand what data they should input
	This objective has been met to a certain extent. My users were satisfied with how the log in and sign up worked, however some said they were slightly confused on how some of the questions were worded. More detail regarding this is in the Interviews with clients section of my evaluation.
2.3	When an account is created, they must be added to the database
	This objective has been met. When a user is created, all their details are stored into the database along with a randomly generated UserId so their details can be accessed later in the program
2.4	Log in only allowed once been checked against the database to see if the user exists
	This objective has been met. The username the user enters is checked to ensure it exists in the database. It also is then checked against the entered password to ensure the correct password is entered as well.
3	The system must predict when the user will get their next period
3.1	Uses data stored in database to make a prediction of when their next period will be
	This objective has been met. • The CurrentDay and AvCycLen values stored in the database are used to predict how many days away the users next period is and displays this information to them as a number of how many days away it is.

	• If it is one or zero days away then it displays that their period is due today or tomorrow. • If they are late to starting it informs the user of how many days late they are to starting their period.
3.2	When a full cycle has passed, the system must use the data in the database and the data from the cycle that's just ended to update its predictions
	This objective has been met. When the user logs that their cycle has ended, data is transferred across to the LastT table to display to the user a summary of the cycle that's just ended. It also updates the CurrentT table to start their cycle back to day one.
4	Displays data about the user's current cycle
4.1	Must display a model to show their cycle in some way
	This objective has been met. A doughnut pie chart is used to display their cycle in proportion of how long their average cycle is to how long their average period is . This model changes based on these values so this may look different for each user and also may look different for one user one cycle compared to another.
4.2	Must display what day the user is on and their current phase
	This objective has been met. • The code uses the StartDate stored in the database and the date of the current day to determine what day the user is on currently and displays this to them • The code then uses this current day, the OnPeriod Boolean and the average cycle length to determine what phase of their period the user is on. It displays this to the user as "menstruating", "luteal", "ovulating" or "follicular"
4.3	Must display if the user is late to start their period
	This objective has been met. If the users calculated current day is greater than the stored average cycle length then this value is displayed to the user to inform them that they are x days late to starting their period
5	Displays data about the user's previous cycle
5.1	Uses data stored in database to display to user how long their previous cycle and period lasted, and if they were, how many days late they were (if they were)
	This objective has been met. • When a user logs their period as ended then LastT is updated with the information stored in CurrentT

	- Recent1 and Recent2 are used to display how many days their complete cycle lasted - Length is used to display how many days their period lasted - LastDaysLate is used to display how many days late they were over their average cycle length (if this is stored to be greater than 0)
6	Use of a help page to display advice to the user
6.1	Use the user's birthday to work out their age and display appropriate information (age specific)
	This objective has been met. A function is called to determine a user's age based upon the stored birthday for them in the database. Then this is used to determine what advice should be displayed to the user on the HelpPage
6.2	Display appropriate advice for if they are on birth control or not
	This objective has been met. This is stored as true or false in the database using the BC Boolean. Depending on if this is true or false, then appropriate information is displayed to the user.
6.3	Display information based upon what phase of their period they are in currently
	This objective has been met. The database stores phase with the value 1, 2, 3 or 4 which corresponds each to a different phase and therefore depending on this, different information is displayed to the user.
7	Users are able to view their personal details
7.1	Have user be able to view their username, first name and password stored on the database
	This objective has been met. Within the UserProfile.xaml the user can view their username, first name, password and also if they are on birth control or not. This information displayed is taken from the information stored on the database for them.
8	Users are able to change their personal details
8.1	Have user be able to change their username, first name and password that's stored on the database
	This objective has been met. - First the user selects the edit details button on the UserProfile.xaml - This redirects them to the EditDetails.xaml in which they can enter in new data for these - If the user enters in data in any of the entries, when they select to update their profile, the data is checked

	Then it is updated in the database
9	Users are able to use a simplified version of the app if they have never had a period
9.1	Have the app ask the user if they have had their first period when they create an account
	This objective has been met. After the user inputs their username, birthday, first name and password they are directed to the BufferPage and here they are asked if they have had a period before.
9.2	If they have not, the app is directed to a temporary homepage
	This objective has been met. If the user selects the button labelled no, then they are directed to the FPFalsePage and from there can log their first period when they get it.
9.3	It also creates a temporary account for them
	This objective has been met. When the user selects the button labelled no and are directed to the FPFalse page, a temporary account is created. This stores their username, first name, password and OnPeriod as false within the database so that they can then log back into their account after closing the app.
9.4	Have a button that leads the user to finish creating their account
	This objective has been met. When the user logs that they have got their first period, the app redirects them to the SignUp2 page and allows the user to finish creating their account Their information is stored on te database.
10	User able to delete account
10.1	If they press the button to delete the account, a warning is displayed to check they definitely want to
	This objective has been met. When the user selects the delete account button on the UserProfile, a warning message asking them to confirm that they actually want to delete their account of not is displayed.
10.2	All their data is permanently deleted from the database
	This objective has been met. If the user selects yes on the warning message, all the data stored for them in the database is deleted and they are navigated back to the WelcomePage.
11	Able to log out without reloading the app
11.1	The user is given the option to log out via a button

	This objective has been met. • A button labelled “log out” is located in the UserProfile. If the user presses this, a message is displayed to confirm if they want to log out or not. • If they press no then they remain on the UserProfile. • If they press yes then they are directed back to the WelcomePage and must reenter their details to get back into their account and view their information.
12	Navigation between all the pages works
12.1	Navigation in log in and sign up system works
	This objective has been met. • The button on the WelcomePage correctly navigates the user to the LogIn or SignUpPage. • The LogIn page correctly navigates the user to the HomePage when they enter their details correctly. • The SignUpPage correctly directs the user to the BufferPage, followed by the SignUp2Page and the HomePage.
12.2	Navigation between the tracker, advice page and user profile works
	This objective has been met. I have implemented a tool bar which allows you to access all three pages with ease. This all works correctly.
12.3	Navigation for a user who has not experienced their first period works
	This objective has been met. The user is correctly directed to the FPFalsePage when necessary and are then correctly navigated to finish setting up their account when they log their first period
13	Users use the app with ease and like the interface
13.1	My clients feel like the app is easy to understand in terms of questions the system asks and the outputs displayed to them etc
	This objective has been met to a certain extent. • They feel like the log in and all the main pages are straightforward and easy to understand and use • However, some feel like the wording of the questions in the SignUp2 page are slightly hard to interpret. This is spoke about further in the summary of my clients interviews
13.2	My clients can easily navigate between the pages in the app
	This objective has been met. All my clients said the app was easy to navigate between pages.

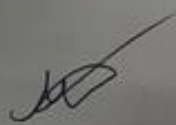
	Advanced objectives
1	Implement a merge sort and binary search
1a	Use of a merge sort to order the user ids that have been used
	This objective has been met. A merge sort of the existing UserIds in the database is performed. This allows for these to then be searched to ensure that the new UserId does not already exist. This takes place in the SignUpPage.
1b	Use of a binary search to check a randomly generated user id doesn't exist in the sorted list
	This objective has been met. A binary search has been implemented to search the list that was sorted using the merge sort. This takes place in the SignUpPage.
2	Implement a hash code for the password
2a	Hash code works
	This objective has been met. A hash code has been implemented and has been tested to show it works in generating a string of integers based on the contents of the password input by the user.
2b	Stored instead of the password for security purposes
	This objective has been met. The hash code created is stored as the password in the database. Then when a user logs into their account, the hash code is performed on their entered password and checked with the hash code stored in the database.
3	Symptom tracker
3a	User can view common symptoms they may experience currently in their cycle
	This objective has been met. Within the HelpPage the user is displayed possible symptoms they may experience based upon what phase of their cycle they are in and if they are on birth control or not.
3b	User can log symptoms
	This objective has not been met because my clients did not rank this as a high priority objective.
3c	User can view their previously logged symptoms
	This objective has not been met because my clients did not set this as an objective, I did, so therefore wasn't a priority.
4	Mood tracker

4a	User can view common moods they may experience currently in their cycle
	This objective has been met. Within the HelpPage the user is displayed possible moods they may experience based upon what phase of their cycle they are in and if they are on birth control or not.
4b	User can log moods
	This objective has not been met because my clients did not rank this as a high priority objective.
4c	User can view their previously logged moods
	This objective has not been met because my clients did not set this as an objective, I did, so therefore wasn't a priority.

Proof of interviews with clients

To what extent do you think the code met your objectives?	
I feel as if it met the objectives and met my goals. I'm happy with this.	
How easy do you think the app was to navigate and use?	
I think it was easy to use. I did feel a bit misled on the questions when creating an account. Other than that it was easy.	
Is there anything you don't like about the system? Please explain.	
I don't like how creating an account works as I found it was misleading.	
Please use this space to suggest any improvements e.g. other features you'd like added or anything you would change.	
make it easier to make an account and also add the ability to add personalised notes regarding mood and symptoms.	
Signature: 	Date: 23/03/25

To what extent do you think the code met your objectives?	
I feel like the code met all of my objectives I really liked the display of the home page and the model app	
How easy do you think the app was to navigate and use?	
I feel like the app was very easy to navigate and use It was easy to understand how the app worked	
Is there anything you don't like about the system? Please explain.	
I didn't like how the previous cycle display looked and worked I feel like it would be better if more detail was added such as more cycle data + the previous symptoms and moods	
Please use this space to suggest any improvements e.g. other features you'd like added or anything you would change.	
Improvements on past visual data would be good such as past symptoms + moods to allow for better tracking + comparison of cycles	
Signature: 	Date: 27 March 2025



To what extent do you think the code met your objectives?	
it met all my objectives and I particularly liked the help page	
How easy do you think the app was to navigate and use?	
it was easy to navigate and use and I liked the design	
Is there anything you don't like about the system? Please explain.	
There's no option to record your symptoms and mood throughout my cycle	
Please use this space to suggest any improvements e.g. other features you'd like added or anything you would change.	
It would be useful to have a feature which was able to record mood and symptoms	
Signature: <i>Billie Tinniswood</i>	Date:

To what extent do you think the code met your objectives?	
I feel like the code has met all of the objectives needed.	
How easy do you think the app was to navigate and use?	
Very easy to use and understood it clearly, navigation wasn't difficult either.	
Is there anything you don't like about the system? Please explain.	
The only thing I don't like is that it only predicts one cycle into the future, so you cannot plan too much in advance.	
Please use this space to suggest any improvements e.g. other features you'd like added or anything you would change.	
I would like the app to predict more than 1 (maybe 2-3) cycles into the future to better understand when my period would occur	
Signature: 	Date: 25/3/25

Overview of client interviews

Question 1 – To what extent do you think the code met your objectives?

All of my clients have felt as if all of their objectives have been met. They are all happy with the app overall. One client mentioned that they liked particularly liked the home page and another the advice page.

Question 2 – How easy do you think the app was to navigate and use?

All of my clients said that the navigation and use of the app was easy and they understood it. This means that the navigation is sufficient for a user to use without getting confused.

For the most part my clients found it easy to understand the questions being asked of them throughout the app, however one client suggested that the questions being asked when they create the account is slightly hard to understand, therefore I will look further into improving this.

Question 3 – Is there anything you don't like about the system? Please explain.

My clients all has slightly different criticisms of the period tracker app, these included:

- As highlighted above, the questions being asked when creating an account were sometimes hard to understand.
- Another client suggested that they didn't like how they could not record their symptoms.
- Another client suggested that they dislike that the app only predicts one cycle into the future and would find it helpful for it to predict two or three.
- Another client suggested that they didn't like the design of the previous cycle.

Question 4 – Please use this space to suggest any improvements e.g. other features you'd like added or anything you would change.

The suggested improvements were mostly closely related to what the clients disliked about the period tracker app. The suggestions from my clients were:

- Make it easier to create an account (in terms of understanding of questions being asked)
- Ability to record mood, symptoms and personalised notes
- App predict 2-3 cycles into the future
- Change the design of the previous cycle section, add more data and have it display previous symptoms and moods.

Improvements for the code

Using my clients suggestions, I have created a plan of improvements for my code

Suggestion 1 - Make it easier to create an account (in terms of understanding of questions being asked)

To fix this I would simply change the wording of the questions within the SignUp2.xaml to ensure its easier for my clients and users to understand. This would ensure that they do not make mistakes or get confused when creating their account. Here is a list of the questions currently being displayed and what I would change them to:

Current label wording	Changed label wording
How long does your cycle last?	How long does your full cycle normally last?
How long does your period normally last?	n/a (I wouldn't change this as I feel as if it is not misleading or easily misunderstood)
What day of your cycle are you currently on?	What day of your current cycle are you on right now?
Are you on any birth control?	n/a (I wouldn't change this as I feel as if it is not misleading or easily misunderstood)

Suggestion 2 - Ability to record mood, symptoms and personalised notes

To meet this suggestion I would start by creating another table within my database with a link to the user using their UserId. I would make use of a button on the home page reading "record symptoms" which would direct the user to a separate page. Here I would give the user the option to select their symptoms and moods from a list or the option to add their own ones (this incorporates the suggestion for personalised notes). If the select to add their own, any input data must be validated to ensure it is a string of letters and not random symbols or numbers. The personlised note or the symptoms or moods selected by the user will then get stored to the new table in the database alongside the phase they were in when they entered this. This will then be used to display the users' common moods around certain times of their cycle back to them. Fo example, when they are in the luteal phase of their next cycle, a section on the help page will display "moods you commonly experience during the luteal phase" and list those. This would help my users understand their cycles better and would be beneficial for them. This would also meet my client's suggestion.

Suggestion 3 - App predict 2-3 cycles into the future

This suggestion can be met by using data that is already stored on the database for the user once they have created an account. A calculation can be performed, similar to the calculation used to predict how many days away their next cycle is, to predict how many days away the next 3 cycles of the user's period is. Another model, such as a calendar, could be used to display this information to the user rather than having a statement that says something like "Cycle after due in: 79 days" as this may be abit misleading. Just giving a number like this would cause confusion and mean the user has to work this out themselves,

hence why a calendar model would be more efficient. This would meet my client's suggestion and be beneficial to the user.

Suggestion 4 - Change the design of the previous cycle section, add more data and have it display previous symptoms and moods.

I would change the previous cycle section to be on a separate page, by having a button on the main page that says, "View previous cycle". I would have a more in depth display on the data from their previous cycle including the length of each phase that occurred throughout. I would use the data stored in the new table of the database, as described above, to also display the common symptoms and moods my user logged associated with each phase.

Summary of project

Testing my project went successfully with the objectives being met by the code. In places where the code did not work, simple changes were applied and the project then worked. For example, on day 7 of testing the functionality of the homepage within the 16 day testing plan, the code failed to calculate the days the user was ovulating, however I amended the code so that it worked. The testing of the functionality of the home page (the 16 day testing pan) showed that the app successfully worked for two different users. It showed that for users who have regular periods, all their information is displayed correctly and accurately. When a full cycle has passed their details stored in the database are not changed (as expected). However, for someone with irregular periods the testing of the functionality showed that their details are changed after a full cycle and that their details are accurately updated. The HomePage was seen to be working fully at displaying the appropriate data to the user every single day. The button within the HomePage was seen to be working correctly as well to log periods for the user.

Overall, my project has met all the core objectives set by my clients. The project reflects the clients' intentions and they are all happy. A couple of advanced objectives did not get met but these were not necessary as my clients did not feel these were a priority. Other suggestions were made by my clients when they reviewed my project are all possible and easy to add within a couple of weeks, making them realistic.

Bibliography

Apple Inc. (2024). *Clue Period Tracker & Calendar Ratings and Reviews*. [Online]. apps.apple.com. Available at: <https://apps.apple.com/us/app/clue-period-tracker-calendar/id657189652?see-all=reviews> [Accessed 11 September 2024].

Clue by Biowink GmbH. (2024). *Clue Ovulation & Period Tracker*. [Online]. helloclue.com. Available at: <https://helloclue.com/> [Accessed 11 September 2024].

Flo Health UK Limited. (2024). *Flo Health*. [Online]. Flo.Health. Last Updated: 2024. Available at: <https://flo.health/about-flo> [Accessed 11 September 2024].

Google Play. (2024). *Luna Period Tracker For Teens*. [Online]. play.google.com. Available at: <https://play.google.com/store/apps/details?id=com.evolutus.luna&hl=en-US&pli=1> [Accessed 11 September 2024].

John D. McKinnon. (2021). *FTC Reaches Settlement With Flo Health Over Fertility-Tracking App*. [Online]. wsj. Last Updated: 14th January 2021. Available at: <https://www.wsj.com/articles/ftc-reaches-settlement-with-flo-health-over-fertility-tracking-app-1161> [Accessed 11 September 2024].

Luna Life Limited. (2023). *Unlock unlimited content with luna premium*. [Online]. weareluna.app. Available at: <https://weareluna.app/premium/> [Accessed 11 September 2024].

Trustpilot A/S. (2024). *flo.health*. [Online]. flo.health Reviews. Available at: <https://uk.trustpilot.com/review/flo.health> [Accessed 11 September 2024].